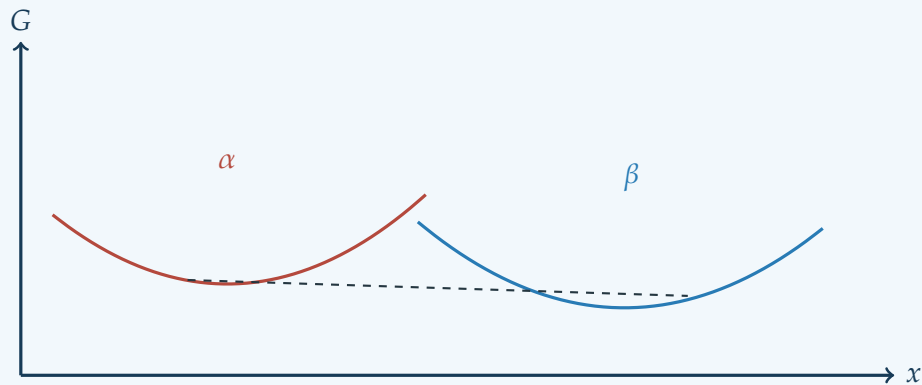


NPTEL COURSE NOTES

Thermodynamics and Kinetics of Materials

A continuous path from energy and entropy
to phase transformations and diffusion



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Understand the physics first; the equations then have a reason.

How to read these notes

These notes accompany the NPTEL lectures rather than replace them. They preserve the sequence and physical intuition of the handwritten classroom development while making the mathematics searchable, accessible and consistently typeset. The lectures provide the spoken explanation and worked discussion; these notes provide a connected derivation, carefully labelled figures, checks and computational exercises. The main argument is set in a readable serif face; equations use matching LaTeX mathematics. Blue handwritten remarks retain the classroom voice: questions, warnings and short physical interpretations.

Typography used in the reconstructed course

Layer	Purpose
LaTeX serif text	Definitions, derivations and explanations
Displayed mathematics	Equations that students should be able to reproduce
<i>Handwritten blue notes</i>	Questions and physical intuition from the classroom
<i>Red handwritten warnings</i>	Common mistakes or assumptions that must be checked

Course references

The NPTEL lectures and the handwritten classroom notes provide the sequence and voice of this course. The following books provide complementary derivations, terminology, data interpretation and problem-solving practice:

1. Peter Atkins and Julio de Paula, *Atkins' Physical Chemistry*, 8th ed., Oxford University Press, 2006.
2. Robert DeHoff, *Thermodynamics in Materials Science*, 2nd ed., CRC Press, 2006.
3. Richard A. Swalin, *Thermodynamics of Solids*, 2nd ed., Wiley, 1972.
4. Boris S. Bokstein, Mikhail I. Mendelev and David J. Srolovitz, *Thermodynamics and Kinetics in Materials Science: A Short Course*, Oxford University Press, 2005.
5. Herbert B. Callen, *Thermodynamics and an Introduction to Thermostatistics*, 2nd ed., Wiley, 1985.
6. Long-Qing Chen, *Thermodynamic Equilibrium and Stability of Materials*, Springer, 2022.
7. Mats Hillert, *Phase Equilibria, Phase Diagrams and Phase Transformations: Their Thermodynamic Basis*, 2nd ed., Cambridge University Press, 2008.
8. David R. Gaskell and David E. Laughlin, *Introduction to the Thermodynamics of Materials*, 6th ed., CRC Press, 2017.
9. Robert W. Balluffi, Samuel M. Allen and W. Craig Carter, *Kinetics of Materials*, Wiley, 2005.

The worked problems in these notes are newly written teaching examples. Their style emphasizes the strategy used by DeHoff and by Bokstein, Mendelev and Srolovitz: identify the controlled variables, select the potential, write the equilibrium condition, calculate with units, and test the physical meaning of the answer.

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Week 1

Systems, energy and the first law

Where we are in the course

state variables → thermodynamic potential → equilibrium → driving force
→ flux or rate → microstructural change

These opening lectures establish the first box: how a material state is described and how its energy changes when heat, work or matter crosses the boundary.

1.1 Thermodynamics of materials

Thermodynamics relates heat and work to measurable properties of materials. Equilibrium thermodynamics is rate independent: it identifies possible equilibrium states and the changes between them without specifying how rapidly those changes occur. Later, kinetics will supply the rates.

The laws introduce three central ideas:

1. energy is conserved;
2. entropy supplies a direction for spontaneous change;
3. the entropy of a perfect crystal approaches zero as the temperature approaches absolute zero.

1.2 Energy from macroscopic and microscopic viewpoints

Energy is the capacity of a physical system to produce change or perform work. Macroscopically we distinguish kinetic, potential and internal energy. The internal energy U collects the microscopic kinetic energies and interactions of the particles constituting the material.

For a monatomic ideal gas, translational motion has three quadratic degrees of freedom. Equipartition assigns an average energy $\frac{1}{2}k_B T$ to each quadratic term, giving

$$\bar{u} = \frac{3}{2}k_B T, \quad U_m(T) = U_m(0) + \frac{3}{2}RT. \quad (1.1)$$

Condensed phases additionally contain substantial interaction energy. Their description therefore cannot generally be reduced to independent particles.

Microscopic pictures explain why energy is stored. Thermodynamics tells us how to account for it without following every atom.

Internal energy is extensive. If two noninteracting systems are combined,

$$U = U^{(1)} + U^{(2)}. \quad (1.2)$$

The molar internal energy $U_m = U/n$ and specific internal energy $u = U/m$ are intensive.

1.3 State functions and path functions

A state is specified by a sufficient set of thermodynamic properties such as U, V, T, p and the component amounts N_i . A state function depends only on the state. Thus

$$\Delta U = U_f - U_i = \int_i^f dU \quad (1.3)$$

is independent of the path. Heat and work describe transfers along a path and are written δq and δw .

For $z(x, y) = xy$,

$$dz = y dx + x dy \quad (1.4)$$

is exact. By contrast, a form such as $\delta w = y dx$ generally gives different integrals along different paths with the same endpoints.

A differential symbol carries physics: dU is a state change; δq and δw are transfers.

1.4 System, surroundings and boundary

The system is the portion of the universe selected for study. The surroundings are everything outside it that can interact with it. A boundary separates them.



The boundary is part of the model.

	Open	Closed	Isolated
Energy transfer	allowed	allowed	forbidden
Matter transfer	allowed	forbidden	forbidden

Wall properties should be stated independently. An adiabatic wall forbids heat transfer; a diathermal wall permits it. An impermeable wall forbids matter transfer. A rigid wall forbids boundary displacement.

1.5 The first law for a closed system

Using the convention that heat supplied to the system and work done on the system are positive,

$$\boxed{dU = \delta q + \delta w.} \quad (1.5)$$

For pressure-volume work,

$$\delta w = -p_{\text{ext}} dV, \quad dU = \delta q - p_{\text{ext}} dV. \quad (1.6)$$

Expansion has $dV > 0$, so the surroundings do negative work on the system. In a reversible, quasistatic change, p_{ext} differs only infinitesimally from the system pressure.

1.5.1 Work depends on the path

The pressure-volume work is the signed area under the path in a p - V diagram:

$$w = - \int_{V_i}^{V_f} p_{\text{ext}} dV. \quad (1.7)$$

Therefore:

$$\text{isochoric: } w = 0, \quad (1.8)$$

$$\text{isobaric: } w = -p_{\text{ext}}(V_f - V_i), \quad (1.9)$$

$$\text{reversible isothermal ideal gas: } w = -nRT \ln\left(\frac{V_f}{V_i}\right). \quad (1.10)$$

For a polytropic path $pV^r = K$, the same integral is evaluated using $p = KV^{-r}$.

1.5.2 Free expansion

When an ideal gas expands into vacuum, $p_{\text{ext}} = 0$ and hence $w = 0$. If the enclosure is adiabatic, $q = 0$, so $\Delta U = 0$. The process is irreversible even though the initial and final states are equilibrium states.

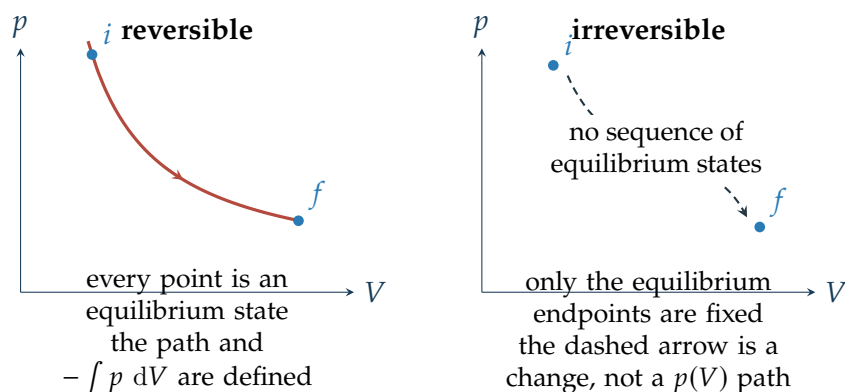
A useful distinction

Quasistatic describes a path through states arbitrarily close to equilibrium. Reversible additionally requires that both system and surroundings can be restored without a net change. Free expansion satisfies neither condition.

1.5.3 When does a process have a path on a p - V diagram?

A point on a thermodynamic diagram represents an equilibrium state. During a reversible process the system passes through a continuous succession of states arbitrarily close to equilibrium. Its pressure is defined at every stage, so the process can be drawn as a path and the area under that path gives the reversible pressure-volume work.

During a finite, irreversible change the system need not have one uniform pressure or temperature. The initial and final equilibrium states may be plotted, and the externally imposed pressure may be known, but joining the endpoints by an ordinary equilibrium curve would falsely assign equilibrium states to the interior of the process.



1.5.4 Equilibrium and spontaneous change: ice and water

At 1 atm and 0 °C, ice and liquid water can coexist at equilibrium. Their temperatures, pressures and chemical potentials are equal, so there is no net thermodynamic driving force for one phase to consume the other. Molecules may still cross the interface in both directions; equilibrium means that the macroscopic rates balance.

At 1 atm and 10 °C, a mixture containing ice and water is not an equilibrium state. Liquid water is the stable phase, the chemical potentials are unequal, and the ice melts spontaneously. That finite change produces entropy and is irreversible. After all the ice has melted and the temperature has become uniform, the system reaches a new equilibrium state.

This state–process distinction and the ice–water example follow the equilibrium-first presentation in Long-Qing Chen’s *Thermodynamic Equilibrium and Stability of Materials*.

0 °C, 1 atm
ice $\rightleftharpoons$ water
 $\mu_{\text{ice}} = \mu_{\text{water}}$
no net transformation

two-phase equilibrium

10 °C, 1 atm
ice $\rightarrow$ water
 $\mu_{\text{ice}} > \mu_{\text{water}}$
spontaneous melting

non-equilibrium, irreversible process

State versus process

Equilibrium describes a state. Reversible and irreversible describe processes. An irreversible process can connect two equilibrium states even though the intervening system states cannot be represented by a single equilibrium path.

1.6 Other forms of work

Pressure-volume work is one member of a broader conjugate-pair structure:

$$\delta w = \sum_j X_j \, dY_j, \quad (1.11)$$

where X_j is a generalized force and Y_j its generalized displacement. Surface work, for example, contains γdA , with surface tension γ conjugate to area A .

1.7 Open systems and flow work

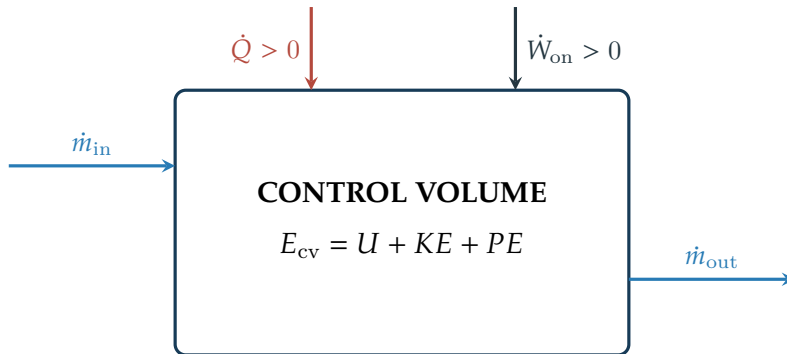
An open system is analysed as a fixed or moving *control volume*. Mass can cross its control surface. Before writing the balance, distinguish the volume and velocity symbols:

Symbol	Name and units	Meaning
V	total volume, m^3	extensive volume occupied by the chosen system or control volume
$v = V/m$	specific volume, $\text{m}^3 \text{kg}^{-1}$	volume per unit mass; equivalently $v = 1/\rho$
$\dot{V} = dV/dt$	volumetric flow rate, $\text{m}^3 \text{s}^{-1}$	volume crossing a section per unit time; for a uniform stream, $\dot{V} = Au_n$
$\mathbf{u}$	velocity, m s^{-1}	local flow-velocity vector; its kinetic energy per unit mass is $ \mathbf{u} ^2/2$

Thus V , v and $\dot{V}$ are different physical quantities. The overdot means a time rate, not “a larger volume.” For a stream of density ρ ,

$$\dot{m} = \rho \dot{V} = \frac{\dot{V}}{v}. \quad (1.12)$$

Each flowing stream transports internal energy u_{int} , kinetic energy $|\mathbf{u}|^2/2$ and potential energy gz per unit mass. It also requires pressure work $p\dot{v}$ to push the stream across the boundary. Combining $u_{\text{int}} + p\dot{v} = h$ prevents that flow work from being counted separately.



Each mass stream carries $h + |\mathbf{u}|^2/2 + gz$; $h = u_{\text{int}} + p\dot{v}$ already includes flow work.

For a control volume, the energy balance has the structure

$$\frac{dE_{\text{cv}}}{dt} = \dot{Q} + \dot{W} + \sum_{\text{in}} \dot{m} \left(h + \frac{|\mathbf{u}|^2}{2} + gz \right) - \sum_{\text{out}} \dot{m} \left(h + \frac{|\mathbf{u}|^2}{2} + gz \right), \quad (1.13)$$

Here $\dot{Q} > 0$ denotes heat entering the control volume and $\dot{W} = \dot{W}_{\text{on}} > 0$ denotes non-flow work done on it, such as shaft or electrical work. For steady operation the accumulation term on the left is zero. The inlet and outlet terms must still be retained unless their contributions cancel.

Avoid double counting

Do not add $p\dot{V}$ flow work once more after using enthalpy in the stream terms. The $p\dot{v}$ contribution is already contained in $h = u_{\text{int}} + p\dot{v}$. Here $p\dot{V} = p\dot{v}\dot{m}$ for a uniform stream.

Measurable: mass flow, temperature, pressure and power. Inferred: the change of the control-volume energy.

1.8 Week 1 learning map**Five ideas to retain**

Choose the boundary before writing the balance. Distinguish extensive from intensive properties. Distinguish state functions from path functions. Declare the work convention. Test every result against a limiting case such as rigid volume, vacuum expansion or zero heat transfer.

Common misconception: zero work does not mean that the volume cannot change; free expansion changes volume while $p_{\text{ext}} = 0$.

Week 2

Heat capacity, enthalpy and measurable heat

The developing storyline

The first law supplied energy bookkeeping. We now connect that bookkeeping with calorimetric measurements and choose a thermodynamic potential suited to constant pressure.

2.1 Heat capacity

Heat capacity measures the heat required for a specified temperature change under stated constraints. Because heat is path dependent, the constraint matters:

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V, \quad C_p = \left(\frac{\partial H}{\partial T} \right)_p. \quad (2.1)$$

At constant volume, pressure-volume work vanishes and

$$\delta q_V = dU = C_V dT. \quad (2.2)$$

For an ideal gas, molecular interactions are neglected and $U = U(T)$. For a monatomic ideal gas, equipartition gives $C_{V,m} = \frac{3}{2}R$. Additional rotational and vibrational modes increase the heat capacity when those modes are thermally accessible.

2.2 Enthalpy

Define

$$H = U + pV. \quad (2.3)$$

Then

$$dH = dU + p dV + V dp. \quad (2.4)$$

For a closed system doing only pressure-volume work,

$$dH = \delta q + V dp. \quad (2.5)$$

At constant pressure,

$$\delta q_p = dH = C_p dT \quad (2.6)$$

when there is no phase change or reaction contribution beyond that represented in H .

Calorimetry measures heat flow. Under constant pressure, thermodynamics identifies it with the enthalpy change.

2.3 Condensed phases and the size of $p\Delta V$

At constant pressure,

$$\Delta H = \Delta U + p\Delta V. \quad (2.7)$$

For melting ice near atmospheric pressure, the molar-volume change is small, making $p\Delta V$ negligible compared with the latent heat. This is typical of many condensed-phase transformations, but it should be checked rather than assumed.

Dimensional and magnitude check

$p\Delta V$ has units $\text{Pa m}^3 = \text{J}$. At 10^5 Pa , a molar-volume change of $10^{-6} \text{ m}^3 \text{ mol}^{-1}$ contributes only about 0.1 J mol^{-1} .

2.4 Reactions containing gases

For ideal gases, $pV = n_g RT$. At fixed temperature,

$$\Delta H = \Delta U + RT\Delta n_g, \quad (2.8)$$

where Δn_g counts gaseous products minus gaseous reactants. Condensed reactants and products are not included in Δn_g .

2.5 Temperature-dependent heat capacity

Heat capacity often varies with temperature. If

$$C_p(T) = a + bT + cT^{-2}, \quad (2.9)$$

then

$$H(T_2) - H(T_1) = \int_{T_1}^{T_2} C_p(T) dT. \quad (2.10)$$

The same principle extends across phase changes by adding the appropriate latent enthalpy at each transition.

At low temperature, a crystalline solid approximately follows the Debye law $C_V \propto T^3$. At high temperature, the lattice heat capacity approaches the Dulong-Petit limit $3R$ per mole of atoms.

2.6 Worked computational experiment

For any tabulated material:

1. select a valid $C_p(T)$ expression and its temperature interval;
2. integrate it numerically and analytically where possible;
3. add latent heats when a phase boundary is crossed;
4. plot $H(T) - H(T_{\text{ref}})$;
5. check units and continuity at every interval boundary.

2.7 Bridge to the fundamental equation

Weeks 1–2 introduce U , H and heat capacities through energy accounting and measurable heat. The next lectures combine the first and second laws:

$$dU = T dS - p dV + \sum_i \mu_i dN_i. \quad (2.11)$$

This equation identifies conjugate variables and leads to thermodynamic potentials, Maxwell relations and equilibrium.



2.8 Week 2 learning map

Questions students should now answer

Why does constant-pressure calorimetry give an enthalpy change? When may $p\Delta V$ be neglected? Why is Δn_g restricted to gases? How does a microscopic degree of freedom affect heat capacity? How is a tabulated $C_p(T)$ converted into an enthalpy change?

Common misconception: heat capacity is not a single unconstrained property. Always state whether volume, pressure or another variable is held fixed.

Week 3

Phase-equilibrium fundamentals: constraints as a tool

How to use this chapter

Do not begin phase equilibrium by memorizing equalities. Draw the system, label every wall, list what the wall permits to cross, write the conserved totals, and then maximize entropy. The equilibrium conditions follow from the constraints.

3.1 System terminology across texts

The physical definition should be expressed through exchanges, because terminology differs among authors.

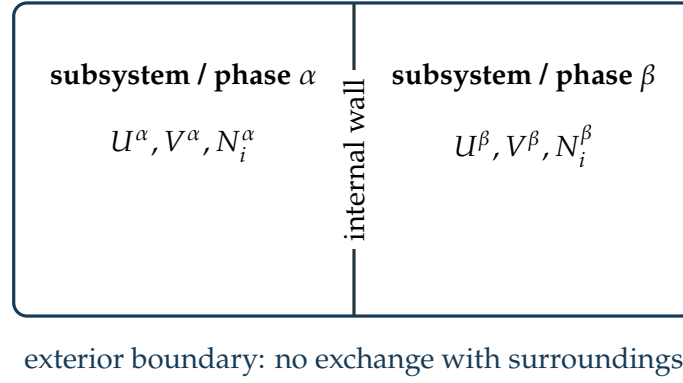
System	Heat/entropy	Volume work	Matter	Common Atkins-style meaning
Isolated	no	no	no	no exchange with surroundings
Closed	possible	possible	no	energy may cross; matter may not
Open	possible	possible	yes	energy and matter may cross

Callen's phrase *closed composite system*, as used in the source notes, means that the complete composite is closed to exterior exchanges. It therefore plays the role commonly called an isolated system in Atkins-style terminology. Long-Qing Chen uses isolated, closed and open in the exchange-based sense summarized above.

Never infer the physics from the word alone. State whether entropy, volume and each species may cross the exterior boundary.

3.2 A closed composite system

The exterior boundary isolates the complete composite. An internal wall divides it into two simple subsystems. In a phase-equilibrium model these subsystems can represent phases α and β .



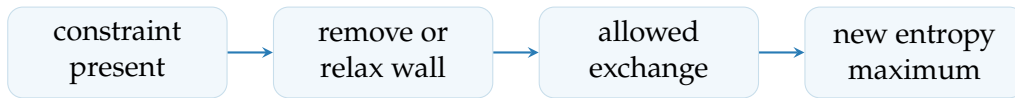
The totals are conserved:

$$U = U^\alpha + U^\beta, \quad V = V^\alpha + V^\beta, \quad N_i = N_i^\alpha + N_i^\beta. \quad (3.1)$$

Removing an internal constraint enlarges the set of states accessible to the composite. The new equilibrium state maximizes total entropy over that enlarged set.

3.3 Walls encode constraints

Wall property	Exchange prevented	Exchange allowed on removal	Equilibrium test
Adiabatic	entropy/heat	energy through diathermal wall	$T^\alpha = T^\beta$
Rigid or fixed	volume	volume through flexible/movable wall	$p^\alpha = p^\beta$
Impermeable to i	component i	N_i through permeable wall	$\mu_i^\alpha = \mu_i^\beta$



Several constraints may be removed together. A movable diathermal wall gives both thermal and mechanical equilibrium. If it is also permeable to every species, full thermal, mechanical and diffusional equilibrium results.

3.4 Deriving the equilibrium conditions

For each subsystem,

$$dS = \frac{1}{T} dU + \frac{p}{T} dV - \sum_i \frac{\mu_i}{T} dN_i. \quad (3.2)$$

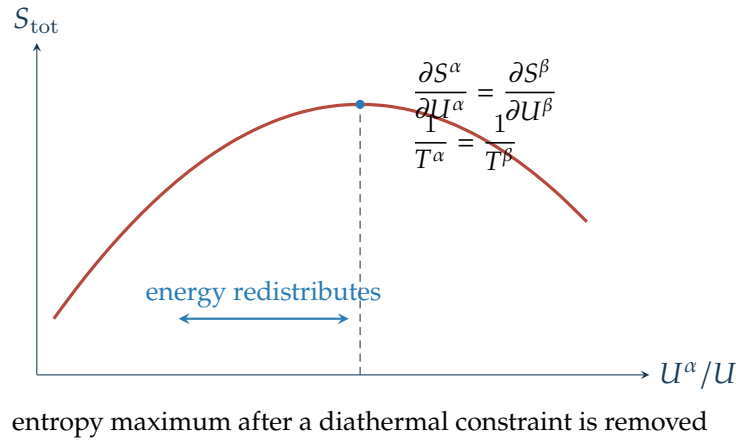
Using conservation to eliminate the variations of phase β gives

$$dS_{\text{tot}} = \left(\frac{1}{T^\alpha} - \frac{1}{T^\beta} \right) dU^\alpha \quad (3.3)$$

$$+ \left(\frac{p^\alpha}{T^\alpha} - \frac{p^\beta}{T^\beta} \right) dV^\alpha - \sum_i \left(\frac{\mu_i^\alpha}{T^\alpha} - \frac{\mu_i^\beta}{T^\beta} \right) dN_i^\alpha. \quad (3.4)$$

Every independently allowed variation has an independently vanishing coefficient at equilibrium. Therefore

$$T^\alpha = T^\beta, \quad p^\alpha = p^\beta, \quad \mu_i^\alpha = \mu_i^\beta. \quad (3.5)$$



The figure is the geometric version of the variational calculation. Energy moves between the two phases while total U remains fixed. At the maximum, the two entropy contributions have equal slopes, which is precisely equality of reciprocal temperature.

3.5 Diffusional equilibrium in solids

If individual species may exchange independently, equality of each chemical potential is the direct condition. In a substitutional solid with a fixed number of lattice sites, composition variations are constrained by $\sum_i x_i = 1$. Only $C - 1$ composition variations are independent. Choosing component K as a reference introduces diffusion potentials

$$M_{iK} = \mu_i - \mu_K, \quad i \neq K. \quad (3.6)$$

Diffusional equilibrium requires equality of the relevant diffusion potentials between phases:

$$M_{iK}^\alpha = M_{iK}^\beta. \quad (3.7)$$

This form is especially useful for constrained solids and phase-field descriptions. It does not say that chemical potential is “a slope”; it identifies the independent potential differences associated with constrained composition changes.

Chemical versus diffusional equilibrium

Use individual chemical potentials when component amounts can vary independently. Use $C - 1$ diffusion potentials when a lattice-site or composition-sum constraint couples their variations.

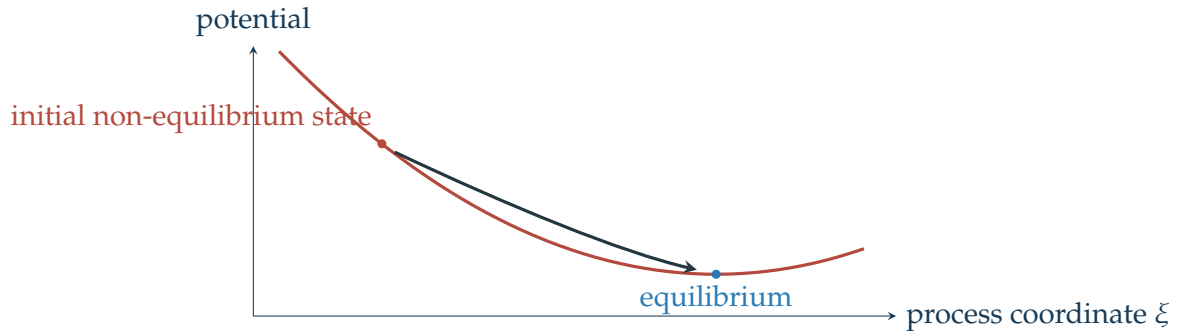
3.6 Equilibrium, process and non-equilibrium

An *equilibrium state* is a constrained entropy maximum and has no finite thermodynamic driving force for an allowed internal change. A *process* is a change from one state to another. A reversible process is an idealized sequence of equilibrium states driven by an infinitesimal force. A spontaneous or irreversible process starts away from equilibrium, has a finite driving force, and produces entropy.

For an internal progress variable ξ ,

$$dU = T dS - p dV + \sum_i \mu_i dN_i - D d\xi, \quad (3.8)$$

where D is the thermodynamic driving force in this sign convention. At equilibrium $D = 0$; during a spontaneous change $D \, d\xi > 0$ and entropy is produced.

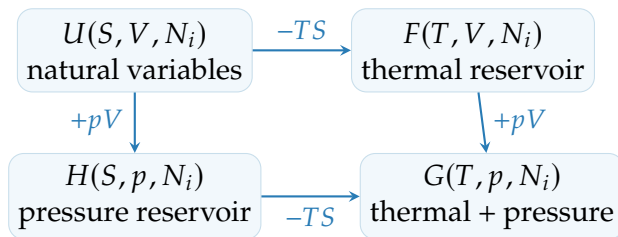


Equilibrium thermodynamics determines the stable state and the available driving force. Non-equilibrium thermodynamics relates forces to rates and entropy production. Kinetic laws determine how rapidly the system follows the path.

3.7 Reservoirs change the useful potential

A reservoir is much larger than the system and maintains an intensive variable while exchanging an extensive quantity. A thermal reservoir fixes T ; a pressure reservoir fixes p ; a chemical reservoir fixes an appropriate chemical potential. Embedding the system and reservoir in a larger isolated composite converts entropy maximization into potential minimization:

$$\begin{aligned}
 T, V, N_i \text{ fixed} & : F \text{ decreases,} \\
 T, p, N_i \text{ fixed} & : G \text{ decreases,} \\
 T, p, \mu_i \text{ fixed} & : \text{the corresponding grand potential decreases.}
 \end{aligned} \tag{3.9}$$



A Legendre transformation replaces an inconvenient controlled extensive variable by the intensive variable fixed by its reservoir.

3.8 A reusable phase-equilibrium workflow

1. Draw the exterior boundary and internal walls.
2. Label each wall as adiabatic/diathermal, rigid/flexible and impermeable/permeable to each species.
3. Identify the extensive quantities conserved for the complete composite.
4. Identify the variations released when a constraint is removed.

5. Write the total entropy variation.
6. Set the coefficient of every independent allowed variation to zero.
7. Check stability using the second variation or curvature of the appropriate potential.
8. Only then select a constitutive or kinetic law for the non-equilibrium process.

3.9 Entropy, thermodynamic potentials and equilibrium

The course pathway

Weeks 1–2 established energy accounting and measurable heat. We now move from state variables to thermodynamic potentials and equilibrium:

$$U(S, V, N_i) \longrightarrow F, H, G \longrightarrow \text{equilibrium conditions} \longrightarrow \text{driving forces.}$$

3.10 Callen's postulatory formulation

The postulatory or axiomatic formulation begins by specifying the class of systems and the equilibrium states to which thermodynamics applies. A *simple system* is an idealized macroscopically homogeneous body whose equilibrium state can be described by energy, volume and component amounts, without additional variables for surfaces, elastic strain, electric polarization, magnetization or other long-range fields. Those effects can be incorporated later by enlarging the state space.

The essential postulates used in these notes are:

1. Equilibrium states exist and are completely characterized by a set of extensive variables: energy, volume and component amounts.
2. There exists an entropy function $S(U, V, N_1, \dots, N_r)$ that is continuous, differentiable, additive and extensive. For fixed V, N_i , it increases monotonically with U .
3. When an internal constraint of an isolated composite system is removed, the equilibrium state is the allowed state that maximizes the total entropy.
4. The entropy reference is fixed by the third law for a perfect crystal at absolute zero.

Terminology warning

Authors use boundary words differently. In the terminology adopted from Callen in these lectures, a *closed composite system* is closed to all exchanges with the exterior and plays the role commonly called an *isolated system* in Atkins-style terminology. We will always state the permitted transfers explicitly rather than rely on the word “closed” alone.

3.11 What thermodynamic equilibrium means

An equilibrium state is time independent at the macroscopic scale and stable against every variation allowed by the constraints. It is not defined merely by “nothing appears to change.” In the entropy representation it is the constrained maximum of total entropy. Removing different internal constraints tests different parts of equilibrium:

- a diathermal wall permits energy exchange and tests thermal equilibrium;
- a movable wall permits volume exchange and tests mechanical equilibrium;
- a permeable or selectively permeable wall permits matter exchange and tests chemical equilibrium.

A composite system is divided conceptually into simple subsystems. In phase-equilibrium applications, each homogeneous phase is treated as a subsystem. More generally, a subsystem is a book-keeping division and need not represent a distinct phase.

A phase is a useful thermodynamic subsystem because it is homogeneous and has its own fundamental relation.

3.12 The fundamental equation

For a reversible change of a simple compressible closed system,

$$\delta q_{\text{rev}} = T dS, \quad \delta w_{\text{rev}} = -p dV. \quad (3.10)$$

Combining the first and second laws gives

$$\boxed{dU = T dS - p dV.} \quad (3.11)$$

Thus $U = U(S, V)$ for fixed composition, with

$$T = \left(\frac{\partial U}{\partial S} \right)_V, \quad p = - \left(\frac{\partial U}{\partial V} \right)_S. \quad (3.12)$$

For a multicomponent system,

$$\boxed{dU = T dS - p dV + \sum_i \mu_i dN_i.} \quad (3.13)$$

Temperature, pressure and chemical potential are conjugate to entropy, volume and component amount.

3.13 Why C_p and C_V differ

Write the volume as $V = V(T, p)$ and define

$$\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_p, \quad \kappa_T = -\frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_T. \quad (3.14)$$

Here α is the volumetric thermal-expansion coefficient and κ_T is the isothermal compressibility. The Maxwell relations derived from thermodynamic potentials lead to

$$\boxed{C_p - C_V = \frac{TV\alpha^2}{\kappa_T}.} \quad (3.15)$$

For a mechanically stable material $\kappa_T > 0$, so $C_p \geq C_V$. At constant pressure some supplied heat accommodates expansion as well as increasing internal energy.

For an ideal gas, $\alpha = 1/T$ and $\kappa_T = 1/p$. Since $pV = nRT$,

$$C_p - C_V = nR, \quad (3.16)$$

or $C_{p,m} - C_{V,m} = R$ per mole.

Checks

The units of $TV\alpha^2/\kappa_T$ are joules per kelvin. If thermal expansion vanished, C_p and C_V would coincide in this expression.

3.14 Measuring entropy from heat capacity

At constant pressure and fixed composition,

$$dH = T dS = C_p dT, \quad dS = \frac{C_p(T)}{T} dT. \quad (3.17)$$

Within a single phase,

$$S(T_2) - S(T_1) = \int_{T_1}^{T_2} \frac{C_p(T)}{T} dT. \quad (3.18)$$

At a reversible phase transition occurring at T_{tr} , the entropy changes discontinuously by

$$\Delta S_{tr} = \frac{\Delta H_{tr}}{T_{tr}}. \quad (3.19)$$

Therefore an entropy calculation crossing melting or boiling must combine heat-capacity integrals with latent-entropy contributions:

$$S(T) = S(0) + \sum_{\text{phases}} \int \frac{C_p}{T} dT + \sum_{\text{transitions}} \frac{\Delta H_{tr}}{T_{tr}}. \quad (3.20)$$

3.15 Low-temperature extrapolation and the third law

At sufficiently low temperature, a crystalline solid approximately obeys $C_p = aT^3$. Hence

$$S(T) - S(0) = \int_0^T aT'^2 dT' = \frac{aT^3}{3} = \frac{C_p(T)}{3}. \quad (3.21)$$

The third law assigns

$$S(0) = 0 \quad (3.22)$$

to a perfect crystal at absolute zero. This reference makes absolute entropies accessible from calorimetric data.

The zero-entropy statement requires a perfect crystal. Disorder frozen into the material may produce residual entropy.

For a reaction or transformation, the standard entropy change is found from absolute molar entropies:

$$\Delta S^\circ = \sum_{\text{products}} \nu_i S_i^\circ - \sum_{\text{reactants}} \nu_i S_i^\circ. \quad (3.23)$$

3.16 Which thermodynamic potential should decrease?

The entropy inequality selects the appropriate potential once the environmental constraints are specified.

3.16.1 Helmholtz free energy

Define

$$F = U - TS. \quad (3.24)$$

For fixed composition,

$$dF = -S dT - p dV. \quad (3.25)$$

At fixed T and V , spontaneous changes satisfy

$$\boxed{\Delta F \leq 0}, \quad (3.26)$$

with equality at equilibrium.

3.16.2 Gibbs free energy

Define

$$G = H - TS = U + pV - TS. \quad (3.27)$$

Then

$$dG = -S dT + V dp + \sum_i \mu_i dN_i. \quad (3.28)$$

At fixed T and p , spontaneous changes satisfy

$$\boxed{\Delta G \leq 0}. \quad (3.29)$$

Constraints	Potential	Equilibrium test	Typical setting
S, V, N_i	U	minimum U	isolated mechanical description
T, V, N_i	F	minimum F	fixed-temperature simulation cell
S, p, N_i	H	minimum H	adiabatic constant-pressure setting
T, p, N_i	G	minimum G	laboratory phase equilibrium

3.17 Extensivity and intensive variables

Entropy is additive and homogeneous of first order:

$$S(\lambda U, \lambda V, \lambda N_1, \dots) = \lambda S(U, V, N_1, \dots). \quad (3.30)$$

Temperature, pressure and chemical potentials are homogeneous of degree zero: scaling the system does not change them. They are intensive variables.

The entropy representation follows from the chain rule:

$$dS = \frac{1}{T} dU + \frac{p}{T} dV - \sum_i \frac{\mu_i}{T} dN_i. \quad (3.31)$$

The quantities $1/T$, p/T and $-\mu_i/T$ are the entropy-representation conjugates.

3.18 Equilibrium by constrained entropy maximization

Consider an isolated composite system containing two subsystems. Total entropy is additive:

$$S = S^{(1)} + S^{(2)}. \quad (3.32)$$

The equilibrium state maximizes S subject to the conserved total quantities.

3.18.1 Thermal equilibrium

Let the separating wall be rigid, impermeable and diathermal. Only energy is exchanged, so

$$dU^{(2)} = -dU^{(1)}. \quad (3.33)$$

Then

$$dS = \left(\frac{1}{T^{(1)}} - \frac{1}{T^{(2)}} \right) dU^{(1)}. \quad (3.34)$$

Since $dU^{(1)}$ is arbitrary, equilibrium requires

$$\boxed{T^{(1)} = T^{(2)}}. \quad (3.35)$$

If $T^{(1)} > T^{(2)}$, spontaneous heat transfer removes energy from subsystem 1 and increases the total entropy.

3.18.2 Mechanical equilibrium

Let the wall also move, so energy and volume may be exchanged while totals remain fixed. After thermal equilibrium is imposed,

$$dS = \frac{p^{(1)} - p^{(2)}}{T} dV^{(1)}. \quad (3.36)$$

Equilibrium therefore requires

$$\boxed{p^{(1)} = p^{(2)}}. \quad (3.37)$$

3.18.3 Chemical equilibrium

Let the wall permit component i to pass. Conservation gives $dN_i^{(2)} = -dN_i^{(1)}$, and at common temperature

$$dS = -\frac{\mu_i^{(1)} - \mu_i^{(2)}}{T} dN_i^{(1)}. \quad (3.38)$$

Thus

$$\boxed{\mu_i^{(1)} = \mu_i^{(2)}} \quad (3.39)$$

for every component that can exchange between the subsystems.

The wall reveals the equilibrium condition

A diathermal wall tests temperature equality, a movable wall tests pressure equality, and a selectively permeable wall tests equality of the corresponding chemical potential. The permitted exchange determines the variable that must equilibrate.

3.19 From an equilibrium condition to a driving force

Away from equilibrium, differences in intensive variables provide thermodynamic driving forces. For two uniform subsystems these appear as differences such as $1/T^{(1)} - 1/T^{(2)}$ or $\mu_i^{(1)} - \mu_i^{(2)}$. In a continuous body they become spatial gradients. Later chapters will connect those gradients to heat, mass and other fluxes.



3.20 Week 3 learning map

Questions students should now answer

How is entropy obtained from heat-capacity data? Why must latent heats be included? Which potential is minimized under fixed T, p conditions? Why do temperature, pressure and chemical potential become equal at equilibrium? How does a difference in an intensive variable become a driving force?

Common misconception: equilibrium is not obtained by declaring every differential zero independently. First identify the permitted exchanges and the conservation constraints.

Week 4

Multiphase equilibrium and thermodynamic structure

Where we are

We now generalize the two-subsystem argument to several phases and components. The central path is

constraints $\rightarrow$ allowed exchanges $\rightarrow$ equilibrium conditions $\rightarrow G(T, p, N_i) \rightarrow \mu_i$.

4.1 Several phases and several components

Consider phases $\alpha, \beta, \dots$ in an isolated composite system. Entropy and the conserved extensive quantities are additive:

$$S = \sum_{\phi} S^{\phi}, \quad U = \sum_{\phi} U^{\phi}, \quad V = \sum_{\phi} V^{\phi}, \quad N_i = \sum_{\phi} N_i^{\phi}. \quad (4.1)$$

For each phase,

$$dS^{\phi} = \frac{1}{T^{\phi}} dU^{\phi} + \frac{p^{\phi}}{T^{\phi}} dV^{\phi} - \sum_i \frac{\mu_i^{\phi}}{T^{\phi}} dN_i^{\phi}. \quad (4.2)$$

Maximizing total entropy subject to the conservation constraints gives

$$T^{\alpha} = T^{\beta} = \dots, \quad p^{\alpha} = p^{\beta} = \dots, \quad \mu_i^{\alpha} = \mu_i^{\beta} = \dots. \quad (4.3)$$

The chemical-potential equality applies to each component that can exchange between the relevant phases.

Count unknowns and independent equations before solving an equilibrium problem.

4.2 Degrees of freedom and the phase rule

For C components and P phases, composition contributes $P(C - 1)$ independent variables; temperature and pressure contribute two. Equality of each component chemical potential between phases

supplies $C(P - 1)$ relations. Hence

$$F = C - P + 2. \quad (4.4)$$

For condensed systems at prescribed pressure, the reduced form is $F = C - P + 1$.

What the phase rule says

F counts the intensive variables that may be changed independently without changing the number of phases at equilibrium. It does not calculate their values.

4.3 Entropy maximum and energy minimum are equivalent

For fixed total U, V, N_i , equilibrium maximizes entropy. An equivalent statement is that for fixed total S, V, N_i , equilibrium minimizes energy. The equivalence follows because $S(U, V, N_i)$ is monotonic in U :

$$\left(\frac{\partial S}{\partial U} \right)_{V, N_i} = \frac{1}{T} > 0. \quad (4.5)$$

If a variation at fixed U could increase S , then the inverse relation $U(S, V, N_i)$ implies that at the original S the corresponding relaxed state has lower U . Thus

$$(\delta S)_{U, V, N_i} = 0, \quad (\delta^2 S)_{U, V, N_i} < 0 \quad \Longleftrightarrow \quad (\delta U)_{S, V, N_i} = 0, \quad (\delta^2 U)_{S, V, N_i} > 0. \quad (4.6)$$

The constraint is part of the principle

“Entropy increases” and “energy decreases” are incomplete statements. Entropy is maximized at fixed U, V, N_i ; energy is minimized at fixed S, V, N_i .

4.4 Homogeneity and Euler’s relation

Internal energy is homogeneous of first order in its extensive arguments:

$$U(\lambda S, \lambda V, \lambda N_1, \dots) = \lambda U(S, V, N_1, \dots). \quad (4.7)$$

Euler’s theorem therefore gives

$$U = TS - pV + \sum_i \mu_i N_i. \quad (4.8)$$

Differentiating this expression and comparing it with the fundamental equation yields the Gibbs-Duhem relation:

$$S dT - V dp + \sum_i N_i d\mu_i = 0. \quad (4.9)$$

At fixed T and p ,

$$\sum_i N_i d\mu_i = 0. \quad (4.10)$$

Thus the chemical potentials in a mixture cannot vary independently.

4.5 Legendre transformations and natural variables

A Legendre transformation replaces an inconvenient natural variable by its conjugate. Applied to $U(S, V, N_i)$, it generates

$$F = U - TS, \quad dF = -S dT - p dV + \sum_i \mu_i dN_i, \quad (4.11)$$

$$H = U + pV, \quad dH = T dS + V dp + \sum_i \mu_i dN_i, \quad (4.12)$$

$$G = U - TS + pV, \quad dG = -S dT + V dp + \sum_i \mu_i dN_i. \quad (4.13)$$

Potential	Natural variables	Useful environment	Equilibrium principle
U	S, V, N_i	isolated constraints	minimum U
F	T, V, N_i	thermal reservoir, fixed V	minimum F
H	S, p, N_i	fixed entropy and pressure	minimum H
G	T, p, N_i	thermal and pressure reservoirs	minimum G

4.6 Reservoirs and the origin of minimum principles

A reservoir is an idealized system so large that a finite exchange changes its extensive variables but not the relevant intensive variable. A heat reservoir maintains T_R ; a pressure reservoir maintains p_R .

Place a system in diathermal contact with a heat reservoir and isolate the composite. Total entropy must increase. Since $\Delta S_R = -\Delta U/T_R$ and equilibrium requires $T = T_R$,

$$\Delta S_{\text{tot}} = \Delta S - \frac{\Delta U}{T} \geq 0 \iff \Delta(U - TS) = \Delta F \leq 0 \quad (4.14)$$

at fixed T, V, N_i . Adding a pressure reservoir gives

$$\Delta G = \Delta(U - TS + pV) \leq 0 \quad (4.15)$$

at fixed T, p, N_i . The free-energy minimum principles are therefore entropy-maximization statements for a larger isolated composite system.

A reservoir changes the convenient constraint; the second law itself has not changed.

4.7 Equations of state and why potentials add no new ones

The fundamental relation $U(S, V, N_i)$ contains the complete equilibrium thermodynamics. Differentiation produces the equations of state

$$T = \left(\frac{\partial U}{\partial S} \right)_{V, N_i}, \quad p = - \left(\frac{\partial U}{\partial V} \right)_{S, N_i}, \quad \mu_i = \left(\frac{\partial U}{\partial N_i} \right)_{S, V, N_{j \neq i}}. \quad (4.16)$$

A Legendre transform changes the independent variables without adding physical information. For example, $G(T, p, N_i)$ yields

$$S = - \left(\frac{\partial G}{\partial T} \right)_{p, N_i}, \quad V = \left(\frac{\partial G}{\partial p} \right)_{T, N_i}, \quad \mu_i = \left(\frac{\partial G}{\partial N_i} \right)_{T, p, N_{j \neq i}}. \quad (4.17)$$

These are alternative representations of the same constitutive content. One does not need an additional independent set of equations of state for every thermodynamic potential. Once one complete fundamental relation is known, the others follow by Legendre transformation.

Why potentials are useful

Potentials do not introduce extra laws. They expose the variables controlled by the environment and make the applicable equilibrium criterion a minimum principle in those variables.

4.8 Chemical potential and partial molar Gibbs free energy

From the differential of G ,

$$\mu_i = \left(\frac{\partial G}{\partial N_i} \right)_{T,p,N_{j \neq i}}. \quad (4.18)$$

Therefore chemical potential is the partial molar Gibbs free energy of component i in the specified phase.

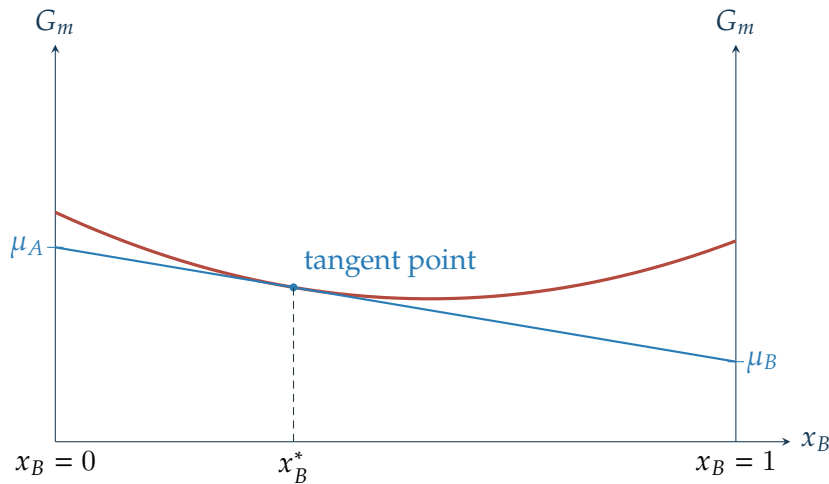
For a binary phase with molar Gibbs free energy $G_m(x_B)$,

$$G = N_A G_A^{\text{partial}} + N_B G_B^{\text{partial}} = N_A \mu_A + N_B \mu_B. \quad (4.19)$$

The tangent at the chosen composition has slope $\mu_B - \mu_A$. Its intercepts at $x_B = 0$ and $x_B = 1$ are μ_A and μ_B respectively.

Read the geometry correctly

The slope of $G_m(x_B)$ is $\mu_B - \mu_A$, not either chemical potential separately. Chemical potentials are obtained from the tangent intercepts.



4.9 Common tangent versus parallel tangents

Two phases α and β at fixed T, p require

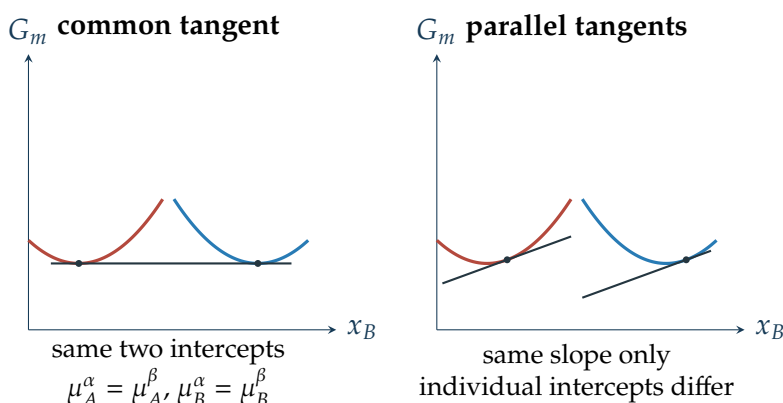
$$\mu_A^\alpha = \mu_A^\beta, \quad \mu_B^\alpha = \mu_B^\beta. \quad (4.20)$$

A single common tangent satisfies both intercept equalities. Equal slopes alone give only

$$\mu_B^\alpha - \mu_A^\alpha = \mu_B^\beta - \mu_A^\beta, \quad (4.21)$$

which does not imply equality of the individual chemical potentials.

Parallel tangents are insufficient: $a - b = c - d$ does not imply $a = c$ and $b = d$.



4.10 Maxwell relations and measurable response functions

Because mixed second derivatives of a smooth potential commute, each potential generates Maxwell relations. For example, from $G(T, p, N_i)$,

$$\left(\frac{\partial S}{\partial p}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_p. \quad (4.22)$$

These relations replace difficult derivatives by experimentally accessible ones and connect heat capacities, thermal expansion and compressibility.

4.11 Bridge to statistical thermodynamics

Macroscopic thermodynamics identifies entropy and chemical potential but does not yet count the microscopic arrangements that produce them. For N sites occupied by $N_A, N_B, \dots$ indistinguishable atoms,

$$\Omega = \frac{N!}{N_A! N_B! \dots}, \quad S_{\text{config}} = k_B \ln \Omega. \quad (4.23)$$

This combinatorial result opens Week 5, where microstates, probability and the partition function connect atomistic descriptions to thermodynamic potentials.

4.12 Week 4 learning map

Essential outcomes

Derive multiphase equilibrium from constrained entropy maximization. Use the phase rule to count freedom. Derive Euler and Gibbs-Duhem relations from extensivity. Choose a potential by its natural variables. Read chemical potentials from tangent intercepts. Distinguish a common tangent from merely parallel tangents.

Computational laboratory. Construct two model molar-free-energy curves. Numerically find two compositions satisfying both chemical-potential equalities, plot the common tangent, and separately display a pair of parallel but non-common tangents to expose the missing condition.

Common misconception: the phase rule counts freedoms; it does not locate the equilibrium compositions.

Week 5

Statistical thermodynamics: from microstates to free energy

Where we are

Macroscopic thermodynamics introduced entropy and free energy without following atoms. Statistical thermodynamics now supplies the microscopic connection:

$$\text{microstates} \rightarrow \Omega \rightarrow S = k_B \ln \Omega \rightarrow Z \rightarrow F = -k_B T \ln Z.$$

5.1 Macrostates and microstates

A macrostate is specified by thermodynamic variables such as U, V, N, T and p . A microstate specifies the microscopic positions and momenta consistent with that macrostate. Many microstates may correspond to the same observable state.

For a monatomic gas containing N atoms, a classical microstate requires $3N$ position coordinates and $3N$ momentum coordinates. Thermodynamics replaces this enormous description with a few state variables.

The macroscopic state is what we measure. The microstates are the atomic arrangements compatible with it.

5.2 Multiplicity and entropy

If a macrostate can be realized in Ω microscopic ways, Boltzmann's relation is

$$S = k_B \ln \Omega. \quad (5.1)$$

For independent systems, multiplicities multiply while entropies add:

$$\Omega_{12} = \Omega_1 \Omega_2, \quad S_{12} = S_1 + S_2. \quad (5.2)$$

The logarithm is therefore required by extensivity.

For N lattice sites occupied randomly by N_A atoms of A and N_B atoms of B,

$$\Omega = \frac{N!}{N_A! N_B!}. \quad (5.3)$$

Using Stirling's approximation, $\ln N! \simeq N \ln N - N$, gives the molar configurational entropy

$$\Delta S_{\text{mix}} = -R(x_A \ln x_A + x_B \ln x_B). \quad (5.4)$$

It vanishes in either pure-component limit and is maximal at $x_A = x_B = 1/2$.

Checks

R supplies units of $\text{J, mol}^{-1}\text{K}^{-1}$. Since $0 < x_i < 1$, each $\ln x_i < 0$, making $\Delta S_{\text{mix}} > 0$. The result assumes random site occupancy and distinguishable lattice sites.

5.3 Probability and the most probable macrostate

For an isolated composite system with fixed total energy,

$$P(U_1) \propto \Omega_1(U_1)\Omega_2(U - U_1). \quad (5.5)$$

Maximizing the total multiplicity is equivalent to maximizing total entropy. Differentiation gives

$$\left(\frac{\partial S_1}{\partial U_1} \right)_{V_1, N_1} = \left(\frac{\partial S_2}{\partial U_2} \right)_{V_2, N_2}, \quad (5.6)$$

or $T_1 = T_2$. The statistical and thermodynamic equilibrium arguments are therefore the same statement at different scales.

5.4 The canonical ensemble

Consider a small system exchanging energy with a much larger heat reservoir at temperature T . The probability of microstate r with energy E_r is

$$P_r = \frac{e^{-\beta E_r}}{Z}, \quad \beta = \frac{1}{k_B T}, \quad (5.7)$$

where

$$Z = \sum_r e^{-\beta E_r} \quad (5.8)$$

is the canonical partition function. For a continuous classical phase space, the sum becomes an appropriately normalized integral.

The Boltzmann factor is not a probability by itself. Division by Z normalizes the distribution.

An energy difference matters relative to $k_B T$. If $\Delta E \gg k_B T$, the higher state is rarely occupied; if $\Delta E \ll k_B T$, both states are appreciably populated.

5.5 Thermodynamic information contained in Z

The Helmholtz free energy follows directly from the partition function:

$$F = -k_B T \ln Z. \quad (5.9)$$

Consequently,

$$U = -\frac{\partial \ln Z}{\partial \beta}, \quad (5.10)$$

$$S = -\left(\frac{\partial F}{\partial T}\right)_{V,N}, \quad (5.11)$$

$$C_V = \left(\frac{\partial U}{\partial T}\right)_{V,N}. \quad (5.12)$$

Energy fluctuations provide another expression:

$$C_V = \frac{\langle E^2 \rangle - \langle E \rangle^2}{k_B T^2}. \quad (5.13)$$

5.6 Equipartition and the ideal gas

Each independent quadratic term in the classical energy contributes $\frac{1}{2}k_B T$ to the mean energy. A monatomic ideal gas therefore has

$$U = \frac{3}{2}Nk_B T = \frac{3}{2}nRT, \quad C_V = \frac{3}{2}Nk_B. \quad (5.14)$$

For one particle of mass m in volume V , the translational partition function is

$$z_{\text{tr}} = \frac{V}{\Lambda^3}, \quad \Lambda = \frac{h}{\sqrt{2\pi m k_B T}}. \quad (5.15)$$

For N indistinguishable noninteracting particles,

$$Z_N = \frac{z_{\text{tr}}^N}{N!}. \quad (5.16)$$

The $N!$ corrects the overcounting of permutations of identical particles.

5.7 Recurring binary-alloy example

For a random A–B substitutional alloy, the configurational partition sum contains the combinatorial multiplicity. If all arrangements have equal energy, mixing is driven entirely by entropy and

$$\Delta G_{\text{mix}} = RT(x_A \ln x_A + x_B \ln x_B) < 0. \quad (5.17)$$

When unlike and like bonds have different energies, an enthalpic term competes with this entropy. That competition will later generate clustering, ordering, miscibility gaps and spinodals.

5.8 The bridge to phase equilibrium

This is the conceptual bridge between Weeks 5 and 6:

$$Z \longrightarrow F = -k_B T \ln Z \longrightarrow G = F + pV \longrightarrow \mu_i = \left(\frac{\partial G}{\partial N_i}\right)_{T,p,N_{j \neq i}} \longrightarrow \text{phase equilibrium.}$$

(5.18)

Microscopic state counting produces a thermodynamic potential. Derivatives of that potential produce chemical potentials. Equality of those chemical potentials determines phase equilibrium.

5.9 Week 5 computational laboratory

For a two-level system with energies 0 and ϵ :

1. calculate $Z = 1 + e^{-\epsilon/(k_B T)}$;
2. plot the two occupation probabilities versus $k_B T/\epsilon$;
3. calculate $U(T)$, $S(T)$ and $C_V(T)$;
4. verify the $T \rightarrow 0$ and $T \rightarrow \infty$ limits;
5. explain the heat-capacity maximum using population transfer.

5.10 Week 5 learning map

Five essential ideas

A macrostate contains many microstates. Entropy measures multiplicity logarithmically. The Boltzmann factor ranks states by energy relative to thermal energy. The partition function normalizes probabilities and generates thermodynamics. Free-energy derivatives connect statistical mechanics to phase equilibrium.

Three equations to reconstruct: $S = k_B \ln \Omega$, $P_r = e^{-\beta E_r}/Z$, and $F = -k_B T \ln Z$.

Common misconception: a positive interaction preference does not automatically mean an ordered phase. The random-alloy approximation contains no sublattice occupancy variable capable of describing long-range order.

Week 6

Phases, transformations and thermodynamic stability

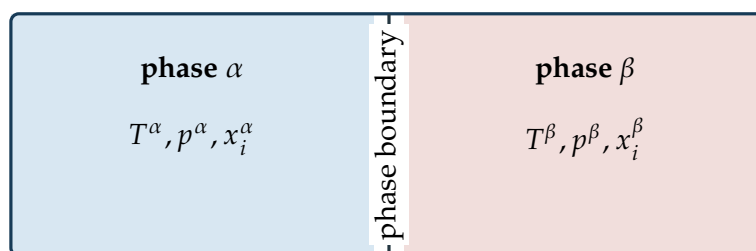
Where we are

The partition function supplied free energy. We now use free energy to decide which phase is stable and whether an internal transformation can proceed:

state $\rightarrow$ potential $\rightarrow$ stability $\rightarrow$ driving force $\rightarrow$ phase transformation.

6.1 What is a phase?

A phase is a portion of matter that is chemically homogeneous, physically distinct and mechanically separable in principle. Its intensive properties, such as composition, density, crystal structure and magnetization, are uniform at equilibrium, apart from interfacial regions and fluctuations.



A phase is a subsystem; the interface is a spatial region, not a third bulk phase.

A component is a chemically independent constituent needed to specify composition. A phase and a component are different ideas. Water and ice are two phases of the same component. A homogeneous A–B solid solution is one phase containing two components.

6.2 Equilibrium and phase transformation

At thermal, mechanical and diffusional equilibrium between phases,

$$T^\alpha = T^\beta, \quad p^\alpha = p^\beta, \quad \mu_i^\alpha = \mu_i^\beta \quad \text{for every transferable component } i. \quad (6.1)$$

A phase transformation is an internal process that changes the amount or internal state of one or more phases. It can involve entropy exchange with the surroundings and entropy production inside the system.

For a process variable ξ ,

$$dU = T dS - p dV + \sum_i \mu_i dN_i - D d\xi, \quad (6.2)$$

where D is the driving force in this sign convention. A spontaneous increment satisfies $D d\xi > 0$. At equilibrium, $D = 0$.

Reversible and irreversible transformation

At the equilibrium transformation temperature, two phases can coexist and an infinitesimal change can be reversed by an infinitesimal change of the constraint. Away from coexistence, a finite driving force produces entropy and the natural transformation is irreversible.

6.3 Which potential should be minimized?

The environment determines the useful potential:

Fixed variables	Potential	Equilibrium criterion	Typical setting
S, V, N_i	U	minimum U	isolated constrained body
T, V, N_i	$F = U - TS$	minimum F	rigid body with heat reservoir
S, p, N_i	$H = U + pV$	minimum H	adiabatic pressure control
T, p, N_i	$G = U - TS + pV$	minimum G	most phase-equilibrium experiments

At fixed T, p, N_i , a possible internal change obeys

$$dG = -D d\xi \leq 0. \quad (6.3)$$

The stable equilibrium is therefore a minimum of G with respect to every allowed internal variable.

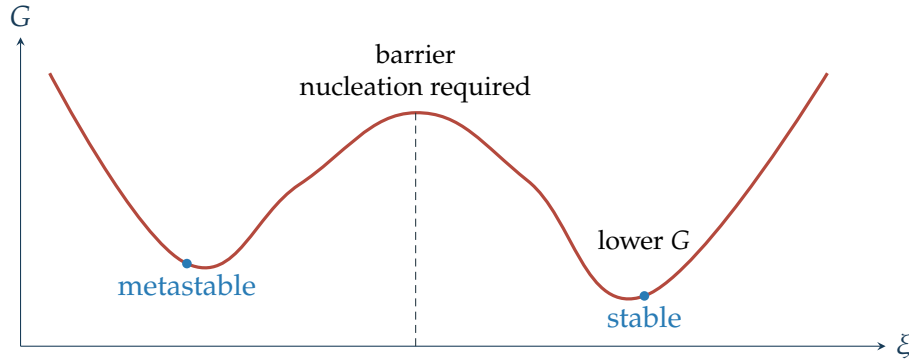
6.4 Local stability, metastability and instability

Consider $G(T, p, N_i, \xi)$ at fixed external variables. At an equilibrium candidate,

$$\left(\frac{\partial G}{\partial \xi} \right)_{T, p, N_i} = 0. \quad (6.4)$$

Its curvature distinguishes the possibilities:

$$\frac{\partial^2 G}{\partial \xi^2} > 0 \text{ stable or metastable minimum,} \quad \frac{\partial^2 G}{\partial \xi^2} < 0 \text{ unstable maximum.} \quad (6.5)$$



A metastable phase is a local minimum separated from the stable phase by a barrier. An unstable state is not a minimum; an arbitrarily small allowed fluctuation lowers the potential. This distinction later separates nucleation from spinodal decomposition.

6.5 Unary phases and molar Gibbs energy

For a one-component phase at fixed pressure,

$$dG_m = -S_m dT, \quad \left(\frac{\partial G_m}{\partial T} \right)_p = -S_m, \quad \left(\frac{\partial^2 G_m}{\partial T^2} \right)_p = -\frac{C_p}{T} < 0. \quad (6.6)$$

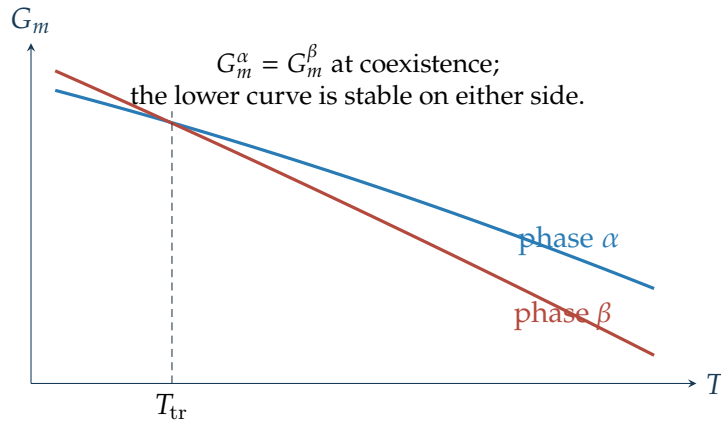
Thus $G_m(T)$ slopes downward and is concave downward for a stable phase with $C_p > 0$. The phase with the lowest molar Gibbs energy is stable.

A straight line is only a tangent

The relation $dG_m = -S_m dT$ does *not* make $G_m(T)$ a straight line. Since S_m changes with temperature, the slope changes too. A straight segment with slope $-S_m(T_0)$ is only the local tangent at T_0 . Over a finite temperature interval the curvature must be retained through

$$\left(\frac{\partial^2 G_m}{\partial T^2} \right)_p = -\frac{C_p}{T}.$$

The curves below are therefore deliberately curved.



At the transformation temperature,

$$G_m^\alpha = G_m^\beta, \quad \Delta G_m = 0, \quad \Delta H_m = T_{tr} \Delta S_m. \quad (6.7)$$

6.6 Computational laboratory

Construct two thermodynamically consistent free energies from reference-state data and heat capacities:

$$H_m^\phi(T) = H_{m,r}^\phi + \int_{T_r}^T C_{p,m}^\phi(T') \, dT', \quad (6.8)$$

$$S_m^\phi(T) = S_{m,r}^\phi + \int_{T_r}^T \frac{C_{p,m}^\phi(T')}{T'} \, dT', \quad (6.9)$$

$$G_m^\phi(T, p_r) = H_m^\phi(T) - TS_m^\phi(T). \quad (6.10)$$

Plot G_m^α and G_m^β , locate T_{tr} from equality, and verify both the slope $\partial G_m / \partial T = -S_m$ and curvature $\partial^2 G_m / \partial T^2 = -C_p / T$. Then add pressure through $\int_{p_r}^p V_m \, dp$ and observe how the crossing moves. This is the same recipe used in Week 7 to construct a unary phase diagram.

6.7 Week 6 learning map

Five essential ideas

A phase is a homogeneous subsystem. Equilibrium conditions follow from allowed exchanges. The controlled variables select the potential. Stable equilibrium is a constrained minimum. Equality of unary molar Gibbs energies defines a transformation temperature.

Common misconception: a local minimum is not necessarily the globally stable phase. It may be metastable because a lower minimum exists beyond a barrier.

Week 7

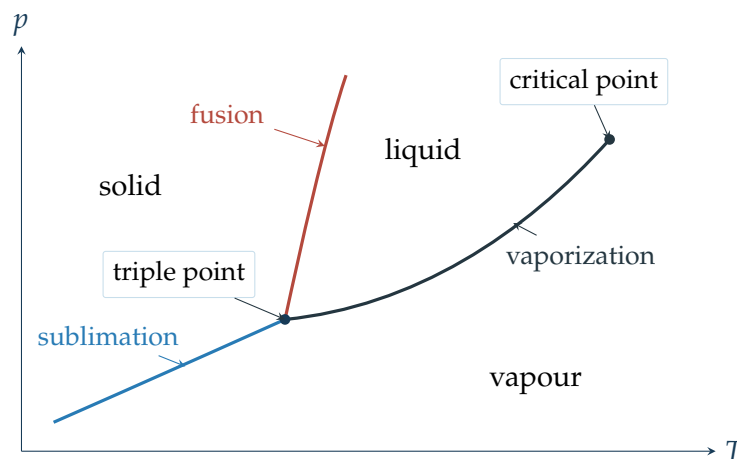
Unary phase diagrams and the Clapeyron equation

Where we are

Week 6 compared phase free energies at one pressure. We now allow pressure and temperature to vary. Equality of chemical potential traces a phase boundary in the p - T plane.

7.1 The unary pressure–temperature diagram

For a pure substance, each single-phase region has two independently variable intensive coordinates, T and p . Along a two-phase boundary, coexistence imposes one relation and leaves one degree of freedom. At the triple point, three coexistence conditions fix both T and p .



The diagram is a map of equilibrium states. Crossing a boundary changes the stable phase. Motion along a boundary changes T and p while preserving equality of the two chemical potentials.

7.2 What the NIST tables give us

The NIST–JANAF tables are a practical source for constructing the standard-pressure part of a unary diagram. For a stated phase and temperature they tabulate quantities such as

$$C_{p,m}^{\circ}, \quad S_m^{\circ}, \quad H_m^{\circ}(T) - H_m^{\circ}(T_r), \quad -\frac{G_m^{\circ}(T) - H_m^{\circ}(T_r)}{T},$$

with $T_r = 298.15$ K and standard pressure $p^{\circ} = 0.1$ MPa. The last column is especially useful because

$$G_m^{\circ}(T) - H_m^{\circ}(T_r) = -T \left[-\frac{G_m^{\circ}(T) - H_m^{\circ}(T_r)}{T} \right]. \quad (7.1)$$

Thus the tabulated Gibbs-energy function can be converted directly into a phase free energy relative to a common enthalpy reference.

The NIST Chemistry WebBook often provides the same temperature dependence as piecewise Shomate polynomials. With $t = T/(1000 \text{ K})$,

$$C_{p,m}^{\circ} = A + Bt + Ct^2 + Dt^3 + \frac{E}{t^2}, \quad (7.2)$$

$$H_m^{\circ} - H_{m,298.15}^{\circ} = At + \frac{Bt^2}{2} + \frac{Ct^3}{3} + \frac{Dt^4}{4} - \frac{E}{t} + F - H, \quad (7.3)$$

$$S_m^{\circ} = A \ln t + Bt + \frac{Ct^2}{2} + \frac{Dt^3}{3} - \frac{E}{2t^2} + G. \quad (7.4)$$

The coefficients are valid only over the temperature interval printed with each set. A phase transition generally requires a new set and the appropriate transition enthalpy and entropy.

Two cautions when reading a database

For an element, $\Delta_f G_m^{\circ} = 0$ for its stable reference form by convention. That zero is not the complete $G_m(T)$ curve and cannot by itself compare allotropes. Compare phase-specific Gibbs functions expressed relative to the same reference. Also, JANAF is primarily a standard-pressure table. Constructing $G_m(T, p)$ away from p° still requires $V_m(T, p)$ or an equation of state through $\int V_m dp$.

For iron, NIST provides separate temperature ranges associated with the α/δ solid, the γ solid and the liquid, including transition entries. This makes iron a useful data-driven laboratory: evaluate all phase functions, plot them on the same axes, and confirm that their crossings reproduce the equilibrium sequence.

NIST data: [JANAF iron tables](#) and [NIST Chemistry WebBook: iron](#).

7.3 Recipe: construct unary equilibrium from phase free energies

For a unary system, do not begin by sketching phase boundaries. Begin with a separate molar Gibbs-energy surface for every candidate phase ϕ :

$$G_m^{\phi}(T, p) = H_m^{\phi}(T, p) - TS_m^{\phi}(T, p). \quad (7.5)$$

At each point (T, p) , the stable phase is the one with the smallest G_m . The following procedure turns thermochemical data into the equilibrium map.

1. **Choose a reference state.** Specify $T_r, p_r, H_{m,r}^{\phi}, S_{m,r}^{\phi}$ for every phase.

2. **Build the temperature dependence.** Use

$$H_m^\phi(T, p_r) = H_{m,r}^\phi + \int_{T_r}^T C_{p,m}^\phi(T') dT', \quad (7.6)$$

$$S_m^\phi(T, p_r) = S_{m,r}^\phi + \int_{T_r}^T \frac{C_{p,m}^\phi(T')}{T'} dT'. \quad (7.7)$$

This produces a curved $G_m^\phi(T, p_r)$; a straight line is only a local tangent.

3. **Add pressure.** At fixed temperature,

$$G_m^\phi(T, p) = G_m^\phi(T, p_r) + \int_{p_r}^p V_m^\phi(T, p') dp'. \quad (7.8)$$

For a condensed phase over a modest pressure range, V_m may be approximated as constant. For gases, an appropriate equation of state is essential.

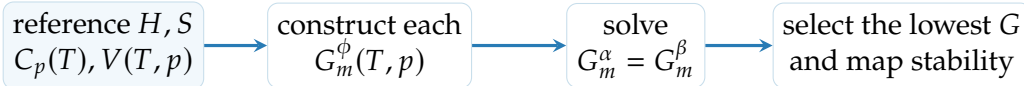
4. **Compare phases.** A two-phase boundary satisfies $G_m^\alpha = G_m^\beta$. On either side, evaluate both energies; the lower one identifies the stable phase.

5. **Find invariant points.** A unary triple point satisfies

$$G_m^\alpha(T_t, p_t) = G_m^\beta(T_t, p_t) = G_m^\gamma(T_t, p_t). \quad (7.9)$$

Solve two independent equality equations for the two unknowns T_t and p_t , then verify the third equality numerically.

6. **Check slopes.** Differentiate the equality of the two Gibbs energies. The boundary must obey $dp/dT = \Delta S_m/\Delta V_m$. A plotted line that violates this sign or magnitude is thermodynamically inconsistent.



What a boundary slope tells us

The sign of a coexistence-line slope reports whether the entropy and volume changes have the same or opposite signs. Its magnitude reports their ratio, not either quantity separately. A steep line can arise from a large ΔS_m , a small ΔV_m , or both. Independent calorimetric or volumetric information is needed to separate them.

7.4 Derivation of the Clapeyron equation

Along coexistence of phases α and β ,

$$\mu^\alpha(T, p) = \mu^\beta(T, p). \quad (7.10)$$

For a pure phase, $d\mu = -S_m dT + V_m dp$. Differentiating the coexistence equality gives

$$-S_m^\alpha dT + V_m^\alpha dp = -S_m^\beta dT + V_m^\beta dp. \quad (7.11)$$

Therefore

$$\boxed{\frac{dp}{dT} = \frac{\Delta S_m}{\Delta V_m} = \frac{\Delta H_m}{T \Delta V_m}}. \quad (7.12)$$

Read the slope physically

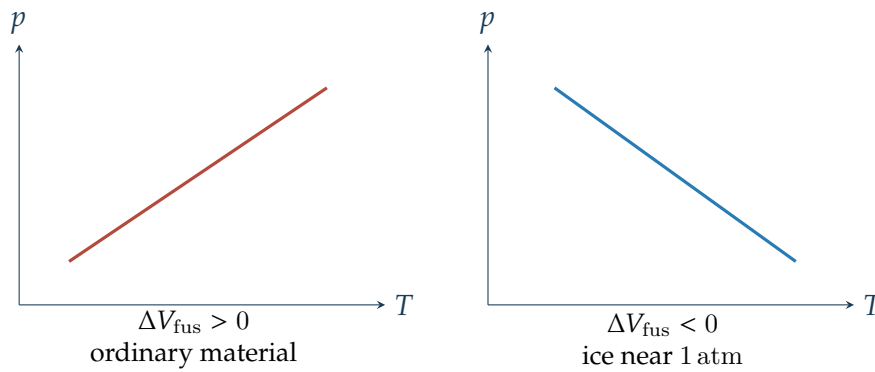
The boundary slope is not an empirical decoration. Its sign and magnitude reflect the entropy and volume changes of the transformation.

7.5 Solid–liquid equilibrium

For melting, $\Delta S_{\text{fus}} > 0$. Most materials expand on melting, so $\Delta V_{\text{fus}} > 0$ and the melting line has positive slope. Ice is a familiar exception near atmospheric pressure because liquid water is denser than ice, so $\Delta V_{\text{fus}} < 0$ and pressure lowers its melting temperature.

If ΔH_m and ΔV_m are approximately constant over a small interval,

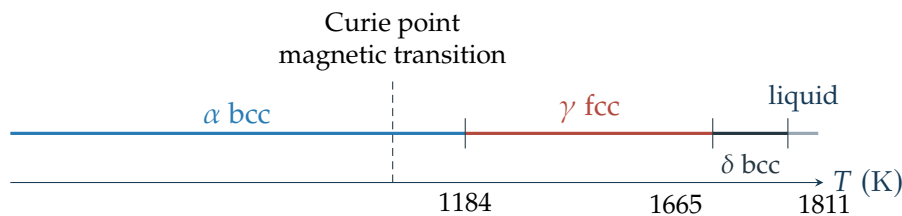
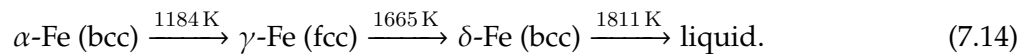
$$\Delta p \simeq \frac{\Delta H_m}{T_{\text{tr}} \Delta V_m} \Delta T, \quad \Delta T \simeq \frac{T_{\text{tr}} \Delta V_m}{\Delta H_m} \Delta p. \quad (7.13)$$



7.6 Two unary materials examples

7.6.1 Iron: several solid phases before melting

Pure iron at approximately atmospheric pressure demonstrates that a unary material need not have only one solid phase. On heating, its equilibrium sequence is approximately



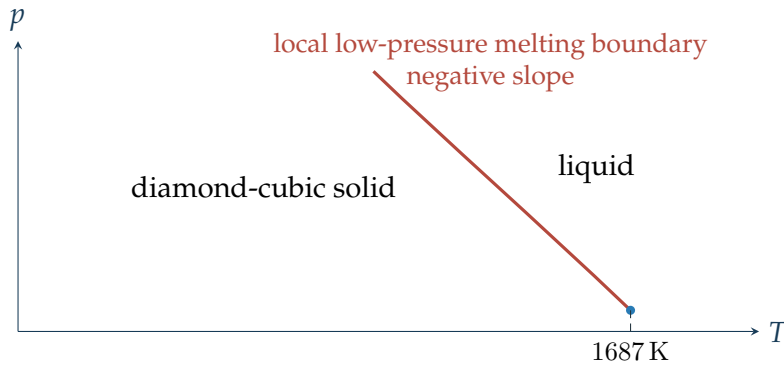
The Curie temperature near 1043 K changes magnetic order within bcc iron. It should not be confused with the bcc-to-fcc structural transformation near 1184 K. Each structural boundary is again an equality of the molar Gibbs energies of the adjacent phases.

7.6.2 Silicon: an initially negative melting-line slope

Diamond-cubic silicon melts near 1687 K (1414 °C) at ambient pressure. The liquid is denser than the open diamond-cubic solid near melting, so

$$\Delta V_{\text{fus}} = V_m^{\text{liquid}} - V_m^{\text{solid}} < 0. \quad (7.15)$$

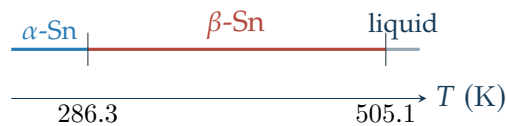
Since $\Delta S_{\text{fus}} > 0$, Clapeyron predicts $(dp/dT)_{\text{coexist}} < 0$: increasing pressure initially lowers the melting temperature.



These sketches are teaching reductions, not complete high-pressure phase diagrams. The iron transition temperatures follow the NIST-JANAF phase data; the silicon melting point and negative initial melting slope are well-established reference results.

7.6.3 Tin: allotropy, kinetics and tin pest

At approximately atmospheric pressure, grey α -Sn has the diamond-cubic structure and is thermodynamically stable below about 286.3 K (13.2 °C). Metallic white β -Sn is tetragonal and is stable above this temperature. It melts near 505.1 K (231.9 °C):



The $\beta \rightarrow \alpha$ transformation can be extremely slow even below the equilibrium temperature. Once nucleated, the lower-density grey phase can cause metallic tin to crumble, a phenomenon called *tin pest*. Tin therefore separates two questions that students often merge: which phase has the lower Gibbs energy, and how rapidly can the transformation occur?

7.7 Liquid–vapour equilibrium and Clausius–Clapeyron

For vaporization, $V_m^{\text{vap}} \gg V_m^{\text{liq}}$ and an ideal vapour has $V_m^{\text{vap}} \approx RT/p$. Then

$$\frac{dp}{dT} \approx \frac{\Delta H_{\text{vap}} p}{RT^2}, \quad \boxed{\frac{d \ln p}{d(1/T)} = -\frac{\Delta H_{\text{vap}}}{R}}. \quad (7.17)$$

If ΔH_{vap} is constant,

$$\ln \frac{p_2}{p_1} = -\frac{\Delta H_{\text{vap}}}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right). \quad (7.18)$$

Clausius-Clapeyron is an approximation to Clapeyron. It uses an ideal vapour, neglects the condensed-phase volume and commonly treats ΔH_{vap} as constant.

7.8 Introduction to partial molar quantities

A solution property is not generally obtained by assigning every component its pure-component molar property. Adding a small amount of component i can change both composition and interactions. For any extensive property $Z(T, p, N_1, \dots, N_C)$, define

$$\bar{Z}_i = \left(\frac{\partial Z}{\partial N_i} \right)_{T, p, N_{j \neq i}}. \quad (7.19)$$

For $Z = G$, $\bar{G}_i = \mu_i$. A partial molar property belongs to component i in the specified phase, composition, temperature and pressure.

Measured and inferred

A phase-boundary slope can be measured. Clapeyron uses it with a volume change to infer a transformation entropy or enthalpy. A partial molar quantity is a derivative inferred from how a total extensive property changes when composition changes.

7.9 Computational laboratory

Use tabulated ΔH and ΔV for one fusion transition. Plot the locally linear melting boundary and test the sign. Then use two vapour-pressure points to estimate ΔH_{vap} from a plot of $\ln p$ against $1/T$. State where the constant-enthalpy approximation is likely to fail.

7.10 Worked examples

7.10.1 Find a triple point from three Gibbs-energy surfaces

Problem. In a model unary system, let T be in kelvin, p in gigapascals and G_m in kJ mol^{-1} . Relative to phase α , three locally valid Gibbs energies are

$$G_m^\alpha = 0, \quad (7.20)$$

$$G_m^\beta = 4.00 - 0.010T + 1.00p, \quad (7.21)$$

$$G_m^\gamma = 9.50 - 0.020T + 0.50p. \quad (7.22)$$

The linearized expressions are local tangent-plane models near the sought point, not global free-energy functions. Find the triple point and the slopes of the three boundaries.

Step 1: impose α - β coexistence.

$$G_m^\alpha = G_m^\beta \implies p = 0.010T - 4.00. \quad (7.23)$$

Step 2: impose α - γ coexistence.

$$G_m^\alpha = G_m^\gamma \implies p = 0.040T - 19.00. \quad (7.24)$$

Step 3: intersect the two boundaries.

$$0.010T - 4.00 = 0.040T - 19.00 \implies \boxed{T_t = 500 \text{ K}}, \quad \boxed{p_t = 1.00 \text{ GPa}}. \quad (7.25)$$

Step 4: verify all three free energies. At (500 K, 1.00 GPa),

$$G_m^\alpha = 0, \quad G_m^\beta = 4 - 5 + 1 = 0, \quad G_m^\gamma = 9.5 - 10 + 0.5 = 0.$$

All three phases coexist.

Step 5: read and check the slopes.

$$\left(\frac{dp}{dT} \right)_{\alpha\beta} = +0.010 \text{ GPa K}^{-1}, \quad (7.26)$$

$$\left(\frac{dp}{dT} \right)_{\alpha\gamma} = +0.040 \text{ GPa K}^{-1}, \quad (7.27)$$

$$\left(\frac{dp}{dT} \right)_{\beta\gamma} = -0.020 \text{ GPa K}^{-1}. \quad (7.28)$$

The negative β - γ slope means that ΔS_m and ΔV_m have opposite signs for the chosen direction of transformation. This construction follows the phase-competition strategy emphasized by DeHoff and Swalin: calculate the thermodynamic surfaces first and let their intersections generate the diagram.

7.10.2 Pressure shift of a melting temperature

Problem. A material melts at $T_m = 1000 \text{ K}$ at pressure p_1 . Its molar enthalpy and volume changes on melting are

$$\Delta H_{\text{fus}} = 10.0 \text{ kJ mol}^{-1}, \quad \Delta V_{\text{fus}} = +0.50 \text{ cm}^3 \text{ mol}^{-1}.$$

Estimate the change in melting temperature when pressure increases by 100 MPa.

Step 1: choose the relation. For a small interval,

$$\Delta T \simeq \frac{T_m \Delta V_{\text{fus}}}{\Delta H_{\text{fus}}} \Delta p.$$

Step 2: convert units.

$$\Delta V_{\text{fus}} = 0.50 \times 10^{-6} \text{ m}^3 \text{ mol}^{-1}, \quad \Delta H_{\text{fus}} = 1.00 \times 10^4 \text{ J mol}^{-1}, \quad \Delta p = 1.00 \times 10^8 \text{ Pa}.$$

Step 3: calculate.

$$\Delta T = \frac{(1000)(0.50 \times 10^{-6})(1.00 \times 10^8)}{1.00 \times 10^4} = 5.0 \text{ K}. \quad (7.29)$$

Step 4: interpret. The positive ΔV_{fus} gives a positive boundary slope. Pressure stabilizes the smaller-volume solid, so a higher temperature is required to melt it.

7.10.3 Vapour pressure from Clausius–Clapeyron

Problem. A liquid has $p_1 = 1.00$ bar at $T_1 = 373.15$ K. Assuming $\Delta H_{\text{vap}} = 40.7$ kJ mol⁻¹ is constant, estimate its vapour pressure at $T_2 = 350.0$ K.

Step 1: write the integrated equation.

$$\ln \frac{p_2}{p_1} = -\frac{\Delta H_{\text{vap}}}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right).$$

Step 2: substitute in SI units.

$$\ln \frac{p_2}{1.00 \text{ bar}} = -\frac{40700}{8.314} \left(\frac{1}{350.0} - \frac{1}{373.15} \right) \quad (7.30)$$

$$= -0.868. \quad (7.31)$$

Step 3: exponentiate and check.

$$p_2 = (1.00 \text{ bar})e^{-0.868} = 0.420 \text{ bar}. \quad (7.32)$$

The lower temperature gives the lower vapour pressure, as required physically.

7.10.4 Equilibrium does not guarantee a rapid transformation

Problem. A specimen of pure white tin is cooled below 13.2 °C but remains metallic for a long time. Does this contradict the unary phase diagram?

Solution. No. The phase diagram identifies grey α -Sn as the equilibrium phase below the transition temperature. It does not specify the rate. Formation of α -Sn requires nucleation and interface motion. A nucleation barrier or slow atomic rearrangement can retain metastable β -Sn. Thermodynamics supplies the direction and driving force; kinetics determines the time required.

7.11 Week 7 learning map

Five essential ideas

A unary diagram maps stable phases in p - T space. A two-phase line is a chemical-potential equality. Clapeyron is exact along coexistence. Clausius–Clapeyron adds approximations for vaporization. Partial molar quantities describe component contributions inside a solution.

Common misconception: a phase-boundary slope does not by itself reveal ΔH ; the transformation volume and temperature are also required.

Week 8

Partial molar properties, Gibbs–Duhem and ideal solutions

Where we are

Unary equilibrium compared one chemical potential between phases. A multicomponent phase requires one chemical potential for each independently transferable component. Partial molar quantities provide that language.

8.1 Extensive properties of a multicomponent phase

Let $Z(T, p, N_1, \dots, N_C)$ be an extensive thermodynamic property. Extensivity means

$$Z(T, p, \lambda N_1, \dots, \lambda N_C) = \lambda Z(T, p, N_1, \dots, N_C). \quad (8.1)$$

Euler's theorem then gives

$$\boxed{Z = \sum_{i=1}^C N_i \bar{Z}_i}, \quad \bar{Z}_i = \left(\frac{\partial Z}{\partial N_i} \right)_{T, p, N_{j \neq i}}. \quad (8.2)$$

Dividing by $N = \sum_i N_i$ gives the molar property

$$Z_m = \sum_i x_i \bar{Z}_i. \quad (8.3)$$

For Gibbs free energy,

$$G = \sum_i N_i \mu_i, \quad G_m = \sum_i x_i \mu_i. \quad (8.4)$$

8.2 Why the partial molar quantities are coupled

The fundamental differential is

$$dG = -S dT + V dp + \sum_i \mu_i dN_i. \quad (8.5)$$

Differentiating $G = \sum_i N_i \mu_i$ and comparing gives the Gibbs–Duhem relation

$$\boxed{S dT - V dp + \sum_i N_i d\mu_i = 0.} \quad (8.6)$$

At fixed T, p ,

$$\sum_i x_i d\mu_i = 0. \quad (8.7)$$

Thus the chemical potentials in one phase cannot be varied independently. In a binary phase,

$$x_A d\mu_A + x_B d\mu_B = 0. \quad (8.8)$$



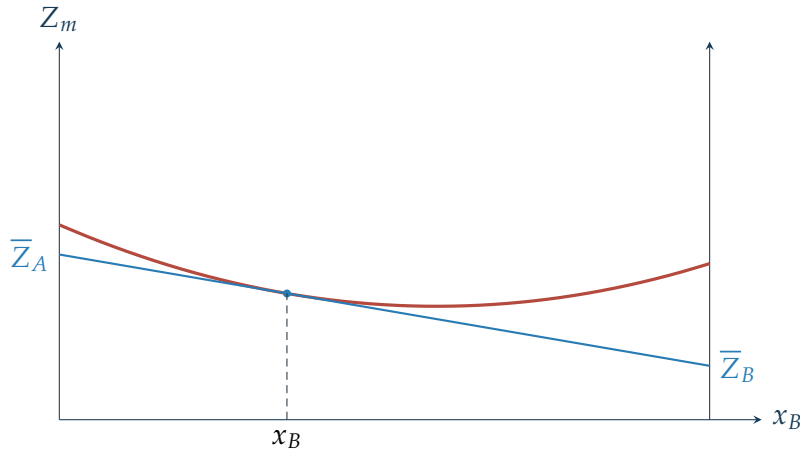
If one chemical potential rises with composition, another must compensate.

8.3 Binary tangent construction

For a binary molar property $Z_m(x_B)$,

$$\bar{Z}_A = Z_m - x_B \frac{dZ_m}{dx_B}, \quad \bar{Z}_B = Z_m + (1 - x_B) \frac{dZ_m}{dx_B}. \quad (8.9)$$

These are exactly the two intercepts of the tangent to $Z_m(x_B)$ at the selected composition.



For $Z = G$, the intercepts are chemical potentials. The slope is $\mu_B - \mu_A$, not either chemical potential separately.

8.4 Partial molar quantities in an n -component phase

For n components, only $n - 1$ mole fractions are independent because

$$x_n = 1 - \sum_{i=1}^{n-1} x_i. \quad (8.10)$$

Write the molar property as $Z_m(T, p, x_1, \dots, x_{n-1})$. At fixed T, p ,

$$dZ_m = \sum_{i=1}^{n-1} \left(\frac{\partial Z_m}{\partial x_i} \right)_{T, p, x_{j \neq i, n}} dx_i. \quad (8.11)$$

The same differential obtained from partial molar quantities is

$$dZ_m = \sum_{i=1}^{n-1} (\bar{Z}_i - \bar{Z}_n) dx_i. \quad (8.12)$$

Therefore

$$\left(\frac{\partial Z_m}{\partial x_i} \right) = \bar{Z}_i - \bar{Z}_n. \quad (8.13)$$

Combining these differences with Euler reconstruction gives

$$\bar{Z}_n = Z_m - \sum_{j=1}^{n-1} x_j \left(\frac{\partial Z_m}{\partial x_j} \right), \quad (8.14)$$

and, for $i = 1, \dots, n-1$,

$$\bar{Z}_i = Z_m + \left(\frac{\partial Z_m}{\partial x_i} \right) - \sum_{j=1}^{n-1} x_j \left(\frac{\partial Z_m}{\partial x_j} \right). \quad (8.15)$$

A tangent plane replaces the tangent line

For a binary solution, the tangent is a line and its two intercepts are the partial molar quantities. For a ternary solution, $Z_m(x_1, x_2)$ is a surface and the tangent is a plane. For an n -component phase, the same construction occurs in an $(n-1)$ -dimensional composition space.

For Gibbs energy, these equations give all chemical potentials. The gradient components are chemical-potential differences:

$$\left(\frac{\partial G_m}{\partial x_i} \right)_{T,p} = \mu_i - \mu_n. \quad (8.16)$$

Thus a multicomponent gradient does not directly give the individual chemical potentials until Euler reconstruction fixes the common offset.

8.5 What information is sufficient?

Before calculating, identify what the problem actually supplies. The following distinctions prevent most errors.

Given	What can be found	What else is required
$Z_m(x_B)$ as a differentiable function	Both $\bar{Z}_A$ and $\bar{Z}_B$ at every interior composition	The derivative dZ_m/dx_B
One value of Z_m and its local slope	Both partial molar quantities at that composition	Nothing further
$\bar{Z}_B(x_B)$ over a range	$\bar{Z}_A(x_B)$ by Gibbs–Duhem integration	One boundary value for $\bar{Z}_A$, usually $\bar{Z}_A(0) = Z_A^{\text{pure}}$
Only one value of $\bar{Z}_B$ at one composition	Not enough to find $\bar{Z}_A$	A composition dependence or derivative, plus an anchor
$\bar{Z}_B - \bar{Z}_A$ and Z_m	Both partial molar quantities	Composition x_B
A table of Z_m versus composition	Approximate both quantities	A justified interpolation or numerical derivative; preferably uncertainty estimates

A value is not a function

Gibbs–Duhem relates *changes* in partial molar quantities. Knowing $\bar{Z}_B$ at one isolated composition does not determine $\bar{Z}_A$. Integration requires the variation of one partial quantity with composition and one reference value for the other.

8.6 Gibbs–Duhem integration in a binary phase

At fixed T, p , the binary relation for any molar property generated from an extensive Z is

$$(1 - x_B) d\bar{Z}_A + x_B d\bar{Z}_B = 0. \quad (8.17)$$

If $\bar{Z}_B(x_B)$ is known as a function, then

$$\frac{d\bar{Z}_A}{dx_B} = -\frac{x_B}{1 - x_B} \frac{d\bar{Z}_B}{dx_B}. \quad (8.18)$$

Using the pure-A endpoint as the integration constant gives

$$\bar{Z}_A(x_B) = Z_A^{\text{pure}} - \int_0^{x_B} \frac{u}{1 - u} \frac{d\bar{Z}_B(u)}{du} du. \quad (8.19)$$

Conversely, if $\bar{Z}_A(x_B)$ is known, integrate from pure B:

$$\bar{Z}_B(x_B) = Z_B^{\text{pure}} + \int_{x_B}^1 \frac{1 - u}{u} \frac{d\bar{Z}_A(u)}{du} du. \quad (8.20)$$

The apparent factors $1/(1 - u)$ or $1/u$ require care near an endpoint. A physically consistent model must have the corresponding derivative approach zero sufficiently rapidly, or the integral must be treated through an appropriate limiting law. For chemical potentials, logarithmic divergence at infinite dilution can be physical; it should not be confused with a numerical failure.

8.6.1 How to use a table at the end of a problem or book

If the data are tabulated rather than given analytically:

1. Confirm that every row refers to the same T, p and single phase.
2. Convert the composition column to mole fraction and verify $x_A + x_B = 1$.
3. Identify whether the table contains $Z_m, \bar{Z}_A, \bar{Z}_B$, an activity, or an excess quantity. They are not interchangeable.
4. Plot the supplied data before differentiating. Look for transcription errors, discontinuities and phase changes.
5. Fit a smooth, thermodynamically sensible interpolant or use carefully chosen finite differences. Differentiation magnifies experimental scatter.
6. Apply either the tangent-intercept equations or Gibbs–Duhem integration.
7. Use a pure-component or independently measured boundary value to fix the integration constant.
8. Reconstruct $Z_m = x_A \bar{Z}_A + x_B \bar{Z}_B$ and test Gibbs–Duhem. Report the effect of the interpolation choice on the result.

Partial-molar calculation checklist

State the phase, T , p , composition variable, units and reference state. Identify the supplied function and the required derivative. Supply an integration anchor when needed. Check endpoint limits, Euler reconstruction and Gibbs–Duhem consistency. Never integrate across a two-phase region as though it were one homogeneous phase.

8.7 Properties and changes of mixing

The change of mixing compares the solution with separated pure components in specified reference states:

$$\Delta Z_{\text{mix}} = Z_{\text{solution}} - \sum_i N_i Z_i^{\text{pure}}. \quad (8.21)$$

The reference temperature, pressure and pure-component states must be stated. Mixing does not mean reaction, and a zero enthalpy of mixing does not imply zero entropy of mixing.

8.8 Ideal-gas mixtures and ideal solutions

For an ideal-gas mixture, component i has partial pressure

$$p_i = x_i p, \quad p = \sum_i p_i. \quad (8.22)$$

Its chemical potential is

$$\mu_i(T, p_i) = \mu_i^\circ(T, p^\circ) + RT \ln \frac{p_i}{p^\circ}. \quad (8.23)$$

An ideal solution is defined thermodynamically by

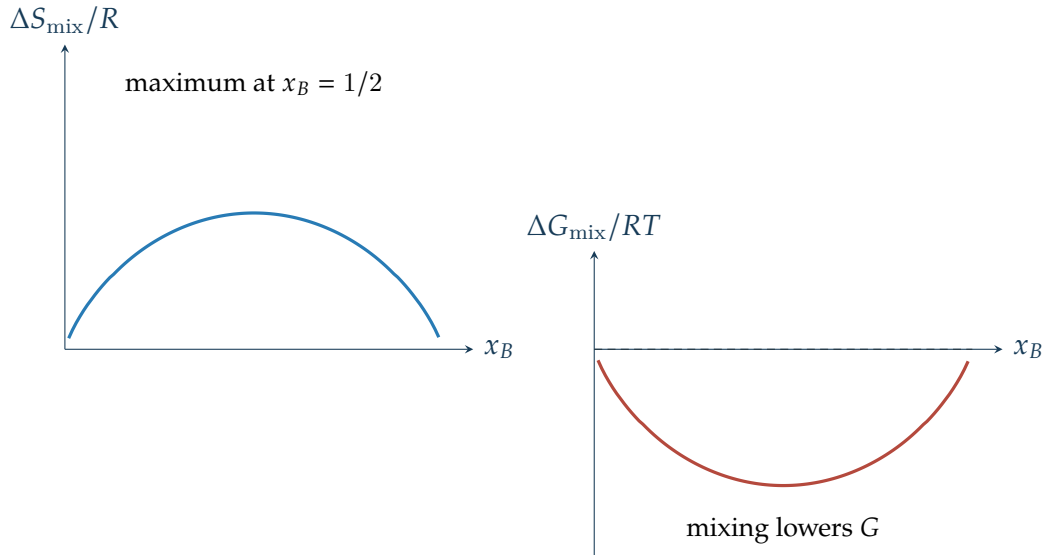
$$\boxed{\mu_i = \mu_i^* + RT \ln x_i}, \quad (8.24)$$

where μ_i^* is the chemical potential of pure i in the same structural state at the same T, p . Therefore

$$\Delta H_{\text{mix}}^{\text{id}} = 0, \quad (8.25)$$

$$\Delta S_{\text{mix}}^{\text{id}} = -R \sum_i x_i \ln x_i, \quad (8.26)$$

$$\Delta G_{\text{mix}}^{\text{id}} = RT \sum_i x_i \ln x_i. \quad (8.27)$$

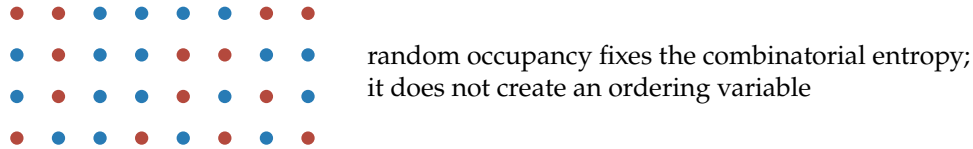


8.9 A lattice picture and its limitation

Randomly distributing N_A A atoms and N_B B atoms over N sites gives

$$\Omega = \frac{N!}{N_A!N_B!}, \quad \Delta S_{\text{mix}} = -R(x_A \ln x_A + x_B \ln x_B). \quad (8.28)$$

This random-alloy approximation describes compositional mixing. It cannot represent long-range ordering unless distinct sublattice occupancies or other order parameters are introduced.



8.10 Computational laboratory

For a binary ideal solution, plot ΔS_{mix} , ΔG_{mix} , $\mu_A - \mu_A^*$ and $\mu_B - \mu_B^*$ from $x_B = 0.001$ to 0.999 . Verify the pure limits, the entropy maximum at $x_B = 1/2$, Gibbs–Duhem numerically, and the tangent-intercept identities.

8.11 Worked examples

8.11.1 Partial molar properties from a tangent

Problem. A binary phase has the model molar property, in kJ mol^{-1} ,

$$Z_m(x_B) = 8 + 12x_B + 20x_B(1 - x_B).$$

Find $\bar{Z}_A$ and $\bar{Z}_B$ at $x_B = 0.30$ and verify Euler reconstruction.

Step 1: evaluate the molar property.

$$Z_m = 8 + 12(0.30) + 20(0.30)(0.70) = 15.8 \text{ kJ mol}^{-1}. \quad (8.29)$$

Step 2: find the tangent slope.

$$\frac{dZ_m}{dx_B} = 12 + 20(1 - 2x_B) = 20.0 \text{ kJ mol}^{-1}. \quad (8.30)$$

Step 3: calculate the two intercepts.

$$\bar{Z}_A = Z_m - x_B \frac{dZ_m}{dx_B} = 15.8 - (0.30)(20.0) = 9.8 \text{ kJ mol}^{-1}, \quad (8.31)$$

$$\bar{Z}_B = Z_m + (1 - x_B) \frac{dZ_m}{dx_B} = 15.8 + (0.70)(20.0) = 29.8 \text{ kJ mol}^{-1}. \quad (8.32)$$

Step 4: verify.

$$x_A \bar{Z}_A + x_B \bar{Z}_B = (0.70)(9.8) + (0.30)(29.8) = 15.8 = Z_m. \quad (8.33)$$

8.11.2 Ideal mixing and chemical-potential shifts

Problem. Mix A and B ideally at $T = 1000 \text{ K}$ with $x_B = 0.25$. Calculate ΔS_{mix} , ΔG_{mix} and the mixing contributions to μ_A and μ_B .

Step 1: entropy of mixing.

$$\Delta S_{\text{mix}} = -R[x_A \ln x_A + x_B \ln x_B] \quad (8.34)$$

$$= -8.314[(0.75) \ln 0.75 + (0.25) \ln 0.25] \quad (8.35)$$

$$= 4.68 \text{ J mol}^{-1} \text{K}^{-1}. \quad (8.36)$$

Step 2: Gibbs energy of mixing. Since $\Delta H_{\text{mix}} = 0$,

$$\Delta G_{\text{mix}} = -T\Delta S_{\text{mix}} = -4.68 \text{ kJ mol}^{-1}. \quad (8.37)$$

Step 3: component chemical potentials.

$$\mu_A - \mu_A^* = RT \ln x_A = -2.39 \text{ kJ mol}^{-1}, \quad (8.38)$$

$$\mu_B - \mu_B^* = RT \ln x_B = -11.53 \text{ kJ mol}^{-1}. \quad (8.39)$$

Step 4: reconstruct the mixing free energy.

$$x_A(\mu_A - \mu_A^*) + x_B(\mu_B - \mu_B^*) = -4.68 \text{ kJ mol}^{-1}. \quad (8.40)$$

8.11.3 Using Gibbs–Duhem as a consistency test

Problem. In a binary phase at fixed T, p and $x_B = 0.40$, measurements give

$$\frac{d\mu_B}{dx_B} = 12.0 \text{ kJ mol}^{-1}.$$

What must $d\mu_A/dx_B$ be?

Solution. At fixed T, p ,

$$x_A d\mu_A + x_B d\mu_B = 0.$$

Dividing by dx_B and using $x_A = 0.60$ gives

$$\frac{d\mu_A}{dx_B} = -\frac{x_B}{x_A} \frac{d\mu_B}{dx_B} = -\frac{0.40}{0.60}(12.0) = -8.0 \text{ kJ mol}^{-1}. \quad (8.41)$$

The opposite signs are required. If independently fitted chemical potentials violate this relation, they cannot describe one thermodynamically consistent phase.

8.11.4 Given one partial molar function, find the other

Problem. At fixed T, p , a single binary phase has

$$\bar{V}_B(x_B) = 10.0 + 2.0x_B \quad \text{cm}^3 \text{mol}^{-1},$$

and $V_A^{\text{pure}} = 8.00 \text{ cm}^3 \text{mol}^{-1}$. Find $\bar{V}_A$ at $x_B = 0.40$, then calculate V_m .

Step 1: identify the required anchor. The supplied function for $\bar{V}_B$ is not enough without one value of $\bar{V}_A$. The pure-A value supplies

$$\bar{V}_A(0) = V_A^{\text{pure}} = 8.00 \text{ cm}^3 \text{mol}^{-1}.$$

Step 2: differentiate the known partial molar volume.

$$\frac{d\bar{V}_B}{dx_B} = 2.0 \text{ cm}^3 \text{mol}^{-1}. \quad (8.42)$$

Step 3: perform the Gibbs–Duhem integration.

$$\bar{V}_A(x_B) = 8.00 - 2.0 \int_0^{x_B} \frac{u}{1-u} du \quad (8.43)$$

$$= 8.00 + 2.0 [x_B + \ln(1 - x_B)]. \quad (8.44)$$

At $x_B = 0.40$,

$$\boxed{\bar{V}_A = 8.00 + 2[0.40 + \ln(0.60)] = 7.778 \text{ cm}^3 \text{mol}^{-1}.} \quad (8.45)$$

Step 4: evaluate the known component and reconstruct the molar volume.

$$\bar{V}_B(0.40) = 10.80 \text{ cm}^3 \text{mol}^{-1}, \quad (8.46)$$

$$V_m = (0.60)(7.778) + (0.40)(10.80) = 8.987 \text{ cm}^3 \text{mol}^{-1}. \quad (8.47)$$

Step 5: check the local Gibbs–Duhem relation.

$$\frac{d\bar{V}_A}{dx_B} = -\frac{0.40}{0.60}(2.0) = -1.333,$$

so $(0.60)(-1.333) + (0.40)(2.0) = 0$ as required.

8.11.5 Determine both partial molar quantities from tabulated data

Problem. At one fixed T, p , a table for a single binary phase contains

x_B	0.35	0.40	0.45
Z_m (kJ mol ⁻¹)	8.92	9.32	9.68

Estimate $\bar{Z}_A$ and $\bar{Z}_B$ at $x_B = 0.40$.

Step 1: estimate the tangent slope with a centred difference.

$$\left. \frac{dZ_m}{dx_B} \right|_{0.40} \simeq \frac{9.68 - 8.92}{0.45 - 0.35} = 7.60 \text{ kJ mol}^{-1}. \quad (8.48)$$

Step 2: calculate both tangent intercepts.

$$\bar{Z}_A = 9.32 - (0.40)(7.60) = \boxed{6.28 \text{ kJ mol}^{-1}}, \quad (8.49)$$

$$\bar{Z}_B = 9.32 + (0.60)(7.60) = \boxed{13.88 \text{ kJ mol}^{-1}}. \quad (8.50)$$

Step 3: verify Euler reconstruction.

$$(0.60)(6.28) + (0.40)(13.88) = 9.32 \text{ kJ mol}^{-1}. \quad (8.51)$$

Step 4: state the limitation. The result depends on the numerical slope. With experimental data, repeat the calculation using a smooth fit and nearby point selections; the spread is part of the uncertainty.

8.11.6 All partial molar quantities in a ternary phase

Problem. A ternary phase has the molar property

$$Z_m(x_1, x_2) = 5 + 2x_1 + 4x_2 + 6x_1x_2 + 3x_1^2 \text{ kJ mol}^{-1}, \quad (8.52)$$

with $x_3 = 1 - x_1 - x_2$. Find $\bar{Z}_1$, $\bar{Z}_2$ and $\bar{Z}_3$ at $x_1 = 0.20$, $x_2 = 0.30$, $x_3 = 0.50$.

Step 1: evaluate the molar property.

$$Z_m = 5 + 2(0.20) + 4(0.30) + 6(0.20)(0.30) + 3(0.20)^2 = 7.08. \quad (8.53)$$

Step 2: calculate the two independent derivatives.

$$q_1 = \frac{\partial Z_m}{\partial x_1} = 2 + 6x_2 + 6x_1 = 5.00, \quad (8.54)$$

$$q_2 = \frac{\partial Z_m}{\partial x_2} = 4 + 6x_1 = 5.20. \quad (8.55)$$

These are $q_1 = \bar{Z}_1 - \bar{Z}_3$ and $q_2 = \bar{Z}_2 - \bar{Z}_3$.

Step 3: calculate the dependent-component partial quantity.

$$\bar{Z}_3 = 7.08 - (0.20)(5.00) - (0.30)(5.20) = \boxed{4.52 \text{ kJ mol}^{-1}}. \quad (8.56)$$

Step 4: calculate the other two.

$$\boxed{\bar{Z}_1 = 4.52 + 5.00 = 9.52}, \quad \boxed{\bar{Z}_2 = 4.52 + 5.20 = 9.72} \text{ kJ mol}^{-1}. \quad (8.57)$$

Step 5: verify Euler reconstruction.

$$(0.20)(9.52) + (0.30)(9.72) + (0.50)(4.52) = 7.08 = Z_m. \quad (8.58)$$

8.12 Week 8 learning map

Five essential ideas

Partial molar properties are composition derivatives of a total extensive property. Euler reconstruction expresses the total through those partial quantities. Gibbs–Duhem couples their changes. Chemical potential is partial molar Gibbs free energy. Ideal mixing combines zero enthalpy of mixing with positive configurational entropy.

Common misconception: ideal does not mean that the components have no energy or no interactions. It means that mixing introduces no excess enthalpy and that the entropy follows random mixing.

Week 9

Activities, real solutions and the quasichemical picture

Where we are

Week 8 defined partial molar quantities and ideal mixing. We now retain the universal form $\mu_i = \mu_i^\circ + RT \ln a_i$ while allowing the activity a_i to record non-ideal interactions. The path is

standard state $\rightarrow a_i \rightarrow \gamma_i \rightarrow G_m^E \rightarrow$ atomic interaction model.

9.1 Activity is defined through chemical potential

For component i in one specified phase,

$$\mu_i(T, p, \{x_j\}) = \mu_i^\circ(T, p) + RT \ln a_i. \quad (9.1)$$

Activity is dimensionless. Its numerical value depends on the chosen standard state, so an activity without a stated standard state is incomplete information.

For a condensed solution using the pure component in the same structural state as the standard state,

$$a_i = \gamma_i x_i, \quad \mu_i = \mu_i^* + RT \ln x_i + RT \ln \gamma_i. \quad (9.2)$$

An ideal solution has $\gamma_i = 1$ throughout. A real solution need not.

What activity means

Activity is the composition-like quantity that produces the correct chemical potential. It is not an effective number of atoms and it is not an arbitrary correction factor. It measures escaping tendency relative to a defined reference.

9.2 Raoult and Henry reference states

For an ideal vapour in equilibrium with a condensed solution,

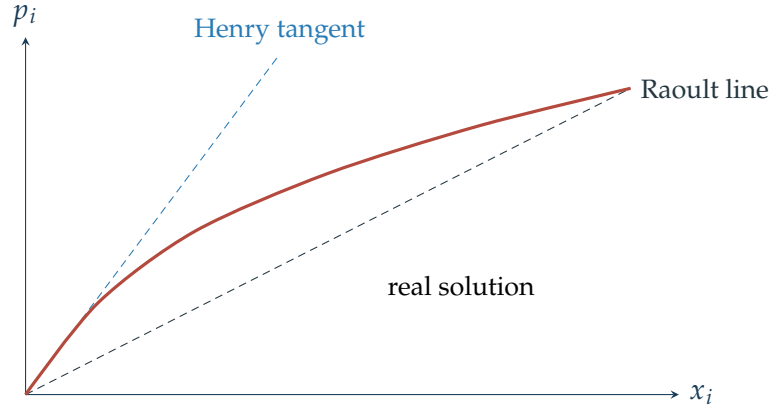
$$\mu_i^{\text{vap}} = \mu_i^{\text{vap},\circ} + RT \ln \frac{p_i}{p^\circ}. \quad (9.3)$$

Near pure i , the Raoult convention uses pure condensed i as standard:

$$p_i = x_i p_i^*, \quad a_i^R = \frac{p_i}{p_i^*} = \gamma_i^R x_i, \quad \gamma_i^R \rightarrow 1 \quad (x_i \rightarrow 1). \quad (9.4)$$

Near infinite dilution, Henry's law is more useful:

$$p_i = K_i^H x_i, \quad a_i^H = \frac{p_i}{K_i^H} = \gamma_i^H x_i, \quad \gamma_i^H \rightarrow 1 \quad (x_i \rightarrow 0). \quad (9.5)$$



Raoult and Henry activities describe the same chemical potential with different reference states. Their numerical activity coefficients must not be mixed in one equation without converting the standard state.

9.3 Gibbs–Duhem constrains activities

At fixed T, p in a binary phase,

$$x_A d\mu_A + x_B d\mu_B = 0. \quad (9.6)$$

Since $d\mu_i = RT d\ln a_i$ at fixed T, p ,

$$\boxed{x_A d\ln a_A + x_B d\ln a_B = 0.} \quad (9.7)$$

Using $a_i = x_i \gamma_i$ and $x_A + x_B = 1$ gives

$$\boxed{x_A d\ln \gamma_A + x_B d\ln \gamma_B = 0.} \quad (9.8)$$

Thus independently fitted activity coefficients must satisfy Gibbs–Duhem. Agreement with vapour-pressure data alone does not excuse a thermodynamically inconsistent pair.

9.4 Excess properties measure departure from ideality

An excess property compares the real solution with an ideal solution at the same T, p and composition:

$$Z_m^E = Z_m - Z_m^{\text{id}}. \quad (9.9)$$

For Gibbs free energy,

$$\boxed{G_m^E = RT \sum_i x_i \ln \gamma_i.} \quad (9.10)$$

The partial molar excess Gibbs energy is

$$\overline{G}_i^E = RT \ln \gamma_i. \quad (9.11)$$

A positive G_m^E raises the real-solution free energy above the ideal curve; a negative value lowers it. Stability still depends on the curvature of the complete G_m , not on the sign of G_m^E alone.

9.4.1 Regular-solution chemical potentials and activities

For

$$G_m = x_A G_A^* + x_B G_B^* + RT(x_A \ln x_A + x_B \ln x_B) + \Omega x_A x_B, \quad (9.12)$$

the tangent-intercept relations give

$$\mu_A = G_A^* + RT \ln x_A + \Omega x_B^2, \quad (9.13)$$

$$\mu_B = G_B^* + RT \ln x_B + \Omega x_A^2. \quad (9.14)$$

Therefore, with pure-component Raoult standards,

$$\boxed{\ln \gamma_A = \frac{\Omega x_B^2}{RT}, \quad \ln \gamma_B = \frac{\Omega x_A^2}{RT}. \quad (9.15)}$$

These expressions automatically satisfy Gibbs–Duhem. They also show why γ_A depends most strongly on the amount of the *other* component.

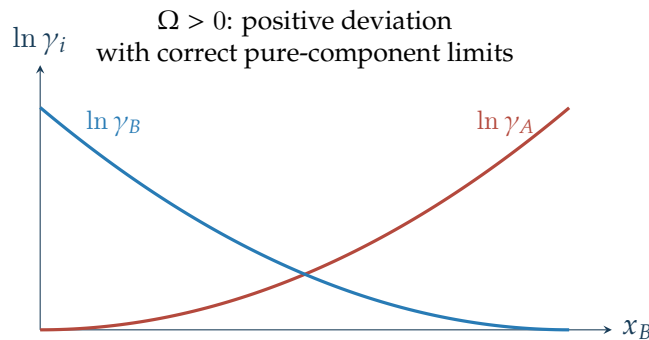
The derivation can be read directly through partial molar excess quantities. Since

$$G_m^E = \Omega x_A x_B, \quad (9.16)$$

the binary tangent-intercept formulas give

$$\overline{G}_A^E = \Omega x_B^2, \quad \overline{G}_B^E = \Omega x_A^2. \quad (9.17)$$

Using $\overline{G}_i^E = RT \ln \gamma_i$ produces the activity-coefficient expressions above.



The endpoint checks are

$$\gamma_A \rightarrow 1 \quad (x_A \rightarrow 1), \quad \gamma_B \rightarrow 1 \quad (x_B \rightarrow 1). \quad (9.18)$$

Differentiating also verifies

$$x_A d \ln \gamma_A + x_B d \ln \gamma_B = 0. \quad (9.19)$$

9.4.2 Multicomponent pairwise regular solution

A common n -component extension uses constant binary interaction parameters:

$$G_m^E = \sum_{i<j} \Omega_{ij} x_i x_j. \quad (9.20)$$

Applying the multicomponent partial-molar formulas gives

$$\bar{G}_i^E = \sum_{j \neq i} \Omega_{ij} x_j - G_m^E, \quad \ln \gamma_i = \frac{\bar{G}_i^E}{RT}. \quad (9.21)$$

For a binary solution this reduces to $\bar{G}_A^E = \Omega_{AB} x_B^2$ and $\bar{G}_B^E = \Omega_{AB} x_A^2$.

Pairwise binary parameters are a model choice, not a general theorem. Real multicomponent solutions may require composition-dependent interactions and explicit ternary or higher-order terms.

9.5 Boiling-point elevation as chemical-potential equality

Let A be a volatile solvent and B a nonvolatile solute. At the boiling point of the solution,

$$\mu_A^L(T_b, x_A) = \mu_A^V(T_b, p). \quad (9.22)$$

For an ideal liquid at the same external pressure as the pure-solvent boiling point T_b^* ,

$$\mu_A^{L,*}(T_b) + RT_b \ln x_A = \mu_A^V(T_b, p). \quad (9.23)$$

Combining this with pure-solvent equilibrium and integrating the Gibbs–Helmholtz relation gives, for approximately constant $\Delta H_{\text{vap},A}$,

$$\ln x_A = -\frac{\Delta H_{\text{vap},A}}{R} \left(\frac{1}{T_b^*} - \frac{1}{T_b} \right). \quad (9.24)$$

For a dilute solution, $\ln x_A \simeq -x_B$ and $\Delta T_b = T_b - T_b^*$ is small:

$$\Delta T_b \simeq \frac{R(T_b^*)^2}{\Delta H_{\text{vap},A}} x_B. \quad (9.25)$$

The elevation follows because mixing lowers the solvent chemical potential in the liquid while the vapour reference at fixed pressure is unchanged.

9.6 Quasichemical bond counting

Consider a binary substitutional alloy on a lattice with coordination number z . Let the energies of nearest-neighbour bonds be ε_{AA} , ε_{BB} and ε_{AB} . Replacing one AA bond and one BB bond by two AB bonds changes the energy by

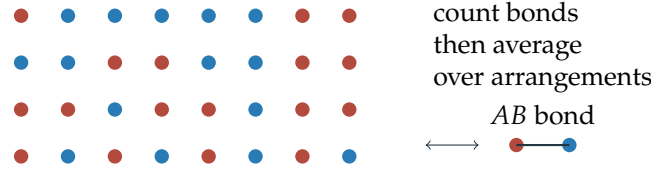
$$\Delta \varepsilon_{\text{pair}} = 2\varepsilon_{AB} - \varepsilon_{AA} - \varepsilon_{BB}. \quad (9.26)$$

Define the molar interaction parameter

$$\Omega = zN_A \left(\varepsilon_{AB} - \frac{\varepsilon_{AA} + \varepsilon_{BB}}{2} \right). \quad (9.27)$$

Under random mixing, the expected bond fractions are x_A^2 , $2x_Ax_B$ and x_B^2 , giving

$$\Delta H_m^{\text{mix}} = \Omega x_A x_B. \quad (9.28)$$



Read the sign of Ω carefully

$\Omega > 0$ means unlike AB bonds are energetically less favourable than the average of AA and BB bonds, so clustering or phase separation is favoured. $\Omega < 0$ favours unlike neighbours. The one-sublattice random-alloy approximation cannot describe long-range order, but it can be extended by resolving crystallographically distinct sublattices and introducing an order parameter.

The simplest regular-solution model combines this enthalpy with the random configurational entropy:

$$\Delta G_m^{\text{mix}} = RT(x_A \ln x_A + x_B \ln x_B) + \Omega x_A x_B. \quad (9.29)$$

Week 10 asks when the curvature of this function produces one stable phase, metastability or spontaneous decomposition.

9.6.1 Assumptions hidden inside the rigid-lattice model

The bond-counting result is not assumption-free. The simplest version assumes:

1. a fixed number of lattice sites and one atom per site;
2. equal site volumes or negligible volume change on mixing;
3. composition-independent nearest-neighbour bond energies;
4. no change of lattice parameter, coordination or local relaxation;
5. random occupation unless sublattice variables are introduced;
6. configurational entropy as the only non-ideal entropy contribution.

Real alloys may have size mismatch, elastic strain, magnetic contributions, vacancies, composition-dependent interactions and different vibrational spectra. These effects can be incorporated, but they are absent from the elementary regular-solution expression.

9.6.2 Configurational and vibrational entropy are different

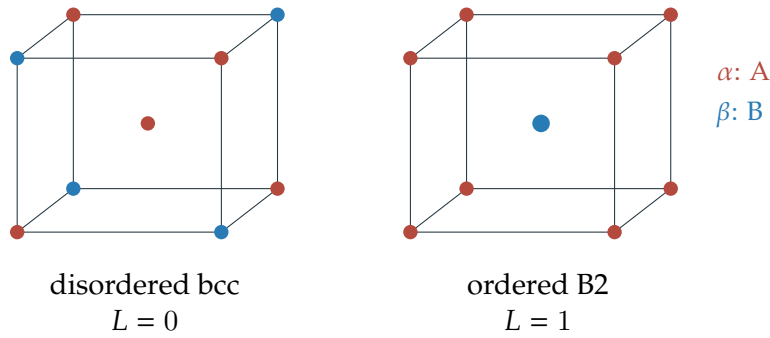
The combinatorial term counts atomic arrangements on sites. Vibrational entropy comes from the spectrum of lattice vibrations. In an Einstein description, a mode of frequency ν has a temperature-dependent entropy obtained from its oscillator partition function. If alloying changes bond stiffnesses or frequencies, then

$$\Delta S_m^{\text{mix}} = \Delta S_m^{\text{conf}} + \Delta S_m^{\text{vib}} + \dots \quad (9.30)$$

The regular-solution approximation usually retains only ΔS_m^{conf} . This is a modeling choice, not a thermodynamic identity.

9.7 A two-sublattice regular-solution model for B2 ordering

A bcc lattice is equivalently two interpenetrating simple-cubic sublattices: corner sites α and body-centre sites β . In disordered bcc, A and B occupy both sublattices with the same probabilities. In equiatomic ordered B2, A preferentially occupies one sublattice and B the other.



For an equiatomic alloy, define the long-range order parameter L through the sublattice site fractions

$$y_A^\alpha = y_B^\beta = \frac{1+L}{2}, \quad y_B^\alpha = y_A^\beta = \frac{1-L}{2}. \quad (9.31)$$

The total composition remains $x_A = x_B = 1/2$ for every L . The parameter changes *where* the atoms sit, not how many A and B atoms exist.

9.7.1 Nearest-neighbour bond fractions

Every nearest-neighbour bond in bcc joins an α site to a β site. Assuming independent occupation within each sublattice,

$$P_{AA} = y_A^\alpha y_A^\beta = \frac{1-L^2}{4}, \quad (9.32)$$

$$P_{BB} = y_B^\alpha y_B^\beta = \frac{1-L^2}{4}, \quad (9.33)$$

$$P_{AB} = y_A^\alpha y_B^\beta + y_B^\alpha y_A^\beta = \frac{1+L^2}{2}. \quad (9.34)$$

These probabilities add to one. If a mole of lattice sites has $zN_A/2$ bonds, then

$$N_{AA} = N_{BB} = \frac{zN_A}{8}(1-L^2), \quad N_{AB} = \frac{zN_A}{4}(1+L^2). \quad (9.35)$$

9.7.2 Enthalpy and configurational entropy

Relative to separated pure A and B in the same lattice, the equiatomic mixing enthalpy becomes

$$\Delta H_m^{\text{mix}}(L) = \frac{\Omega}{4}(1 + L^2). \quad (9.36)$$

At $L = 0$ this recovers the random regular-solution value $\Omega/4$. When $\Omega < 0$, increasing L creates more favourable AB bonds and lowers the enthalpy.

Each sublattice contains half the sites. Its ideal site-mixing entropy gives

$$\Delta S_m^{\text{conf}}(L) = -R \left[\frac{1+L}{2} \ln \frac{1+L}{2} + \frac{1-L}{2} \ln \frac{1-L}{2} \right]. \quad (9.37)$$

Thus $\Delta S_m^{\text{conf}}(0) = R \ln 2$ and $\Delta S_m^{\text{conf}}(1) = 0$.

The ordering free energy is therefore

$$\Delta G_m(L) = \frac{\Omega}{4}(1 + L^2) + RT \left[\frac{1+L}{2} \ln \frac{1+L}{2} + \frac{1-L}{2} \ln \frac{1-L}{2} \right]. \quad (9.38)$$

9.7.3 Equilibrium order parameter

At fixed $T, p, x_A = x_B = 1/2$, minimize with respect to the internal variable L :

$$\frac{\partial \Delta G_m}{\partial L} = \frac{\Omega L}{2} + \frac{RT}{2} \ln \frac{1+L}{1-L} = 0. \quad (9.39)$$

Equivalently,

$$L = \tanh \left(-\frac{\Omega L}{2RT} \right). \quad (9.40)$$

For $\Omega < 0$, a nonzero solution appears below the mean-field transition temperature

$$T_{\text{ord}} = -\frac{\Omega}{2R}. \quad (9.41)$$

What the order parameter adds

Composition x_B distinguishes A-rich from B-rich material. The long-range order parameter L distinguishes two arrangements at the same composition. A phase-field or thermodynamic model of B2 ordering therefore needs both kinds of variables when composition and ordering can evolve independently.

9.8 A calculation checklist for activities

1. State the phase, temperature, pressure and composition scale.
2. State the standard state: Raoult, Henry, gas at p° , or another declared reference.
3. Write $\mu_i = \mu_i^\circ + RT \ln a_i$ before introducing γ_i .
4. Use $a_i = \gamma_i x_i$ only for the selected composition convention.

5. Check the correct limiting behaviour: $\gamma_i^R \rightarrow 1$ as $x_i \rightarrow 1$ or $\gamma_i^H \rightarrow 1$ as $x_i \rightarrow 0$.
6. Test Gibbs–Duhem for every fitted pair of activities.
7. Reconstruct $G_m^E = RT \sum x_i \ln \gamma_i$ and inspect its magnitude and curvature.
8. Distinguish an activity measurement from the model used to interpret its composition dependence.

9.9 Worked examples

9.9.1 Activity and activity coefficient from vapour pressure

Problem. At a fixed temperature, pure A has vapour pressure $p_A^* = 80.0$ kPa. Above a liquid with $x_A = 0.60$, the measured partial pressure is $p_A = 42.0$ kPa. Find the Raoult activity and activity coefficient of A.

Step 1: use the declared Raoult standard.

$$a_A^R = \frac{p_A}{p_A^*} = \frac{42.0}{80.0} = 0.525. \quad (9.42)$$

Step 2: separate activity from composition.

$$\gamma_A^R = \frac{a_A^R}{x_A} = \frac{0.525}{0.60} = \boxed{0.875}. \quad (9.43)$$

The negative deviation from Raoult's law means A has a lower escaping tendency than an ideal solution of the same composition.

9.9.2 Recover the other activity coefficient by Gibbs–Duhem

Problem. A binary solution obeys

$$\ln \gamma_B = Ax_A^2, \quad (9.44)$$

where A is constant at the selected T, p . Determine $\ln \gamma_A$ using Gibbs–Duhem and the Raoult endpoint $\gamma_A \rightarrow 1$ as $x_A \rightarrow 1$.

Step 1: choose $x_B = x$ so $x_A = 1 - x$.

$$\ln \gamma_B = A(1 - x)^2, \quad \frac{d \ln \gamma_B}{dx} = -2A(1 - x).$$

Step 2: apply Gibbs–Duhem.

$$(1 - x) \frac{d \ln \gamma_A}{dx} + x \frac{d \ln \gamma_B}{dx} = 0, \quad (9.45)$$

$$\frac{d \ln \gamma_A}{dx} = 2Ax. \quad (9.46)$$

Step 3: integrate and use the endpoint.

$$\ln \gamma_A = Ax^2 + C. \quad (9.47)$$

At $x = 0$, pure A requires $\ln \gamma_A = 0$, hence $C = 0$ and

$$\boxed{\ln \gamma_A = Ax_B^2, \quad \ln \gamma_B = Ax_A^2.} \quad (9.48)$$

9.9.3 From bond energies to the regular-solution parameter

Problem. An alloy has $z = 8$ and

$$\varepsilon_{AA} = -0.30 \text{ eV}, \quad \varepsilon_{BB} = -0.20 \text{ eV}, \quad \varepsilon_{AB} = -0.22 \text{ eV}$$

per bond. Find the sign and magnitude of Ω per mole of atoms. Use $1 \text{ eV atom}^{-1} = 96.485 \text{ kJ mol}^{-1}$.

Step 1: compare unlike and like bonds.

$$\varepsilon_{AB} - \frac{\varepsilon_{AA} + \varepsilon_{BB}}{2} = -0.22 - \frac{-0.30 - 0.20}{2} = +0.030 \text{ eV}. \quad (9.49)$$

Step 2: include the coordination number.

$$\Omega = z(0.030)(96.485) = \boxed{23.2 \text{ kJ mol}^{-1}}. \quad (9.50)$$

The positive value favours fewer unlike bonds and therefore a tendency to cluster. Whether macroscopic separation occurs also depends on the entropy and temperature.

9.9.4 Boiling-point elevation of a dilute solution

Problem. A solvent boils at $T_b^* = 400 \text{ K}$ and has $\Delta H_{\text{vap}} = 32.0 \text{ kJ mol}^{-1}$. Estimate the boiling-point elevation for an ideal solution with nonvolatile-solute mole fraction $x_B = 0.020$.

$$\Delta T_b \approx \frac{(8.314)(400)^2}{32000}(0.020) = \boxed{0.831 \text{ K}}. \quad (9.51)$$

The dilute approximation should be checked against the unexpanded logarithmic equation when x_B is not small.

9.9.5 Activities predicted by a regular solution

Problem. At $T = 1000 \text{ K}$, a binary regular solution has $\Omega = 12.0 \text{ kJ mol}^{-1}$ and $x_B = 0.30$. Calculate both Raoult activity coefficients, activities, and the excess Gibbs energy.

Step 1: calculate the activity coefficients.

$$\ln \gamma_A = \frac{(12000)(0.30)^2}{(8.314)(1000)} = 0.1299, \quad \gamma_A = 1.139, \quad (9.52)$$

$$\ln \gamma_B = \frac{(12000)(0.70)^2}{(8.314)(1000)} = 0.7072, \quad \gamma_B = 2.028. \quad (9.53)$$

Step 2: calculate activities.

$$a_A = (0.70)(1.139) = \boxed{0.797}, \quad a_B = (0.30)(2.028) = \boxed{0.609}. \quad (9.54)$$

Activities need not sum to one.

Step 3: calculate and cross-check G_m^E .

$$G_m^E = RT[x_A \ln \gamma_A + x_B \ln \gamma_B] \quad (9.55)$$

$$= (8.314)(1000)[(0.70)(0.1299) + (0.30)(0.7072)] \quad (9.56)$$

$$= \boxed{2.52 \text{ kJ mol}^{-1}}. \quad (9.57)$$

This equals $\Omega x_A x_B = (12.0)(0.70)(0.30) = 2.52 \text{ kJ mol}^{-1}$.

9.9.6 The same dilute solution in Henry and Raoult conventions

Problem. For component B, $K_B^H = 120$ kPa and $p_B^* = 80$ kPa. At $x_B = 0.050$, the measured partial pressure is $p_B = 6.0$ kPa. Express the activity and activity coefficient in both conventions.

Henry convention.

$$a_B^H = \frac{6.0}{120} = 0.050, \quad \gamma_B^H = \frac{0.050}{0.050} = \boxed{1.00}. \quad (9.58)$$

Raoult convention.

$$a_B^R = \frac{6.0}{80} = 0.075, \quad \gamma_B^R = \frac{0.075}{0.050} = \boxed{1.50}. \quad (9.59)$$

The physical vapour pressure and chemical potential are unchanged. Only the standard state, and therefore the numerical activity, has changed.

9.10 Multiphase equilibrium and the phase rule

For P phases and C components at equilibrium,

$$T^1 = \dots = T^P, \quad p^1 = \dots = p^P, \quad \mu_i^1 = \dots = \mu_i^P \quad (i = 1, \dots, C). \quad (9.60)$$

Counting independent intensive variables and these constraints gives the Gibbs phase rule

$$\boxed{F = C - P + 2.} \quad (9.61)$$

For condensed phase diagrams drawn at a fixed pressure, pressure is no longer a freedom:

$$\boxed{F = C - P + 1.} \quad (9.62)$$

Thus a binary two-phase field at fixed pressure has one degree of freedom: choosing T fixes the two equilibrium compositions. A binary three-phase equilibrium at fixed pressure has $F = 0$ and is invariant.

A section through a larger diagram

A familiar T - x phase diagram is a constant-pressure section of a p - T - x equilibrium map. Changing pressure generally moves the phase boundaries and invariant temperatures; the two-dimensional diagram silently holds one intensive variable fixed.

9.11 Computational laboratory

Plot $p_i(x_i)$ for an ideal Raoult solution and for a smooth real-solution curve. Draw the Raoult line at $x_i = 1$ and the Henry tangent at $x_i = 0$. Calculate a_i , γ_i and G_m^E , then verify Gibbs–Duhem numerically. Change $\Omega/(RT)$ and observe how energetic preference competes with ideal configurational entropy.

9.12 Week 9 learning map

Five essential ideas

Activity carries chemical potential. Its value requires a standard state. Raoult and Henry conventions use different limits. Activity coefficients in one phase are coupled by Gibbs–Duhem. Bond-energy preferences can generate excess Gibbs energy, but a random solution model cannot describe long-range order.

Common misconception: $\Omega < 0$ means unlike neighbours are favoured; it does not automatically mean that an ordered crystal has been described. Ordering requires site or sublattice information.

Week 10

Regular solutions, stability and binary phase diagrams

Where we are

Week 9 connected activities to atomic interactions. We now ask whether a homogeneous solution is stable and how competing free-energy curves generate a phase diagram:

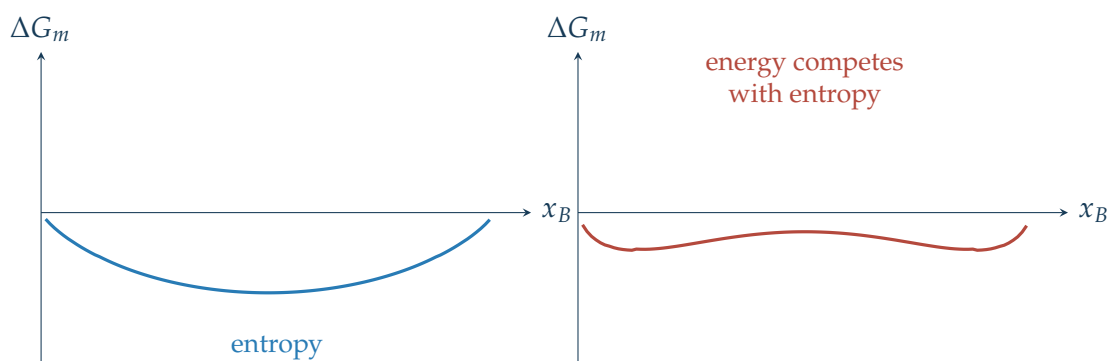
$G_m(x, T) \rightarrow$ curvature $\rightarrow$ binodal and spinodal $\rightarrow$ phase fractions and transformations.

10.1 The symmetric regular solution

For $x = x_B$ and $x_A = 1 - x$, the free energy of mixing is

$$\Delta G_m^{\text{mix}} = RT[(1 - x) \ln(1 - x) + x \ln x] + \Omega x(1 - x). \quad (10.1)$$

The first term always favours mixing. A positive Ω penalizes unlike neighbours and may overcome entropy at low temperature. A negative Ω lowers the random-solution energy but still does not introduce a long-range ordering variable.



10.2 Local stability and the spinodal

A homogeneous phase is locally stable against an infinitesimal composition fluctuation when

$$\left(\frac{\partial^2 G_m}{\partial x^2} \right)_{T,p} > 0. \quad (10.2)$$

For the symmetric regular solution,

$$\frac{\partial^2 \Delta G_m^{\text{mix}}}{\partial x^2} = RT \left(\frac{1}{x} + \frac{1}{1-x} \right) - 2\Omega. \quad (10.3)$$

The spinodal is the locus where this curvature vanishes:

$$x_{\text{sp}}^{\pm} = \frac{1}{2} \left[1 \pm \sqrt{1 - \frac{2RT}{\Omega}} \right]. \quad (10.4)$$

It exists only for $T \leq T_c$, with

$$x_c = \frac{1}{2}, \quad T_c = \frac{\Omega}{2R}. \quad (10.5)$$

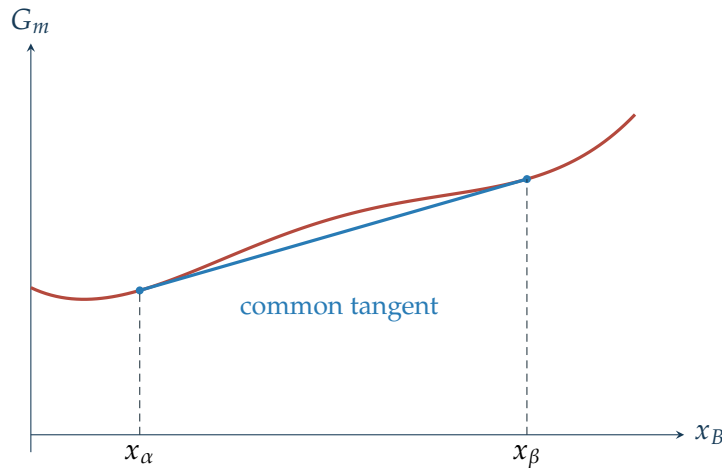
Spinodal is a local criterion

Inside the spinodal, arbitrarily small long-wavelength fluctuations lower the homogeneous free energy. Between the spinodal and binodal, the homogeneous phase is metastable: separation lowers the global free energy, but a finite fluctuation and an interface are needed.

10.3 Binodal from the common tangent

Two compositions x_α and x_β coexist when both component chemical potentials are equal. Geometrically, one line must be tangent to $G_m(x)$ at both compositions:

$$G'_m(x_\alpha) = G'_m(x_\beta) = \frac{G_m(x_\beta) - G_m(x_\alpha)}{x_\beta - x_\alpha}. \quad (10.6)$$



For the symmetric regular solution, $x_\beta = 1 - x_\alpha$ and the common-tangent condition reduces to

$$\ln \frac{x_\alpha}{1 - x_\alpha} + \frac{\Omega}{RT}(1 - 2x_\alpha) = 0. \quad (10.7)$$

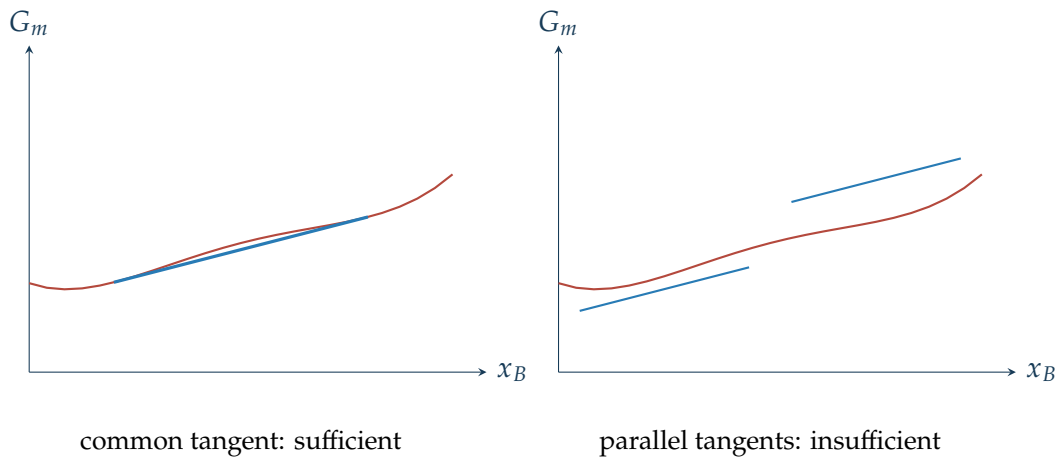
This transcendental equation is solved numerically at each temperature.

10.3.1 Why parallel tangents are insufficient

Equal tangent slopes impose only

$$\mu_B^\alpha - \mu_A^\alpha = \mu_B^\beta - \mu_A^\beta. \quad (10.8)$$

This is one difference equality. It does not separately guarantee $\mu_A^\alpha = \mu_A^\beta$ and $\mu_B^\alpha = \mu_B^\beta$. The same tangent line, including the same intercepts on both component axes, supplies both equalities.



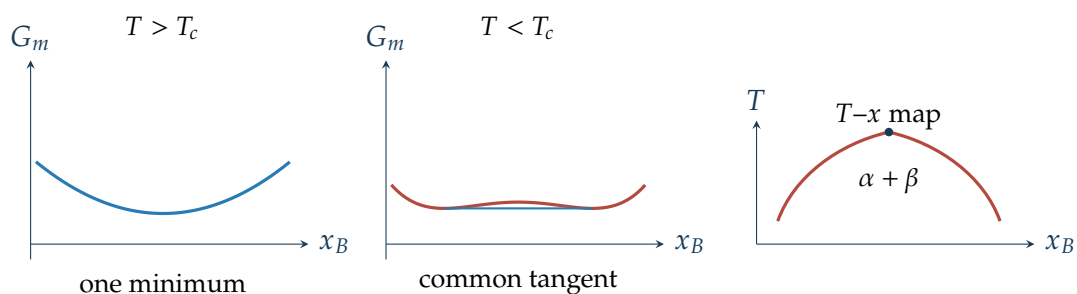
10.4 From free-energy curves to a phase diagram

At every temperature:

1. construct the free energy of every candidate phase over its composition range;
2. find the lower convex envelope of all curves;
3. retain single-phase portions that lie on the envelope;
4. locate common tangencies between two curves or two parts of one curve;
5. record the touching compositions as phase-boundary points;
6. repeat in temperature and join the points smoothly;
7. check limiting temperatures, stoichiometric constraints and the phase rule.



10.4.1 Read a miscibility gap from a sequence of free-energy curves



Above T_c , every overall composition is represented by one stable homogeneous minimum. Below T_c , the common-tangent contact compositions become the two binodal points at that temperature. Repeating the construction traces the coexistence dome. The spinodal lies inside this dome and comes from zero local curvature rather than from the tangent.

10.5 Solid–liquid equilibrium in an ideal binary system

Equality of chemical potential for component i between ideal liquid and ideal solid gives

$$\mu_i^{L,*} + RT \ln x_i^L = \mu_i^{S,*} + RT \ln x_i^S. \quad (10.9)$$

Approximating the pure-component fusion enthalpy as constant,

$$\ln \frac{x_i^L}{x_i^S} = \frac{\Delta H_{\text{fus},i}}{R} \left(\frac{1}{T_{m,i}} - \frac{1}{T} \right). \quad (10.10)$$

One equation is written for A and another for B. Together with $x_A^\phi + x_B^\phi = 1$, they determine the conjugate solid and liquid compositions at each temperature.

10.5.1 Where the liquidus relation comes from

Define the pure-component fusion Gibbs energy

$$\Delta G_{\text{fus},i}^*(T) = \mu_i^{L,*}(T) - \mu_i^{S,*}(T). \quad (10.11)$$

The Gibbs–Helmholtz equation gives

$$\left[\frac{\partial(\Delta G_{\text{fus},i}^*/T)}{\partial T} \right]_p = -\frac{\Delta H_{\text{fus},i}^*}{T^2}. \quad (10.12)$$

At the pure melting point, $\Delta G_{\text{fus},i}^*(T_{m,i}) = 0$. Assuming constant fusion enthalpy and integrating from $T_{m,i}$ to T produces the preceding logarithmic liquid–solid composition relation. The commonly used liquidus equation is therefore an integrated chemical-potential equality, not an empirical curve fit.

10.6 Common binary phase-diagram topologies

The topology depends on the relative free energies and composition ranges of the liquid, terminal solids and intermediate phases.

10.6.1 Complete liquid and solid solubility

If both liquid and solid are stable over the full composition range, common tangents between $G_m^L(x)$ and $G_m^S(x)$ generate liquidus and solidus curves. Between them, liquid and solid coexist with different compositions. This is the isomorphous case.

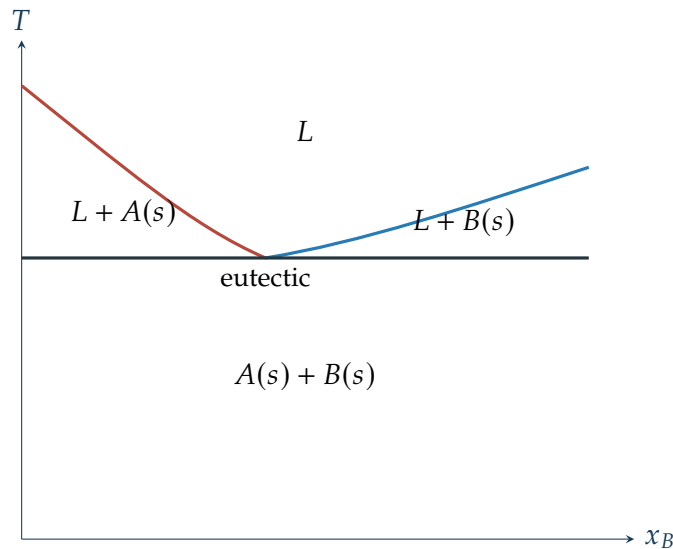
10.6.2 Complete liquid solubility but negligible terminal solid solubility

If solid A dissolves essentially no B and solid B dissolves essentially no A, the two solid free energies reduce to endpoint states. For ideal liquid, the two liquidus branches are

$$\ln x_A^L = \frac{\Delta H_{\text{fus},A}}{R} \left(\frac{1}{T_{m,A}} - \frac{1}{T} \right), \quad (10.13)$$

$$\ln x_B^L = \frac{\Delta H_{\text{fus},B}}{R} \left(\frac{1}{T_{m,B}} - \frac{1}{T} \right). \quad (10.14)$$

Their intersection satisfies $x_A^L + x_B^L = 1$ and defines the ideal eutectic. At fixed pressure the eutectic has three phases, $L + A(s) + B(s)$, so $F = 2 - 3 + 1 = 0$.



10.6.3 Limited terminal solubility

When A-rich α and B-rich β solids accept finite solute concentrations, their free-energy curves extend inward from the endpoints. Common tangencies generate solvus boundaries as well as solidus and liquidus boundaries. The eutectic reaction becomes



with three distinct equilibrium compositions at the invariant temperature.

10.6.4 Intermediate compounds

A stoichiometric compound A_mB_n is represented ideally by a free-energy value at the fixed composition $x_B = n/(m + n)$, often drawn as a line phase in a T - x diagram. A nonstoichiometric compound has a finite homogeneity range and therefore requires a composition-dependent free-energy model. Ordered intermediate phases commonly require sublattice models rather than a single random-solution curve.

Do not infer the model from the drawing alone

A narrow phase field may be a genuinely stoichiometric compound, a nonstoichiometric phase with a small homogeneity range, or a plotting simplification. The thermodynamic

model must state which interpretation is intended.

10.7 Free-energy atlas of binary phase diagrams

A phase diagram is a compact record of many Gibbs-free-energy diagrams. We consider a binary condensed system at fixed pressure, so vapour is absent and pressure is not an independent variable. At every temperature, draw $G_m^p(x_B)$ for every possible condensed phase p and retain the lower convex envelope. A curve segment on that envelope is a stable single phase; a straight segment joining two tangency points is a stable two-phase mixture.

The central construction rule

At fixed T and p , phases p and q coexist when

$$\mu_A^p = \mu_A^q, \quad \mu_B^p = \mu_B^q.$$

On a binary G_m - x_B diagram, one line must therefore be tangent to both curves. At an invariant temperature three phases coexist, so the same line touches all three relevant free-energy curves at their three equilibrium compositions.

10.7.1 Begin with the phase free energies, not a remembered diagram

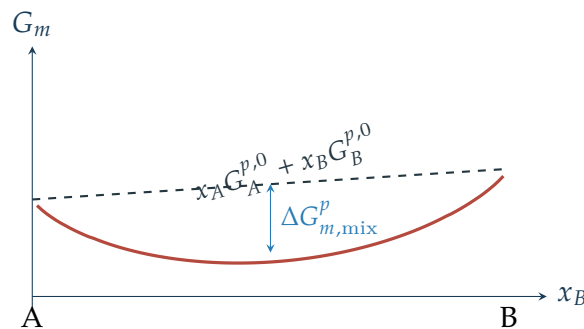
For each liquid or solid phase p ,

$$G_m^p(x_B, T) = x_A G_A^{p,0}(T) + x_B G_B^{p,0}(T) + \Delta G_{m,\text{mix}}^p(x_B, T). \quad (10.16)$$

The first two terms form the straight reference-state line between pure A and pure B in phase p . The mixing term bends that line. The relative vertical positions of liquid and solid reference states change with temperature through the fusion free energies,

$$\Delta G_{\text{fus},i}(T) = G_i^{L,0}(T) - G_i^{S,0}(T), \quad (10.17)$$

which changes sign at $T_{m,i}$.



What temperature changes

Changing temperature does more than change the depth of an ideal-mixing curve. It changes the pure-component reference-state free energies and may change the mixing free energy differently in each phase. Those movements decide which curve enters the lower envelope

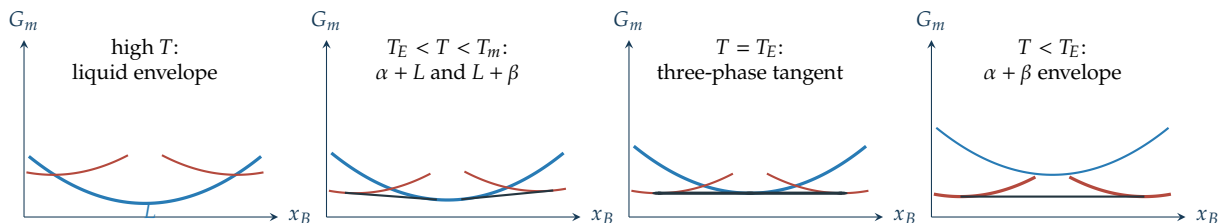
and where common tangencies occur.

10.7.2 The method used to build a condensed binary diagram

1. List all possible condensed phases: L , terminal solutions α and β , and any intermediate phases.
2. Write every $G_m^p(x_B, T)$ using the same elemental reference states.
3. Select one temperature and draw all curves on the same G_m scale.
4. Replace non-convex portions by their lower convex envelope. Mark every tangent point.
5. Repeat at a nearby temperature. The tangent points move with temperature.
6. Plot each tangent-point composition at its temperature. Their loci are liquidus, solidus, solvus or liquid-miscibility boundaries.
7. At a temperature where one tangent touches three equilibrium compositions, draw the invariant horizontal in the T - x_B diagram.

10.7.3 A four-temperature construction: the eutectic as an example

The following sequence should be read from left to right. It is schematic, but every displayed invariant tangent is an actual tangent to the curves shown.



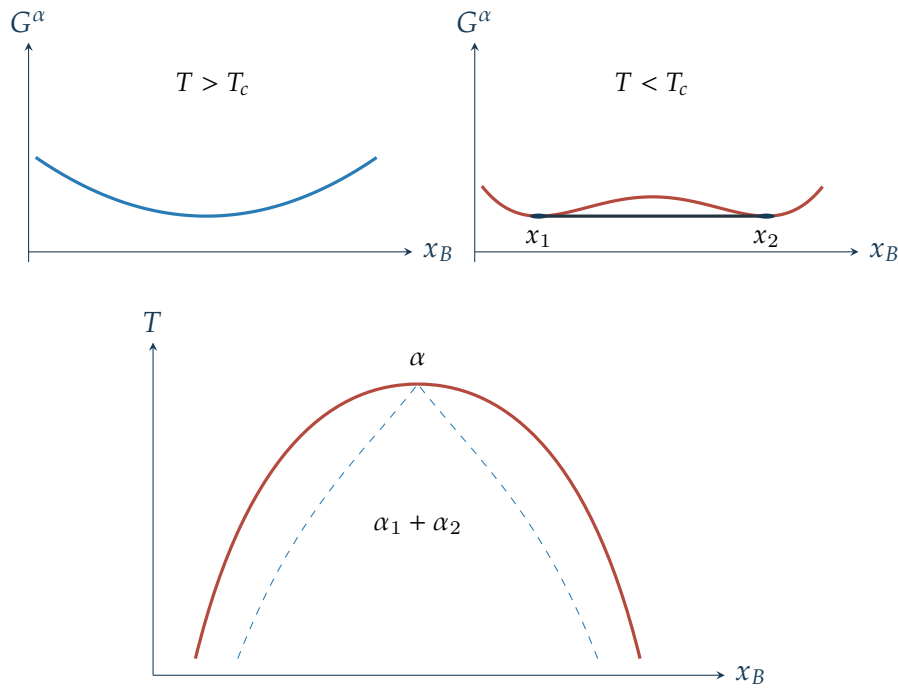
Reading the sequence gives the diagram rather than assuming it. The liquidus records the liquid tangent compositions; the solidus records the solid tangent compositions; the common three-phase temperature becomes the eutectic horizontal.

Stable, metastable and unstable extensions

Only the lower convex envelope is globally stable. A locally convex curve lying above that envelope is metastable. A portion with $\partial^2 G_m / \partial x_B^2 < 0$ is locally unstable. Phase-boundary lines may be extended as useful metastable constructions, but those extensions must be drawn differently and labelled.

10.7.4 Miscibility gap: one phase curve becomes non-convex

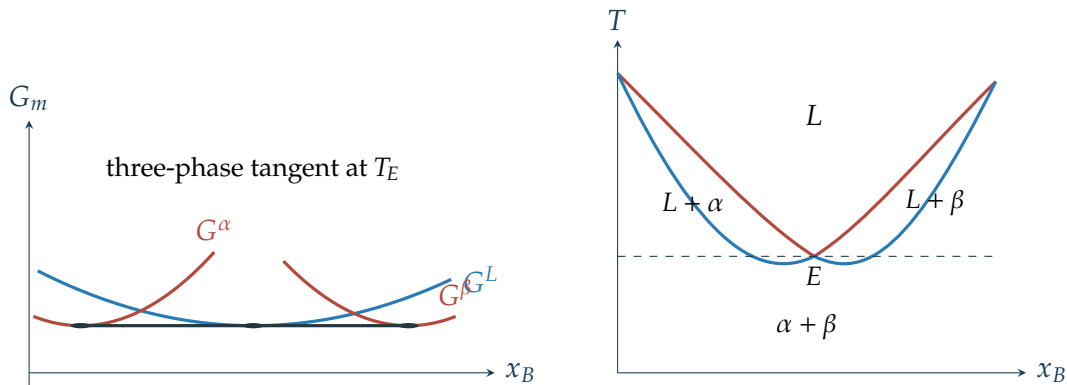
Above the critical temperature the free energy of phase α is convex. Below it, one common tangent touches the same curve twice and selects the compositions of α_1 and α_2 .



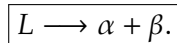
The binodal is obtained from global common tangency. The spinodal is obtained from local curvature, $\partial^2 G / \partial x_B^2 = 0$. The solid lines in the T - x diagram are binodals; the dashed inner curves are spinodals.

10.7.5 Eutectic: one liquid becomes two solids

Compare G^L , G^α and G^β as temperature is lowered. Above the liquidus, liquid is on the lower envelope. In an $L + \alpha$ or $L + \beta$ field, a tangent connects the liquid and one solid. At T_E , one line touches all three curves. Below T_E , the α - β tangent is the stable envelope.

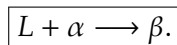


Thus the invariant reaction on cooling is

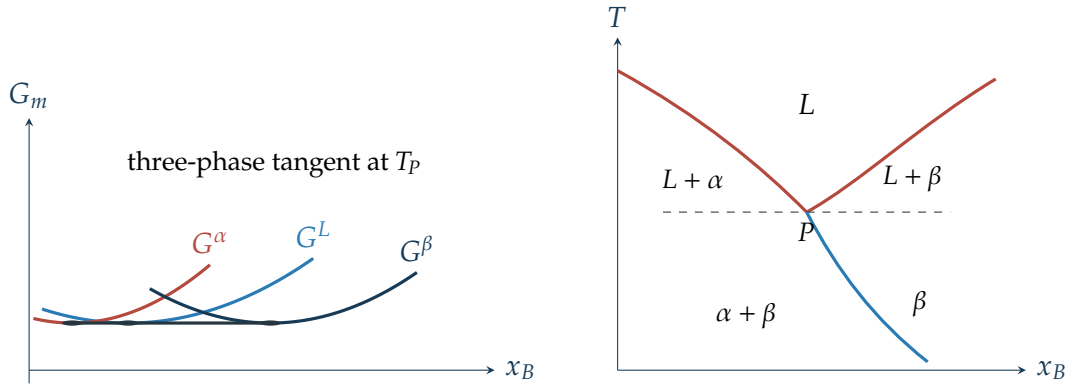


10.7.6 Peritectic: a liquid and a solid form another solid

At T_P , one tangent touches G^L at x_L^P , G^α at x_α^P and G^β at x_β^P . Immediately below T_P , β moves onto the lower envelope:

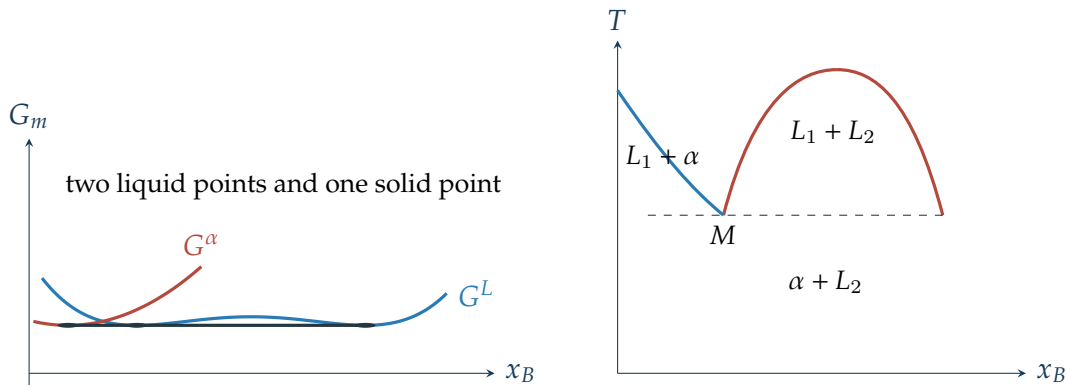
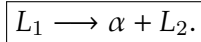


The product often forms between liquid and α , so diffusion through the β layer may prevent complete equilibrium in an experiment.



10.7.7 Monotectic: liquid demixing meets a solid

A monotectic requires a liquid miscibility gap. The non-convex G^L permits two liquids, L_1 and L_2 . At T_M , one line touches G^L twice and G^α once:



10.7.8 A unified invariant-reaction table

Name	Reaction on cooling	Free-energy interpretation at the invariant temperature
Eutectic	$L \rightarrow \alpha + \beta$	one tangent touches one liquid and two solid curves
Peritectic	$L + \alpha \rightarrow \beta$	one tangent touches the liquid, reactant-solid and product-solid curves
Monotectic	$L_1 \rightarrow \alpha + L_2$	one tangent touches a non-convex liquid curve twice and a solid curve once
Syntectic	$L_1 + L_2 \rightarrow \alpha$	the same three-phase tangency, with the solid stable below the invariant temperature
Eutectoid	$\gamma \rightarrow \alpha + \beta$	solid-state analogue of a eutectic: three solid curves share one tangent
Peritectoid	$\alpha + \beta \rightarrow \gamma$	solid-state analogue of a peritectic

The arrow requires temperatures above and below

The common tangent establishes three-phase equilibrium exactly at the invariant temperature. The direction written on cooling is determined by which curve moves onto the lower convex envelope immediately below that temperature. A single tangency drawing does not determine the reaction arrow.

10.7.9 Worked example: amounts after a monotectic reaction

Suppose an alloy has $x_B^0 = 0.30$. At T_M , the common tangent gives $x_B^\alpha = 0.05$ and $x_B^{L_2} = 0.80$. After the monotectic reaction reaches equilibrium,

$$f_\alpha = \frac{x_B^{L_2} - x_B^0}{x_B^{L_2} - x_B^\alpha} = \frac{0.80 - 0.30}{0.80 - 0.05} = 0.667,$$

$$f_{L_2} = \frac{x_B^0 - x_B^\alpha}{x_B^{L_2} - x_B^\alpha} = 0.333.$$

The tangent supplies equilibrium compositions; the lever rule supplies amounts.

10.7.10 Construction recipe from a thermodynamic model

1. Write $G_m^p(x_B, T)$ for every candidate phase using consistent reference states.
2. At a chosen T , calculate every curve on the same vertical scale.
3. Retain single-phase intervals on the lower convex envelope.
4. Solve pairwise common-tangent equations for two-phase endpoints.
5. Locate temperatures at which one tangent touches three curves, or one curve twice and another once.
6. Repeat over temperature and join the endpoints to form liquidus, solidus, solvus and miscibility-gap boundaries.
7. Label the stable fields first; then name the invariant reactions.

10.8 Phase amounts: the lever rule

For an overall composition x_0 lying between x_α and x_β ,

$$f_\alpha = \frac{x_\beta - x_0}{x_\beta - x_\alpha}, \quad f_\beta = \frac{x_0 - x_\alpha}{x_\beta - x_\alpha}, \quad f_\alpha + f_\beta = 1. \quad (10.18)$$

The lever rule follows from material balance. It does not determine the equilibrium compositions; those come from chemical-potential equality or the common tangent.

10.9 Ordering, clustering and phase separation

Concept	Required description	Diagnostic
Clustering tendency	Preference for like neighbours; may be represented by $\Omega > 0$	Positive mixing enthalpy and enhanced like-pair probability
Macroscopic phase separation	Non-convex total free energy plus global equilibrium construction	Common tangent and two distinct equilibrium compositions
Long-range ordering	Distinct sublattices or site-occupancy order parameter	Unequal sublattice occupancies and an order-disorder transition

A random regular solution can predict clustering and phase separation. It cannot establish long-range order because all lattice sites are treated as equivalent.

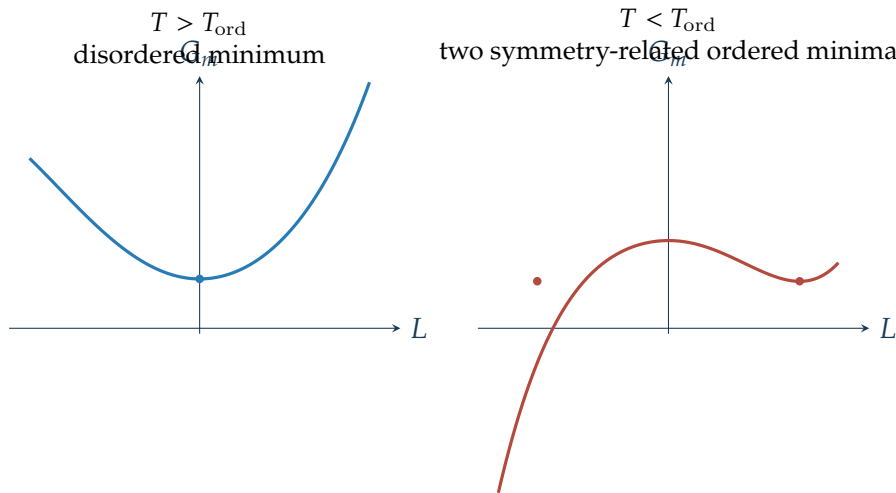
10.10 Order-disorder transition in the B2 sublattice model

Week 9 derived the equiatomic two-sublattice free energy $G_m(L)$. Its curvature at the disordered state is

$$\left. \frac{\partial^2 G_m}{\partial L^2} \right|_{L=0} = \frac{\Omega}{2} + RT. \quad (10.19)$$

For $\Omega < 0$, the disordered state changes stability when this curvature vanishes:

$$T_{\text{ord}} = -\frac{\Omega}{2R}. \quad (10.20)$$



The two minima $+L$ and $-L$ correspond to interchanging which species prefers the α sublattice. They are distinct antiphase variants of the same B2 ordered phase. An antiphase boundary separates regions with opposite signs of L .

10.10.1 Ordering is not phase separation

For $\Omega > 0$, the important instability is often in composition space: the curvature $\partial^2 G_m / \partial x_B^2$ can become negative and drive decomposition into A-rich and B-rich phases. For $\Omega < 0$ in the two-sublattice model, the instability can instead be in ordering space: $\partial^2 G_m / \partial L^2$ becomes negative at $L = 0$ while the overall composition remains equiatomic.

Question	Phase separation	B2 ordering
Variable that changes Conserved?	Composition x_B Yes for a closed substitutional alloy	Sublattice order L No; atoms exchange sublattice preference locally while total composition is fixed
Typical energetic sign Equilibrium signature	$\Omega > 0$ favours like neighbours Common tangent between different compositions	$\Omega < 0$ favours unlike α - β neighbours Minimum of $G_m(L)$ at nonzero L
Microstructural object	A-rich and B-rich domains	Ordered variants and antiphase boundaries

10.10.2 Worked ordering calculation

Problem. An equiatomic B2 model has $\Omega = -20.0 \text{ kJ mol}^{-1}$. Find the mean-field ordering temperature. At $T = 800 \text{ K}$, solve

$$L = \tanh\left(-\frac{\Omega L}{2RT}\right)$$

approximately.

Step 1: transition temperature.

$$T_{\text{ord}} = \frac{20000}{2(8.314)} = \boxed{1203 \text{ K}}. \quad (10.21)$$

Step 2: form the dimensionless interaction.

$$-\frac{\Omega}{2RT} = \frac{20000}{2(8.314)(800)} = 1.5035. \quad (10.22)$$

Step 3: fixed-point iteration. Starting with $L_0 = 0.8$,

$$L_1 = \tanh(1.5035L_0) = 0.835, \quad L_2 = 0.850, \quad L_3 = 0.856,$$

and continued iteration gives

$$\boxed{L_{\text{eq}} \simeq 0.859}. \quad (10.23)$$

Step 4: interpret site occupancies.

$$y_A^\alpha = y_B^\beta = \frac{1+L}{2} = 0.9295, \quad y_B^\alpha = y_A^\beta = 0.0705. \quad (10.24)$$

The alloy remains 50–50 overall, but approximately 93% of each sublattice is occupied by its preferred species.

10.11 Worked examples

10.11.1 Critical temperature, spinodal and binodal

Problem. A symmetric regular solution has $\Omega = 20.0 \text{ kJ mol}^{-1}$. At $T = 800 \text{ K}$, find the critical temperature and spinodal compositions. Then determine the binodal compositions numerically.

Step 1: critical point.

$$T_c = \frac{\Omega}{2R} = \frac{20000}{2(8.314)} = \boxed{1203 \text{ K}}. \quad (10.25)$$

Since $800 < T_c$, a miscibility gap exists.

Step 2: spinodal.

$$x_{\text{sp}}^{\pm} = \frac{1}{2} \left[1 \pm \sqrt{1 - \frac{2(8.314)(800)}{20000}} \right] \quad (10.26)$$

$$= \boxed{0.211 \text{ and } 0.789}. \quad (10.27)$$

Step 3: binodal equation. For the A-rich composition $x_{\alpha} < 1/2$,

$$f(x_{\alpha}) = \ln \frac{x_{\alpha}}{1 - x_{\alpha}} + \frac{20000}{(8.314)(800)}(1 - 2x_{\alpha}) = 0. \quad (10.28)$$

Bisection or Newton iteration gives $x_{\alpha} = 0.0701$. By symmetry,

$$\boxed{x_{\alpha} = 0.0701, \quad x_{\beta} = 0.9299}. \quad (10.29)$$

The binodal encloses the spinodal. Compositions between 0.211 and 0.789 are unstable; the remaining portions inside the binodal are metastable.

10.11.2 Lever-rule phase fractions

Problem. At a selected temperature, $x_{\alpha} = 0.20$ and $x_{\beta} = 0.80$. A specimen has $x_0 = 0.35$. Find the equilibrium phase fractions.

$$f_{\alpha} = \frac{0.80 - 0.35}{0.80 - 0.20} = \boxed{0.75}, \quad (10.30)$$

$$f_{\beta} = \frac{0.35 - 0.20}{0.80 - 0.20} = \boxed{0.25}. \quad (10.31)$$

Check: $(0.75)(0.20) + (0.25)(0.80) = 0.35$.

10.11.3 Melting-point depression of a dilute ideal solution

Problem. Component A melts at $T_{m,A} = 1000 \text{ K}$ with $\Delta H_{\text{fus},A} = 12.0 \text{ kJ mol}^{-1}$. At equilibrium with nearly pure solid A, let $x_A^S \simeq 1$ and $x_A^L = 0.95$. Estimate the liquidus temperature.

Step 1: write the ideal equilibrium relation.

$$\ln(0.95) = \frac{12000}{R} \left(\frac{1}{1000} - \frac{1}{T} \right). \quad (10.32)$$

Step 2: solve for T .

$$\frac{1}{T} = \frac{1}{1000} - \frac{R}{12000} \ln(0.95) = 0.0010355 \text{ K}^{-1}, \quad (10.33)$$

so

$$\boxed{T \simeq 966 \text{ K}}. \quad (10.34)$$

The solute lowers the chemical potential of A in the liquid, so equilibrium with solid A occurs below the pure melting point.

10.11.4 Calculate an ideal eutectic

Problem. A and B form an ideal liquid and mutually insoluble pure solids. Let

$$T_{m,A} = 1000 \text{ K}, \quad \Delta H_{\text{fus},A} = 12.0 \text{ kJ mol}^{-1},$$

$$T_{m,B} = 800 \text{ K}, \quad \Delta H_{\text{fus},B} = 10.0 \text{ kJ mol}^{-1}.$$

Find the eutectic temperature and liquid composition.

Step 1: write both liquidus branches.

$$x_A^L(T) = \exp \left[\frac{12000}{R} \left(\frac{1}{1000} - \frac{1}{T} \right) \right], \quad (10.35)$$

$$x_B^L(T) = \exp \left[\frac{10000}{R} \left(\frac{1}{800} - \frac{1}{T} \right) \right]. \quad (10.36)$$

Step 2: impose the eutectic composition constraint.

$$x_A^L(T_E) + x_B^L(T_E) = 1. \quad (10.37)$$

Numerical bisection gives

$$\boxed{T_E = 603.4 \text{ K}}, \quad \boxed{x_A^E = 0.3873, \text{quad} x_B^E = 0.6127}. \quad (10.38)$$

Step 3: check the physics. The eutectic temperature lies below both pure melting points, and the two liquid mole fractions add to one. At this point liquid, pure solid A and pure solid B coexist.

10.11.5 Bi–Cd eutectic: a handbook-data example

This example follows the Type I phase-diagram exercise in Bokstein, Mendeleev and Srolovitz. Within the model, liquid Bi–Cd obeys Raoult’s law at every composition and the solid phases are pure Bi and pure Cd. The pure-component data are

Component	T_m (K)	ΔH_{fus} (kJ mol ^{−1})
Bi	544.5	10.88
Cd	594.2	6.40

Problem. Calculate the eutectic temperature and liquid composition predicted by these assumptions.

Step 1: write the Bi liquidus branch. Equilibrium between liquid Bi in the solution and pure solid Bi gives

$$x_{\text{Bi}}^L(T) = \exp \left[\frac{10880}{R} \left(\frac{1}{544.5} - \frac{1}{T} \right) \right]. \quad (10.39)$$

Step 2: write the Cd liquidus branch.

$$x_{\text{Cd}}^L(T) = \exp \left[\frac{6400}{R} \left(\frac{1}{594.2} - \frac{1}{T} \right) \right]. \quad (10.40)$$

Step 3: impose the eutectic constraint. At the intersection of the two branches, the liquid is one binary phase, so

$$f(T) = x_{\text{Bi}}^L(T) + x_{\text{Cd}}^L(T) - 1 = 0. \quad (10.41)$$

Both exponentials increase monotonically with T below their respective melting points. Bisection is therefore straightforward:

T (K)	x_{Bi}^L	x_{Cd}^L	$f(T)$
400	0.420	0.533	-0.047
410	0.455	0.559	+0.013
407.81	0.44684	0.55316	approximately 0

Hence

$$\boxed{T_E = 407.81 \text{ K}}, \quad \boxed{x_{\text{Bi}}^E = 0.44684, \quad x_{\text{Cd}}^E = 0.55316.} \quad (10.42)$$

Step 4: interpret and check. The predicted eutectic temperature lies below both pure melting points, the two liquid mole fractions sum to one, and three phases coexist:

$$L(x_{\text{Cd}} = 0.55316) + \text{Bi}(s) + \text{Cd}(s).$$

At fixed pressure this is invariant because $F = C - P + 1 = 2 - 3 + 1 = 0$.

Model result versus assessed diagram

This calculation demonstrates how pure-component fusion data and an ideal-liquid model generate a eutectic without drawing it by hand. Its numerical result belongs to the assumptions of the textbook exercise. Comparison with an assessed experimental Bi–Cd diagram tests the adequacy of Raoult behaviour, negligible solid solubility and constant fusion enthalpies.

10.11.6 Cooling a hypoeutectic Bi–Cd alloy

Problem. Use the idealized Bi–Cd diagram just calculated. An alloy has overall composition $x_{\text{Cd}} = 0.300$, on the Bi-rich side of the eutectic $x_{\text{Cd}}^E = 0.55316$. Immediately above T_E , approximate the solid as pure Bi and the liquid as the eutectic liquid. Find the phase fractions and describe the cooling curve.

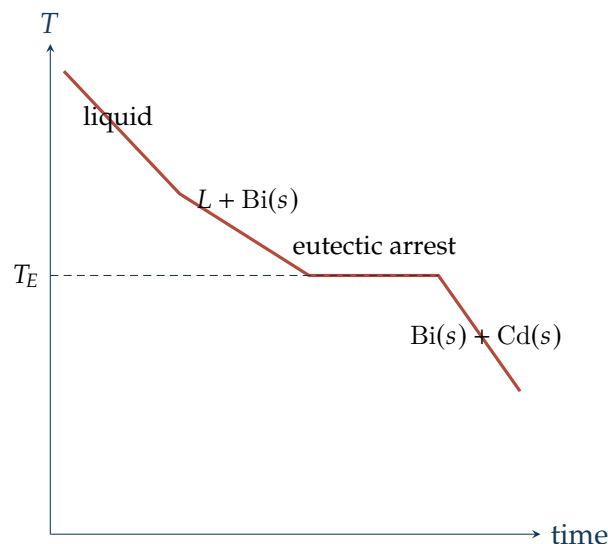
Step 1: apply the lever rule immediately above T_E .

$$f_L = \frac{x_0 - x_{\text{Bi}(s)}}{x_L - x_{\text{Bi}(s)}} = \frac{0.300 - 0}{0.55316 - 0} = \boxed{0.542}, \quad (10.43)$$

$$f_{\text{Bi}(s)} = 1 - f_L = \boxed{0.458}. \quad (10.44)$$

Step 2: follow the transformations.

1. Above the liquidus, the alloy is entirely liquid and cools without a phase transformation.
2. At the liquidus, primary Bi begins to crystallize. Latent heat reduces the cooling-curve slope, while the remaining liquid becomes richer in Cd and moves along the liquidus toward x_{Cd}^E .
3. At T_E , the remaining fraction $f_L = 0.542$ transforms isothermally into Bi(s)+Cd(s). The ideal equilibrium cooling curve therefore shows a horizontal eutectic arrest.
4. Below T_E , the fully solid two-phase mixture cools again, generally with a different slope because its heat capacity and thermal transport differ from those of the liquid-containing state.



What a cooling curve measures

The discontinuity in slope or an isothermal arrest is an experimental thermal signature. A family of cooling curves at known compositions can locate liquidus and invariant temperatures, but identifying equilibrium boundaries also requires control of undercooling, transformation kinetics and heat-extraction conditions.

10.12 Computational laboratory

For several values of $\Omega/(RT)$, plot ΔG_m^{mix} , its first derivative and its second derivative. Locate spinodals from zero curvature and binodals from a numerical common-tangent calculation. Assemble the results into a T - x miscibility-gap diagram. Add an overall composition and verify the lever-rule reconstruction. Finally compare a common-tangent solver with a deliberately incorrect parallel-slope test.

10.13 Phase-diagram construction checklist

1. List every candidate phase and its valid composition range.
2. Use one common reference for all Gibbs-energy functions.
3. Check endpoint values, slopes, units and temperature dependence.
4. Construct the lower convex envelope, not merely the lowest value at isolated grid points.
5. Require a common tangent for two-phase equilibrium.
6. Distinguish binodal, spinodal and metastable regions.
7. Use the phase rule to check degrees of freedom.
8. Apply the lever rule only after equilibrium compositions are known.
9. Treat stoichiometric compounds and sublattice-ordered phases with appropriate models.
10. Verify limiting cases and compare calculated invariant points with independent data.

10.14 Week 10 learning map

Five essential ideas

Entropy stabilizes mixing at high temperature. Curvature determines local stability. Common tangents determine global equilibrium. Spinodal and binodal are different boundaries. A phase diagram is a compact record of competing Gibbs-energy surfaces, not a picture drawn independently of them.

Common misconception: equal slopes at two points do not establish equilibrium. The same tangent must touch both free-energy curves so that both component chemical potentials are equal.

Week 11

Experimental phase diagrams and reacting systems

Where we are

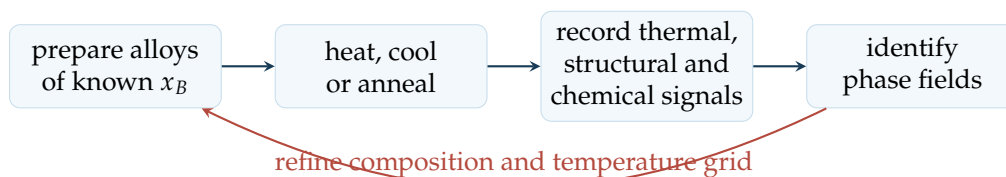
Weeks 8–10 built solution free energies and obtained phase boundaries by minimizing Gibbs energy. We now reverse the direction of reasoning:

measured signal $\longrightarrow$ phase boundary or reaction state $\longrightarrow$ thermodynamic interpretation.

The first half of this week asks how a phase diagram is actually determined. The second half extends equilibrium thermodynamics to systems in which chemical reactions change the amounts of the species.

11.1 A phase diagram is assembled from experiments

A phase diagram is not read directly from a single cooling curve. It is a synthesis of complementary observations made on alloys of known overall composition after controlled thermal histories. Each method answers a different question.



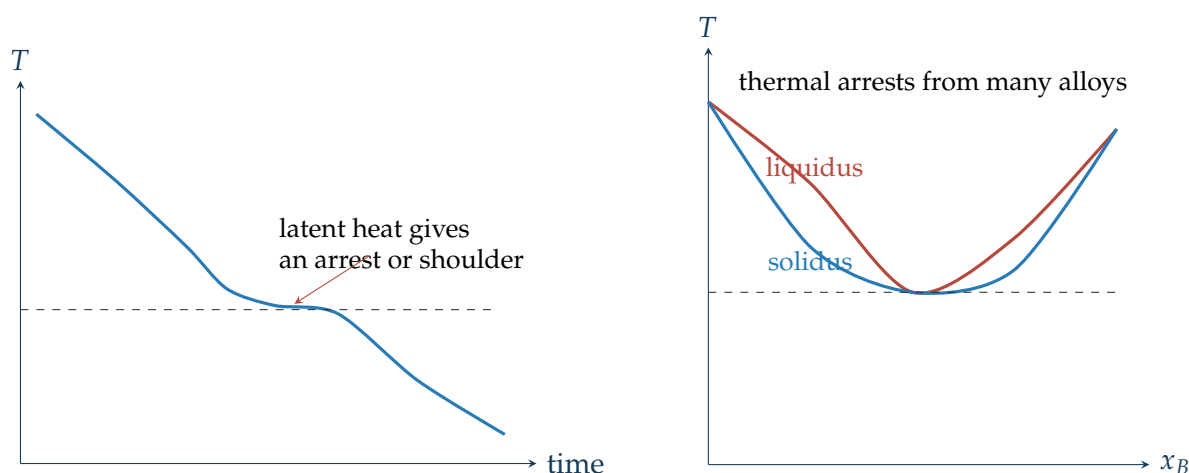
Method	Observable	What it establishes
Cooling or heating curve	arrest or change of slope in $T(t)$	temperature at which latent heat is released or absorbed
DSC/DTA	heat-flow peak or temperature difference	transition temperatures and transformation enthalpy
X-ray diffraction	peaks, intensities and lattice parameters	phase identity, crystal structure and sometimes phase fraction
Microscopy	morphology and contrast	number and spatial arrangement of phases
Microanalysis	composition inside each phase	tie-line endpoints and solubility limits
Diffusion couple	composition and phase sequence after annealing	phases and diffusion paths across a composition range

Equilibrium must be approached, not assumed

An observed boundary is an equilibrium boundary only if the specimen has had enough time to equilibrate and the result is independent of the path within experimental uncertainty. Heating and cooling arrests can differ because of undercooling, overheating, sluggish diffusion or nucleation barriers.

11.2 Cooling curves and thermal analysis

For a specimen losing heat to its surroundings, a single-phase region normally cools continuously. A transformation releases latent heat and slows the temperature change. An invariant reaction can produce an arrest because the phase rule fixes the equilibrium temperature at fixed pressure.



11.2.1 How to extract a boundary

1. Calibrate the thermocouple or calorimeter and state the heating or cooling rate.
2. Determine onset temperatures by a consistent baseline construction.
3. Repeat at several rates and extrapolate toward slow-rate behaviour when appropriate.
4. Compare heating and cooling. Their separation is evidence of kinetic delay, not a two-valued equilibrium temperature.

5. Confirm the phases by diffraction or microscopy. A thermal event alone does not identify the products.

11.2.2 Worked example: latent heat from a cooling arrest

A 20 g alloy has an effective heat capacity of $0.50 \text{ J g}^{-1} \text{ K}^{-1}$. Away from a transformation it cools at 5.0 K min^{-1} . During an arrest lasting 2.0 min, assume the same average heat-loss rate that would otherwise cool the specimen.

The heat-loss rate is

$$\dot{Q} = (20)(0.50)(5.0) = 50 \text{ J min}^{-1}.$$

The released latent heat is therefore approximately

$$Q_{\text{tr}} = \dot{Q}\Delta t = (50)(2.0) = 100 \text{ J}, \quad \Delta H_{\text{tr}} \approx 5.0 \text{ J g}^{-1}.$$

This estimate requires a baseline correction in a real experiment because heat loss and heat capacity vary with temperature.

11.3 Structure and composition complete the diagram

Thermal analysis locates events; diffraction and chemical analysis determine what phases are present and where the tie-line endpoints lie.

11.3.1 Lattice parameter as a solubility indicator

In a substitutional solid solution, Vegard's law is often a useful first approximation:

$$a(x_B) \simeq (1 - x_B)a_A + x_Ba_B. \quad (11.1)$$

If the measured lattice parameter stops changing as more solute is added, the matrix composition may have reached a solubility limit while a second phase grows. Vegard's law is empirical; ordering, elastic strain and non-ideal bonding can produce curvature.

11.3.2 Worked example: identify a solvus composition

At 700 K, a matrix has $a_A = 0.3600 \text{ nm}$ and a dilute Vegard slope $da/dx_B = 0.0400 \text{ nm}$. Samples with overall $x_B = 0.08, 0.12$ and 0.20 all show the same matrix lattice parameter 0.3640 nm , while the amount of a second phase increases.

$$x_B^\alpha = \frac{0.3640 - 0.3600}{0.0400} = 0.10.$$

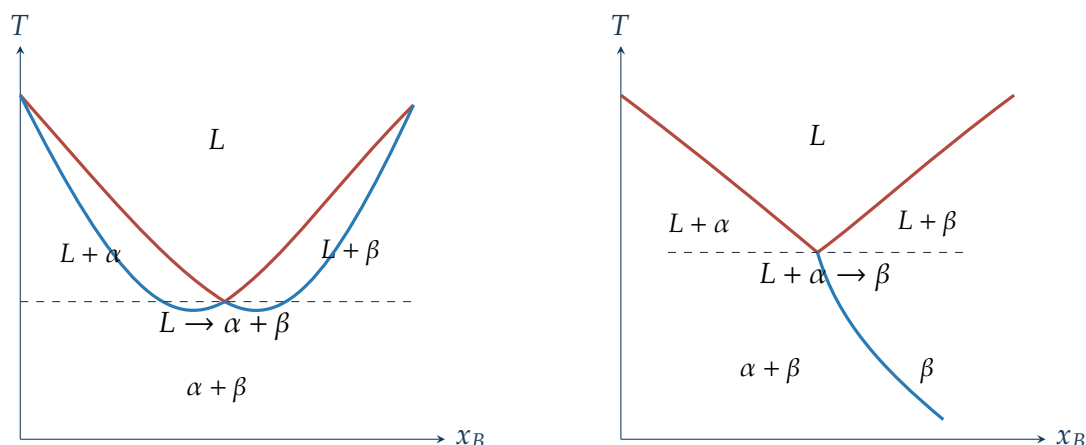
The repeated matrix lattice parameter indicates that the α phase remains at about 10% B; added B changes the phase fraction rather than the equilibrium matrix composition.

11.4 Recognizing invariant reactions

At constant pressure a binary system has

$$F = C - P + 1 = 3 - P.$$

Three coexisting phases give $F = 0$, so an invariant reaction occurs at a fixed temperature and fixed phase compositions.



Eutectic (left) and peritectic (right). The topology and phase labels, rather than the visual shape alone, identify the reaction.

A diagram gives equilibrium amounts, not morphology

The lever rule gives phase fractions after equilibrium has been reached. It does not predict lamellar spacing, particle size, dendrite morphology or the time needed to transform. Those require kinetics and interfacial physics.

11.5 Reacting systems: species, components and reactions

A *species* is a chemically distinguishable entity. A *component* is one member of a smallest independent set needed to describe all phase compositions. If S species participate in R independent reactions, then

$$C = S - R, \quad \boxed{F = S - P - R + 2 = C - P + 2.} \quad (11.2)$$

The number R is the rank of the stoichiometric reaction set. Writing more equations does not create more independent reactions.

11.5.1 Example: calcium carbonate decomposition

For



$S = 3$, $R = 1$ and $P = 3$, so $F = 1$. Specifying the temperature fixes the equilibrium p_{CO_2} ; alternatively, specifying the pressure fixes the decomposition temperature.

11.5.2 Example: carbon-oxygen system

With species C, O_2 , CO and CO_2 , the elements C and O form a component basis, so $C = 2$ and $R = S - C = 2$. A possible independent pair is



The reaction $\text{C} + \text{CO}_2 \rightleftharpoons 2\text{CO}$ is a linear combination and must not be counted again.

11.6 Extent of reaction and material balances

For one reaction, assign stoichiometric coefficients $\nu_i < 0$ to reactants and $\nu_i > 0$ to products. The extent ξ is defined by

$$dN_i = \nu_i d\xi, \quad N_i = N_{i,0} + \nu_i \xi. \quad (11.3)$$

For m independent reactions,

$$N_i = N_{i,0} + \sum_{r=1}^m \nu_{ir} \xi_r. \quad (11.4)$$

The admissible range of every ξ_r is constrained by $N_i \geq 0$.

11.6.1 Worked example: ammonia synthesis

For $N_2 + 3H_2 \rightleftharpoons 2NH_3$, start with 1.00 mol N_2 and 3.00 mol H_2 :

$$N_{N_2} = 1 - \xi, \quad N_{H_2} = 3 - 3\xi, \quad N_{NH_3} = 2\xi, \quad N_{\text{tot}} = 4 - 2\xi.$$

Thus $0 \leq \xi \leq 1$. At $\xi = 0.40$, the mole fractions are

$$y_{N_2} = \frac{0.60}{3.20} = 0.1875, \quad y_{H_2} = 0.5625, \quad y_{NH_3} = 0.2500.$$

11.7 Reaction Gibbs energy, reaction quotient and equilibrium

At fixed T and p ,

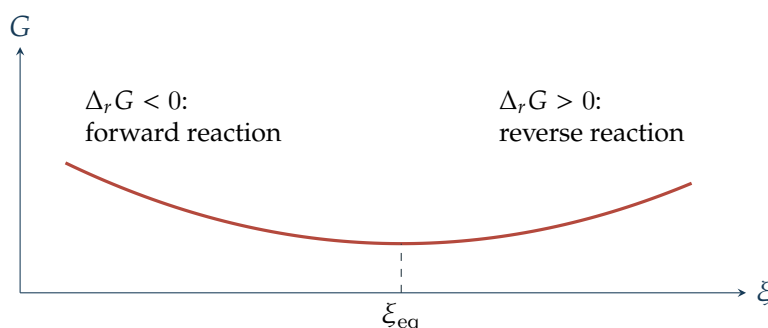
$$dG = \sum_i \mu_i dN_i = \left(\sum_i \nu_i \mu_i \right) d\xi = \Delta_r G d\xi. \quad (11.5)$$

Therefore

$$\Delta_r G = \sum_i \nu_i \mu_i = \Delta_r G^\circ + RT \ln Q. \quad (11.6)$$

The equilibrium condition is $\Delta_r G = 0$, giving

$$\Delta_r G^\circ = -RT \ln K, \quad Q = K. \quad (11.7)$$



Do not compare Q and K backwards

If $Q < K$, then $\Delta_r G = RT \ln(Q/K) < 0$ and the forward reaction lowers G . If $Q > K$, the reverse reaction lowers G .

11.7.1 Worked example: direction of reaction

For $A(g) + B(g) \rightleftharpoons 2C(g)$, let $K = 25$ and the partial pressures, divided by the standard pressure, be $a_A = 0.40$, $a_B = 0.20$, $a_C = 0.50$. Then

$$Q = \frac{a_C^2}{a_A a_B} = \frac{0.25}{0.08} = 3.125 < K.$$

At 1000 K,

$$\Delta_r G = RT \ln(Q/K) = (8.314)(1000) \ln(3.125/25) = -17.3 \text{ kJ mol}^{-1}.$$

The forward reaction is spontaneous at that composition.

11.8 Temperature, pressure and equilibrium

Differentiating $\Delta_r G^\circ = -RT \ln K$ gives the van't Hoff equation

$$\boxed{\frac{d \ln K}{dT} = \frac{\Delta_r H^\circ}{RT^2}} \quad (11.8)$$

If $\Delta_r H^\circ$ is approximately constant,

$$\ln \frac{K_2}{K_1} = -\frac{\Delta_r H^\circ}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right). \quad (11.9)$$

11.8.1 Worked example: an exothermic reaction

An exothermic reaction has $\Delta_r H^\circ = -80 \text{ kJ mol}^{-1}$ and $K_1 = 100$ at 700 K. Estimate K_2 at 800 K:

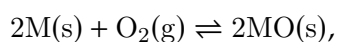
$$\ln \frac{K_2}{100} = -\frac{-80000}{8.314} \left(\frac{1}{800} - \frac{1}{700} \right) = -1.718,$$

so $K_2 \approx 17.9$. Heating reduces the equilibrium constant of an exothermic reaction.

For an ideal-gas reaction, increasing total pressure favours the side with fewer gas molecules only when the composition can respond and $\Delta \nu_{\text{gas}} \neq 0$. This statement follows from the pressure dependence of Q ; it is not an independent rule.

11.9 Heterogeneous reactions and Ellingham diagrams

For pure condensed phases their activities are unity. In the oxidation reaction



$$Q = \frac{1}{p_{O_2}/p^\circ}, \quad \Delta_r G = \Delta_r G^\circ - RT \ln \frac{p_{O_2}}{p^\circ}.$$

At equilibrium,

$$\boxed{\frac{p_{O_2}^{\text{eq}}}{p^\circ} = \exp \left(\frac{\Delta_r G^\circ}{RT} \right)} \quad (11.10)$$

An Ellingham diagram plots $\Delta_r G^\circ(T) \approx \Delta_r H^\circ - T \Delta_r S^\circ$. Its slope is approximately $-\Delta_r S^\circ$; a change of slope signals a phase transformation or a change in stoichiometry.

11.9.1 Normalize every oxidation reaction consistently

Ellingham lines must be compared for reactions consuming the same amount of oxygen. For an oxide M_xO_y , write the reaction per mole of O_2 :



The vertical axis is then $\Delta_r G^\circ$ per mole of O_2 . Comparing unnormalized reactions can reverse an apparent stability ranking.

At a reference temperature T_0 ,

$$\Delta_r H^\circ(T) = \Delta_r H^\circ(T_0) + \int_{T_0}^T \Delta C_p^\circ dT, \quad (11.12)$$

$$\Delta_r S^\circ(T) = \Delta_r S^\circ(T_0) + \int_{T_0}^T \frac{\Delta C_p^\circ}{T} dT, \quad (11.13)$$

$$\Delta_r G^\circ(T) = \Delta_r H^\circ(T) - T\Delta_r S^\circ(T). \quad (11.14)$$

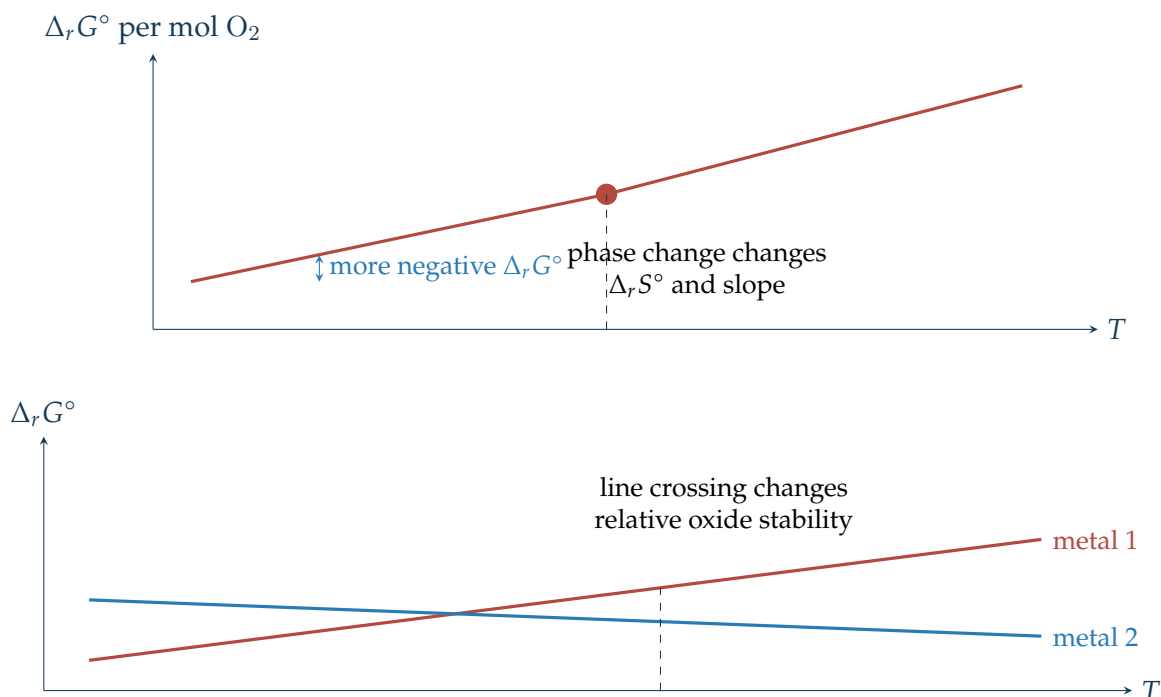
If ΔC_p° is neglected over a limited range, the Ellingham line is approximately straight:

$$\boxed{\text{intercept} \simeq \Delta_r H^\circ, \quad \text{slope} \simeq -\Delta_r S^\circ.} \quad (11.15)$$

Most metal-oxidation reactions consume one mole of gas and produce condensed oxide. Their reaction entropy is therefore usually negative and their Ellingham lines usually slope upward. Similar loss of gaseous entropy gives many oxide lines similar slopes.

11.9.2 Why lines kink but Gibbs energy remains continuous

When the metal or oxide melts, boils or changes crystal structure, its entropy changes. The reaction entropy and therefore the slope $-\Delta_r S^\circ$ change. The two line segments meet continuously at the transformation temperature because the two forms of that substance have equal Gibbs energy at their equilibrium transformation.



11.9.3 Read oxidation and reduction from oxygen potential

For pure metal and pure oxide, both condensed-phase activities are unity. For the oxygen-normalized reaction,

$$Q = \frac{1}{p_{\text{O}_2}/p^\circ}, \quad \Delta_r G = \Delta_r G^\circ - RT \ln \frac{p_{\text{O}_2}}{p^\circ}. \quad (11.16)$$

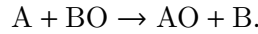
Consequently,

$$\boxed{\frac{p_{\text{O}_2}^{\text{eq}}}{p^\circ} = \exp\left(\frac{\Delta_r G^\circ}{RT}\right)}. \quad (11.17)$$

- If $p_{\text{O}_2} > p_{\text{O}_2}^{\text{eq}}$, then $\Delta_r G < 0$ and oxidation is favoured.
- If $p_{\text{O}_2} < p_{\text{O}_2}^{\text{eq}}$, the reverse reaction is favoured and the oxide tends to dissociate or be reduced.
- A lower Ellingham line means a more stable oxide relative to its metal and oxygen, provided reactions use the same oxygen normalization.

11.9.4 Why one metal can reduce another metal oxide

Suppose oxidation of A lies below oxidation of B. Subtract the B-oxidation reaction from the A-oxidation reaction. The oxygen cancels, leaving an exchange reaction of the form



Its standard free energy is the vertical separation

$$\Delta_r G_{\text{exchange}}^\circ = \Delta_r G_{\text{ox,A}}^\circ - \Delta_r G_{\text{ox,B}}^\circ. \quad (11.18)$$

If the A line is lower, this difference is negative and A can reduce B oxide under standard-state conditions. At a line crossing the exchange reaction has $\Delta_r G^\circ = 0$; the preferred direction reverses across the crossing.

11.9.5 Worked example: interpret a line crossing

Let two oxygen-normalized reactions have, in kJ mol^{-1} of O_2 ,

$$\Delta_r G_{\text{ox,A}}^\circ = -500 + 0.200T, \quad \Delta_r G_{\text{ox,B}}^\circ = -420 + 0.120T.$$

Equating the lines gives $T_x = 1000$ K. At 800 K,

$$\Delta_r G_{\text{ox,A}}^\circ = -340 \text{ kJ}, \quad \Delta_r G_{\text{ox,B}}^\circ = -324 \text{ kJ}.$$

A oxidation is lower by 16 kJ mol^{-1} of O_2 , so A can reduce B oxide in their standard states. Above 1000 K, the B oxidation line is lower and the standard-state preference reverses.

11.9.6 Richardson oxygen-pressure scale

At equilibrium, $\Delta_r G^\circ = RT \ln(p_{\text{O}_2}^{\text{eq}}/p^\circ)$. For a fixed oxygen pressure this is a straight line through the origin with slope

$$R \ln\left(\frac{p_{\text{O}_2}}{p^\circ}\right). \quad (11.19)$$

Drawing a ray from the origin through a point on an oxidation line therefore converts $\Delta_r G^\circ$ into equilibrium oxygen pressure. Richardson's nomographic scale collects these rays. Analogous scales use $p_{\text{CO}_2}/p_{\text{CO}}$ and $p_{\text{H}_2\text{O}}/p_{\text{H}_2}$ for carbon- and hydrogen-containing reducing atmospheres.

11.9.7 Non-standard condensed phases

If the metal is dissolved in an alloy or the oxide is a solution, condensed-phase activities are not unity. For

$$2M + O_2 \rightleftharpoons 2MO,$$

$$\Delta_r G = \Delta_r G^\circ + RT \ln \left[\frac{a_{MO}^2}{a_M^2 (p_{O_2}/p^\circ)} \right]. \quad (11.20)$$

Thus

$$\frac{p_{O_2}^{eq}}{p^\circ} = \frac{a_{MO}^2}{a_M^2} \exp \left(\frac{\Delta_r G^\circ}{RT} \right). \quad (11.21)$$

A standard Ellingham chart alone cannot decide a non-standard alloy/slag equilibrium; the activities must be supplied.

What an Ellingham diagram does not tell us

It predicts equilibrium driving forces and comparative stability. It does not give the reaction rate, nucleation barrier, transport through an oxide scale, scale adhesion or whether reactants can make contact. A thermodynamically possible reduction may remain experimentally slow.

Cambridge interactive experiment

Use the University of Cambridge DoITPoMS [Ellingham-diagram teaching package](#) and [interactive Ellingham diagram](#). Select two metals together with carbon or hydrogen, move the temperature crosshair, and record $\Delta_r G^\circ$, K , $p_{O_2}^{eq}$ and any slope change. Then change the metal activity and explain the shift using the non-standard reaction quotient above. The site identifies the tool as educational; use assessed thermodynamic databases for quantitative process design.

11.9.8 Worked example: equilibrium oxygen pressure

If $\Delta_r G^\circ = -400$ kJ per mole of O_2 at 1000 K, then

$$\ln \frac{p_{O_2}^{eq}}{p^\circ} = \frac{-400000}{(8.314)(1000)} = -48.11,$$

and $p_{O_2}^{eq} \approx 1.28 \times 10^{-21} p^\circ$. The very low value means that the oxide is stable over a wide range of ordinary oxygen pressures.

11.10 Numerical equilibrium for several reactions

For multiple reactions, use the extents ξ_r as independent unknowns:

$$N_i = N_{i,0} + \sum_r \nu_{ir} \xi_r, \quad a_i = a_i(\{N_j\}, T, p), \quad \Delta_r G_r^\circ + RT \ln Q_r = 0.$$

A robust calculation must enforce $N_i \geq 0$, use dimensionless activities, and verify elemental balances after convergence.

Calculation checklist

1. List species, phases and conserved elements.
2. Determine the rank of the stoichiometric matrix; remove dependent reactions.
3. Choose independent reaction extents and write every $N_i(\xi)$.
4. Select activity models appropriate to each phase.
5. Solve $Q_r = K_r$ subject to non-negative amounts.
6. Recalculate Q_r/K_r , element balances and phase stability.

11.11 Computational laboratory

1. Generate synthetic cooling curves for five alloy compositions. Add a latent-heat peak and finite instrument response. Compare onset, peak and extrapolated arrest temperatures.
2. Fit a phase-boundary dataset with uncertainty bars. Test whether a smooth fit is consistent with every datum rather than drawing a line through selected points.
3. For ammonia synthesis, solve $Q(\xi) = K$ for several pressures and plot equilibrium conversion against pressure.
4. Construct an Ellingham-style plot from tabulated $\Delta_r H^\circ$ and $\Delta_r S^\circ$. Mark crossings and calculate the corresponding equilibrium oxygen pressures.

11.12 Week 11 learning map**Five essential ideas**

1. A phase diagram combines thermal, structural and compositional evidence.
2. Kinetic delay must be separated from an equilibrium boundary.
3. The reacting-system phase rule counts independent reactions, not written equations.
4. $\Delta_r G = RT \ln(Q/K)$ gives the reaction direction at the present state.
5. Temperature and pressure effects follow from $K(T)$ and the activity expression.

Can I move from an experimental signal to a thermodynamic statement without hiding the assumptions in between?

Week 12

Interfaces, irreversible thermodynamics and diffusion

Where we are

Equilibrium thermodynamics identifies stable phases and reaction states. Real transformations also create interfaces and transport heat or matter. This week completes the course storyline:

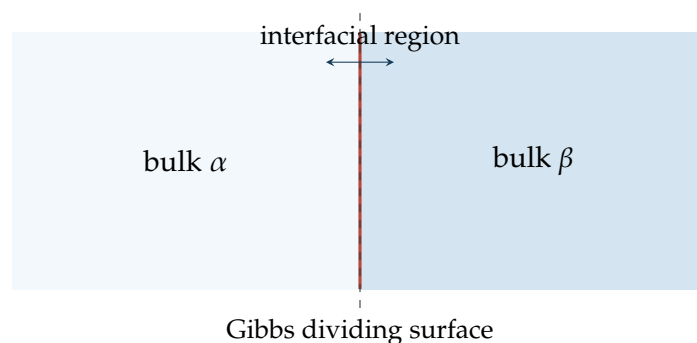
free-energy landscape $\rightarrow$ driving force $\rightarrow$ flux or interface motion $\rightarrow$ microstructural change.

12.1 Why an interface changes equilibrium

A bulk phase has an extensive Gibbs energy. An interface adds an energy proportional to area:

$$G = G^\alpha + G^\beta + \gamma A, \quad (12.1)$$

where γ is the interfacial free energy. When A/V is large, as in nanoparticles, fine precipitates or porous materials, the interface can measurably shift equilibrium.



12.2 Gibbs dividing surface and surface excess

Replace the real, diffuse interface by a mathematical dividing surface. For any extensive property Z ,

$$Z^\sigma = Z - Z^\alpha - Z^\beta, \quad \Gamma_Z = \frac{Z^\sigma}{A}. \quad (12.2)$$

The numerical value of a surface excess depends on the chosen dividing surface. One can choose a particular component k to have $N_k^\sigma = 0$; this is the Gibbs dividing surface relative to component k . At fixed T , p and composition, the reversible work needed to create area is

$$\gamma = \left(\frac{\partial G}{\partial A} \right)_{T, p, N_i} \quad (12.3)$$

For multicomponent interfaces, adsorption couples composition and interfacial energy through the Gibbs adsorption equation. At constant T and p ,

$$d\gamma = - \sum_i \Gamma_i d\mu_i. \quad (12.4)$$

Interface versus dividing surface

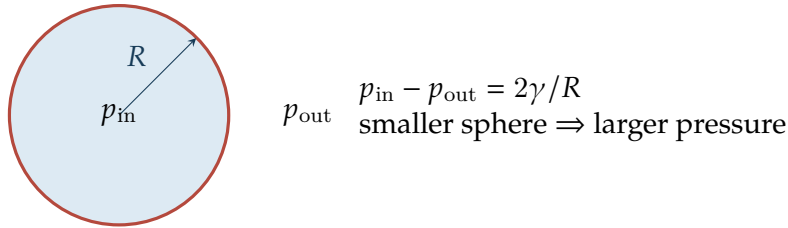
The physical interface has finite thickness and spatially varying structure. The Gibbs dividing surface is an accounting construction that makes bulk and excess contributions precise. Moving the construction changes individual excesses, but it does not move the physical interface.

12.3 Curvature and the Laplace pressure

For a surface with principal radii R_1 and R_2 , mechanical equilibrium gives

$$\Delta p = p_{\text{inside}} - p_{\text{outside}} = \gamma \left(\frac{1}{R_1} + \frac{1}{R_2} \right). \quad (12.5)$$

For a sphere, $R_1 = R_2 = R$ and $\Delta p = 2\gamma/R$. For a cylinder, one radius is infinite and $\Delta p = \gamma/R$.



12.3.1 Worked example: a nanoscale precipitate

For $\gamma = 0.20 \text{ J m}^{-2}$ and $R = 10 \text{ nm}$,

$$\Delta p = \frac{2(0.20)}{10^{-8}} = 4.0 \times 10^7 \text{ Pa} = 40 \text{ MPa}.$$

Capillary pressure is therefore not a small correction at nanometre length scales.

12.4 Kelvin, Gibbs-Thomson and Ostwald-Freundlich relations

At equilibrium the chemical potential must be equal across a curved interface. For an incompressible condensed phase with molar volume V_m , the pressure shift changes its chemical potential by

$V_m \Delta p$. Comparing a spherical particle with a planar interface gives

$$\mu(R) - \mu(\infty) = \frac{2\gamma V_m}{R}. \quad (12.6)$$

For vapour pressure over a spherical droplet,

$$\ln \frac{p(R)}{p(\infty)} = \frac{2\gamma V_m}{RRT}. \quad (12.7)$$

For a dilute solute in equilibrium with a spherical precipitate,

$$c_{\text{eq}}(R) = c_{\text{eq}}(\infty) \exp\left(\frac{2\gamma V_m}{RRT}\right). \quad (12.8)$$

Small particles have a higher surrounding equilibrium concentration. Matter can therefore diffuse from smaller particles toward larger ones: the thermodynamic origin of Ostwald ripening.

12.4.1 Worked example: curvature-raised solubility

Let $\gamma = 0.50 \text{ J m}^{-2}$, $V_m = 7.0 \times 10^{-6} \text{ m}^3 \text{ mol}^{-1}$, $R = 5 \text{ nm}$ and $T = 800 \text{ K}$. Then

$$\frac{2\gamma V_m}{RRT} = \frac{2(0.50)(7.0 \times 10^{-6})}{(5 \times 10^{-9})(8.314)(800)} = 0.2105,$$

so $c_{\text{eq}}(R)/c_{\text{eq}}(\infty) = e^{0.2105} = 1.234$. The equilibrium solubility beside the small particle is about 23% higher than beside a planar interface.

For solid-liquid equilibrium, curvature produces a melting-point depression. With $\Delta H_m > 0$,

$$T_m(R) \simeq T_m(\infty) \left(1 - \frac{2\gamma_{sl} V_m}{R \Delta H_m}\right). \quad (12.9)$$

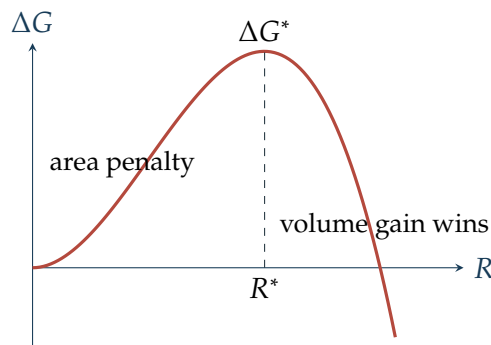
12.5 Classical nucleation as competition between volume and area

For a spherical nucleus with bulk driving force $\Delta g_v < 0$ per unit volume,

$$\Delta G(R) = 4\pi R^2 \gamma + \frac{4\pi R^3}{3} \Delta g_v. \quad (12.10)$$

The critical radius and barrier are

$$R^* = \frac{2\gamma}{|\Delta g_v|}, \quad \Delta G^* = \frac{16\pi\gamma^3}{3|\Delta g_v|^2}. \quad (12.11)$$



Thermodynamic possibility is not transformation rate

A phase can lower the bulk Gibbs energy and still require a finite nucleus. Interfacial energy creates the barrier; atomic mobility controls how often the system crosses it and how fast a supercritical nucleus grows.

12.5.1 Worked example: critical nucleus

For $\gamma = 0.10 \text{ J m}^{-2}$ and $|\Delta g_v| = 2.0 \times 10^7 \text{ J m}^{-3}$,

$$R^* = 10 \text{ nm}, \quad \Delta G^* = 4.19 \times 10^{-17} \text{ J}.$$

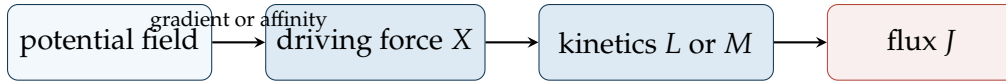
At $T = 800 \text{ K}$, this is about $3795 k_B T$. Homogeneous nucleation is therefore extremely unlikely unless the driving force increases or a heterogeneous site reduces the interfacial cost.

12.6 Driving force, flux and dissipation

Equilibrium thermodynamics tells us whether a change can lower the appropriate potential. It does not by itself give the rate. Kinetics begins by separating two ideas:

$$\boxed{\text{driving force: departure from equilibrium,} \quad \text{flux: rate of response.}} \quad (12.12)$$

The word *force* here means a generalized thermodynamic force. It need not have units of newtons. A chemical-potential gradient, a temperature gradient and a reaction affinity can all drive change.



conservation then converts the flux divergence
into accumulation: $\partial c / \partial t = -\nabla \cdot \mathbf{J} + R$

12.6.1 Entropy production finds the correct force–flux pairs

Near equilibrium, each small region is assumed large enough to possess T , p and μ_i , but small compared with the distance over which those fields change. This is the *local-equilibrium assumption*. Combining local thermodynamics with the balance laws separates entropy change into transported entropy and entropy produced within the material. A useful schematic form of the local entropy-production density is

$$\sigma = \mathbf{q} \cdot \nabla \left(\frac{1}{T} \right) - \sum_i \mathbf{J}_i \cdot \nabla \left(\frac{\mu_i}{T} \right) - \sum_r \frac{r_r \Delta_r G_r}{T} + \frac{1}{T} \boldsymbol{\tau} : \nabla \mathbf{v} \geq 0. \quad (12.13)$$

Here σ has units of entropy produced per volume per time. Each term is a product “flux $\times$ conjugate force.” The signs are chosen so that a spontaneous response gives non-negative entropy production.

Process	Flux or rate	conjugate thermodynamic force
Heat conduction	heat flux $\mathbf{q}$	$\nabla(1/T)$
Diffusion of i	species flux $\mathbf{J}_i$	$-\nabla(\mu_i/T)$
Electrical conduction	charge flux $\mathbf{J}_q$	electric-field contribution divided by T
Chemical reaction r	reaction rate r_r	$-\Delta_r G_r/T$
Interface migration	normal velocity v_n	free-energy reduction per swept volume, divided by T
Viscous flow	viscous stress τ	velocity-gradient combination divided by T

Driving force and flux are not interchangeable

A large driving force may produce a small flux when mobility is low. A small driving force may produce an appreciable flux when mobility is high. Thermodynamics supplies the direction; the kinetic coefficient supplies the scale of the rate.

12.6.2 Linear response near equilibrium

For one uncoupled process, the simplest admissible relation is

$$J = LX, \quad L \geq 0, \quad (12.14)$$

so that $JX = LX^2 \geq 0$. For several simultaneous processes,

$$J_i = \sum_j L_{ij} X_j. \quad (12.15)$$

The diagonal terms describe direct response; the off-diagonal terms describe cross effects. Under the conditions for microscopic reversibility, Onsager reciprocity gives $L_{ij} = L_{ji}$ for properly selected fluxes and forces. Non-negative dissipation requires the symmetric part of $\mathbf{L}$ to be positive semidefinite. For two processes this implies

$$L_{11} \geq 0, \quad L_{22} \geq 0, \quad L_{11}L_{22} - L_{12}^2 \geq 0. \quad (12.16)$$

Thus a temperature gradient can contribute to mass flux and a chemical-potential gradient can contribute to heat flux. These are the Soret and Dufour cross effects. They are omitted only after an assumption, not because coupling is forbidden.

12.6.3 Diffusion: from chemical potential to Fick's law

At uniform temperature and without electrical, stress or other contributions, write the constitutive law as

$$\mathbf{J} = -\mathcal{M} \nabla \mu, \quad \mathcal{M} \geq 0. \quad (12.17)$$

Then $-\mathbf{J} \cdot \nabla \mu / T = \mathcal{M} |\nabla \mu|^2 / T \geq 0$. If $\mu = \mu(c)$,

$$\mathbf{J} = -\mathcal{M} \frac{\partial \mu}{\partial c} \nabla c \equiv -D_{\text{chem}} \nabla c, \quad D_{\text{chem}} = \mathcal{M} \frac{\partial \mu}{\partial c}. \quad (12.18)$$

Fick's first law is therefore a special force–flux law. The kinetic factor $\mathcal{M}$ and the thermodynamic factor $\partial \mu / \partial c$ play different roles. Inside a chemical spinodal, $\partial \mu / \partial c < 0$; the resulting negative

chemical diffusivity signals uphill diffusion and growth of composition fluctuations, not a negative atomic mobility.

For an ideal solution, $\mu = \mu^0 + RT \ln c$ when c is used consistently with its standard state. Hence

$$\nabla\mu = \frac{RT}{c}\nabla c, \quad D = \frac{MRT}{c}. \quad (12.19)$$

If one instead defines $\mathbf{J} = -Mc\nabla\mu$, then $D = MRT$. The apparent difference is only a mobility convention; the definition of the kinetic coefficient must always be stated.

12.6.4 The same structure beyond diffusion

The force–flux pattern is reusable:

$$\mathbf{q} = -k\nabla T = kT^2\nabla(1/T), \quad \text{heat conduction,} \quad (12.20)$$

$$r = -L_r\Delta_r G, \quad \text{reaction near equilibrium,} \quad (12.21)$$

$$v_n = M_\Gamma \mathcal{F}_\Gamma, \quad \text{interface migration,} \quad (12.22)$$

$$\dot{\eta} = -L_\eta \frac{\delta G}{\delta \eta}, \quad \text{non-conserved order parameter,} \quad (12.23)$$

$$\frac{\partial c}{\partial t} = \nabla \cdot \left(M_c \nabla \frac{\delta G}{\delta c} \right), \quad \text{conserved composition field.} \quad (12.24)$$

The last two equations are the Allen–Cahn and Cahn–Hilliard forms. Both say that a thermodynamic derivative supplies the driving force and a positive kinetic coefficient turns it into a rate, while conservation determines whether the evolution law contains a divergence.

12.6.5 Worked check: can the response law violate the second law?

Consider two scalar processes with

$$\begin{pmatrix} J_1 \\ J_2 \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}.$$

The leading minors are $4 > 0$ and $4(2) - 1^2 = 7 > 0$, so the coefficient matrix is positive definite. For $X_1 = 1$ and $X_2 = -1$, $J_1 = 3$, $J_2 = -1$, and

$$\sigma = J_1 X_1 + J_2 X_2 = 3 + 1 = 4 > 0.$$

Notice that the cross term changes both fluxes, yet the total dissipation remains positive.

A practical checklist

Before writing a kinetic equation, ask: What extensive quantity is transported? What potential changes when it moves? What is the conjugate force? What mobility or conductivity relates force to flux? What conservation law converts the flux into accumulation? Which reference frame and boundary conditions define the measured flux?

This progression follows the irreversible-thermodynamics treatment used in Bokstein–Mendeleev–Srolovitz and in Balluffi–Allen–Carter. The accompanying MIT OpenCourseWare sequence begins with fields, gradients, fluxes and continuity, then develops entropy production and conjugate driving forces before applying them to diffusion.

12.7 Conservation comes before a constitutive law

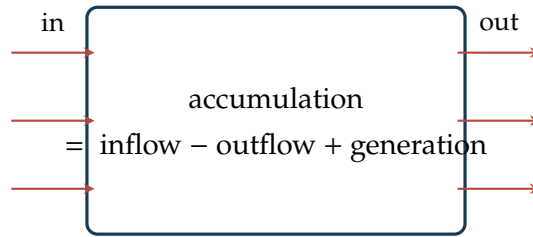
For concentration c in a fixed control volume V ,

$$\frac{d}{dt} \int_V c \, dV = - \int_{\partial V} \mathbf{J} \cdot \mathbf{n} \, dA + \int_V R \, dV. \quad (12.25)$$

Using the divergence theorem gives the local continuity equation

$$\boxed{\frac{\partial c}{\partial t} = -\nabla \cdot \mathbf{J} + R.} \quad (12.26)$$

This statement is independent of the material model for $\mathbf{J}$. It remains valid when Fick's law fails.



12.8 Fick's laws and their assumptions

For a binary isotropic medium with constant diffusivity,

$$\boxed{\mathbf{J} = -D \nabla c} \quad (12.27)$$

is Fick's first law. Combining it with conservation and $R = 0$ gives Fick's second law:

$$\boxed{\frac{\partial c}{\partial t} = D \nabla^2 c.} \quad (12.28)$$

If $D = D(c)$,

$$\frac{\partial c}{\partial t} = \nabla \cdot [D(c) \nabla c], \quad (12.29)$$

and one must not pull D outside the divergence.

For a semi-infinite solid initially at c_0 and held at surface composition c_s ,

$$\boxed{\frac{c(x, t) - c_s}{c_0 - c_s} = \operatorname{erf} \left(\frac{x}{2\sqrt{Dt}} \right).} \quad (12.30)$$

The diffusion length scales as $\sqrt{Dt}$.

12.8.1 Worked example: carburization depth

Let $D = 1.0 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$ and $t = 4.0 \text{ h} = 1.44 \times 10^4 \text{ s}$. The distance at which $(c - c_s)/(c_0 - c_s) = 0.5$ satisfies $\operatorname{erf}(z) = 0.5$, so $z = 0.4769$:

$$x = 2z\sqrt{Dt} = 2(0.4769)\sqrt{(10^{-11})(1.44 \times 10^4)} = 3.62 \times 10^{-4} \text{ m}.$$

The corresponding depth is about 0.36 mm.

12.9 Which diffusivity is being measured?

Quantity	Typical experiment	Reference frame or meaning	What it should not be confused with
Tracer D_i^*	isotope penetration	random motion of labelled atoms	chemical interdiffusion in a concentration gradient
Intrinsic D_i	diffusion couple plus Kirkendall markers	flux of species i relative to the lattice	interdiffusion coefficient
Interdiffusion $\bar{D}$	concentration profile of a diffusion couple	evolution of composition in a chosen volume-fixed frame	either individual intrinsic coefficient
Chemical diffusivity	relaxation of a composition perturbation	mobility multiplied by a thermodynamic factor	tracer diffusivity alone
Mobility M_i	inferred using transport and thermodynamics	kinetic response to a chemical-potential gradient	a diffusivity without a thermodynamic factor

In a binary substitutional alloy, a common form of the Darken relation is

$$\bar{D} = (x_B D_A^* + x_A D_B^*) \Phi, \quad \Phi = \frac{\partial \ln a_A}{\partial \ln x_A} = \frac{\partial \ln a_B}{\partial \ln x_B}, \quad (12.31)$$

under the assumptions used to connect tracer mobilities to intrinsic diffusion. Vacancy-wind corrections, non-ideal volume effects and frame conventions must be stated when greater accuracy is required.

A fitted profile does not identify every coefficient

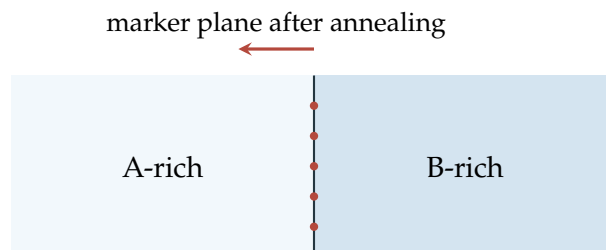
A concentration profile constrains the combined transport response. Different intrinsic mobilities can sometimes reproduce the same profile. Marker motion, tracer data or other independent measurements are needed to distinguish them.

12.10 Kirkendall effect and marker motion

If A and B have unequal intrinsic fluxes relative to the lattice, inert markers move relative to a laboratory-fixed frame. In one dimension,

$$J_A^{\text{int}} = -D_A \frac{\partial c_A}{\partial x}, \quad J_B^{\text{int}} = -D_B \frac{\partial c_B}{\partial x}. \quad (12.32)$$

The lattice drift required by site conservation produces the Kirkendall marker displacement. Marker motion is therefore direct evidence that atoms do not simply exchange in equal counter-fluxes.



12.11 Boltzmann scaling and extracting composition-dependent diffusivity

For a one-dimensional diffusion couple with $D = D(c)$, no sources and time-independent boundary compositions, introduce

$$\eta = \frac{x}{\sqrt{t}}. \quad (12.33)$$

Profiles at different annealing times collapse when plotted against $x/\sqrt{t}$ if the assumptions hold. The Matano construction then converts a measured profile into an interdiffusion coefficient. In one common form,

$$\tilde{D}(c^*) = -\frac{1}{2t} \left(\frac{dx}{dc} \right)_{c^*} \int_{c^-}^{c^*} [x(c) - x_M] dc, \quad (12.34)$$

where x_M is chosen from mass balance.

What time scaling can and cannot show

In a Boltzmann-scaling regime at fixed temperature, diffusivity is a function of composition rather than annealing time. Additional times test self-similarity and experimental reproducibility. Additional couples expand composition coverage. Different temperatures are required to determine temperature dependence.

12.12 Temperature dependence and activation energy

Diffusivity is often approximated by

$$D = D_0 \exp \left(-\frac{Q}{RT} \right). \quad (12.35)$$

Thus a plot of $\ln D$ against $1/T$ has slope $-Q/R$ when one mechanism and one Arrhenius range apply.

12.12.1 Worked example: obtain Q

Suppose $D_1 = 1.0 \times 10^{-15} \text{ m}^2 \text{ s}^{-1}$ at $T_1 = 800 \text{ K}$ and $D_2 = 2.0 \times 10^{-14} \text{ m}^2 \text{ s}^{-1}$ at $T_2 = 900 \text{ K}$. Then

$$Q = R \frac{\ln(D_2/D_1)}{1/T_1 - 1/T_2} = 8.314 \frac{\ln 20}{1/800 - 1/900} = 179 \text{ kJ mol}^{-1}.$$

After finding Q , either datum gives D_0 . A curved Arrhenius plot warns of changing mechanism, phase, defect population or composition.

12.13 Steady state, transient state and boundary conditions

Steady state means $\partial c / \partial t = 0$; it does not mean zero flux. For constant D in a planar membrane of thickness L with fixed endpoint concentrations,

$$c(x) = c_1 + \frac{c_2 - c_1}{L} x, \quad J = -D \frac{c_2 - c_1}{L}. \quad (12.36)$$

At equilibrium the chemical potential is spatially uniform and the flux vanishes. A steady state with $J \neq 0$ requires maintained boundary conditions and continuously produces entropy.

12.14 Worked synthesis: precipitation and coarsening

Consider a supersaturated alloy aged inside a two-phase region.

1. The common tangent identifies equilibrium matrix and precipitate compositions.
2. A finite interfacial energy creates a nucleation barrier.
3. Curvature raises the chemical potential of small precipitates.
4. Diffusion transports solute down the chemical-potential gradient.
5. Small particles dissolve and large particles grow during coarsening.

This one example joins the equilibrium, interfacial and kinetic parts of the course without treating them as unrelated formulae.

12.15 Computational laboratories

1. Plot $\Delta G(R)$ for several driving forces. Track how R^* and ΔG^* change.
2. Plot $c_{\text{eq}}(R)/c_{\text{eq}}(\infty)$ and find the radius below which curvature changes solubility by more than 10%.
3. Solve the one-dimensional diffusion equation for a step profile and verify collapse against $x/\sqrt{t}$.
4. Generate an error-function profile, add realistic noise and estimate D . Compare differentiation-based and direct-fit sensitivity to noise.
5. Plot $\ln D$ against $1/T$ for two mechanisms and identify the temperature at which a single Arrhenius fit becomes misleading.

12.16 Week 12 learning map

Five essential ideas

1. An interface adds free energy and curvature shifts equilibrium.
2. Conservation is general; a flux law supplies the material response.
3. Entropy production pairs fluxes with thermodynamic forces.
4. Tracer, intrinsic, interdiffusion and chemical diffusivities answer different experimental questions.
5. Thermodynamics supplies the direction and driving force; mobility and interfaces determine the path and rate.

Can I identify the conserved quantity, the driving force, the kinetic coefficient and the boundary conditions before solving the equation?